



INFRASTRUCTURE DEPARTMENT STANDARD OPERATING PROCEDURAL MANUAL

Version 1.2

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Reviewed By

Name	Designation	Signature	Date
Temitope Olanipekun	System Administrator		08/Nov/2023

Approved

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Wale Olayisade	Chief Information Officer		11/Dec/2023

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1.0 Data Center Equipment Maintenance Process

1.1 Background

Through quarterly inspections, cleaning, lubricating, and making minor adjustments on our Data Center Equipment, minor problems can be detected and corrected before they become a major problem that can shut down the data center.

1.2 Process Objective (Purpose)

The main purpose of quarterly maintenance is to ensure that all our data center equipment is always operating at 100% efficiency. This process is a guideline for performing our quarterly maintenance on the Data Center equipment. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to resolve the challenges.

1.3 Process Owner

Data Center Equipment Maintenance Process is owned by Group Head, Information Technology while Lead, Infrastructure is saddled with responsibility of implementation.

1.4 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Support /Shift Team	Data center equipment maintenance (FM200 and Smoke detectors)	1.1 Power off the FM200 switch in the console room 1.2 Monitor the vendor while he/she perform the require maintenance 1.3 Allow the dust to subside for about 20 minutes 1.4 Power on the FM200 switch in the console room 1.5 Get a job completion card from the vendor.	30 minutes

Support /Shift Team	Data center equipment maintenance (Cooling Systems, UPS and Inverters)	1.1 Monitor the vendor while he/she perform the require maintenance 1.2 Vendor to review the operations of the HVAC AC, UPS, Standby AC, Inverters in the data center, Inverters in the 5 th floor store room and the batteries of these inverters. 1.3 Vendor to carry out preventive maintenance on the devices Get a job completion card from the vendor	
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1.5 Key Risks and Controls

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/workaround, this should be stated as Not Applicable.

#	Risk	Control
1	Expired FM200 will render the extinguisher useless and will not protect the Data Center in case of a fire.	Supervise maintenance of the FM200 and ensure Job Completion Card is signed by IT and Internal Control.
2	Damage to Data Center Equipment leading to service failure.	Admin sending early notifications to vendors to remind them of the maintenance.
3	Vendors unable to gain access to the data center	Getting approval before maintenance date for vendors access

2.0 Disabling E-mail and Domain Accounts for Ex-employees

2.1 Background

A domain user is one whose username and password are stored on a domain controller rather than the computer the user is logging into. When you log in as a domain user, the computer asks the domain controller what privileges are assigned to you. When the computer receives an appropriate response from the domain controller, it logs you in with the proper permissions and restrictions.

2.2 Process Objective (Purpose)

The purpose of this is to restrict permissions and monitor activities of the end users by the administrator. This process is a guideline to assist the Infrastructure team to disable an existing user on the Active directory who has left the company. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to resolve the challenges.

2.3 Process Owner

Group Head, IT.

2.4 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	Disable User on domain	1.1 Get an approval by email from HCM 1.2 Remote to the domain server 1.3 Launch the Active Directory Users and computer 1.4 Right Click xpresspay.com.ng and click on Find 1.5 Type the username 1.6 Right click the user and click on disable 1.7 Take a screenshot of the disabled page 1.8 Send the evidence to the HCM	10 Minutes

Role	Task	Process Steps	Estimated TAT
Server Administrator	Disable User on Office 365 portal	2.1 Login to the Office 365 admin portal 2.2 Click on search and type the username 2.3 Click on sign in block and block the user from signing in 2.4 Remove the user license as guided in the screenshot 2.5 Take the screenshot and send to HCM	20 Minutes

2.5 Key Risks and Controls.

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

#	Risk	Control
1	Privilege escalation leading to loss of data, fraud etc.	IS Control performing daily review of exited Staff to ensure they have been disabled from the Active Directory/Office 365
2	Access to unauthorized users	Prompt sending of exited Staff by HCM to IT Department for prompt blocking.

3.0 Creation of New User on Active Directory

3.1 Background

A domain user is one whose username and password are stored on a domain controller rather than the computer the user is logging into. This process is a guideline required to create a new user on the Active directory. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to resolve the challenges.

3.2 Process Objective (Purpose)

When you log in as a domain user, the computer asks the domain controller what privileges are assigned to you. When the computer receives an appropriate response from the domain controller, it logs you in with the proper permissions and restrictions. The purpose of this is to create new end users by the administrator.

3.3 Process Owner

Group Head, Information Technology

3.4 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	Enable User on Office 365 portal	1.0 Login to the Office 365 admin portal 1.1 Click on new users 1.3 Supply the user profile 1.4 Add user to group 1.5 Click ok	20 Minutes

3.5 Key Risks and Controls

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/workaround, this should be stated as Not Applicable.

Risk	Control
Work delay for the Staff not created leading to High TAT	IS Control to ensure that new Staff are profiled on the Active Directory immediately after resumption.

4.0 How to replicate VM servers to DR site-Nutanix

4.0 Background

A disaster recovery test (DR test) is the examination of each step in a disaster recovery plan as outlined in an organization's business continuity/disaster recovery (BCDR) planning process. Evaluating the DR plan helps ensure that an organization can recover data, restore business critical applications, and continue operations after an interruption of services.

4.1 Process Objective (Purpose)

This process is a guideline for performing failover testing and replication of virtual machines to the DR site. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to resolve the challenges.

4.2 Process Owner

Group Head, IT

4.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
System Administrator	Pre-failover meeting	1.1 Schedule meetings with the stakeholders to carry everybody along	As estimated
	Pre-Fail Checks Over	2.1 All PSSP, PTSP and Agency Banking applications were accessed via the internet & internal links 2.2 All PSSP, PTSP and Agency Banking applications were accessed via the LAN	As Schedule

		2.3 Database key record counts & size were taken	
IT Personnel (Infrastructure, Network, Database, Developer)	Scenarios Simulated	3.1 Primary HQ internet went down. 3.2 Application failover to DR 3.3 Accessing application (on the LAN) while on DR 3.4 Accessing application (on the internet) while on DR	As Schedule
IT Personnel (Infrastructure, Network, Database, Developer)	Post DR checks: Failback to HQ	4.1 Failback of the Network, applications and DB was done.	As Schedule

4.4 Key Risks/Controls

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

Key Risks	Controls
Hardware/software failure	Scheduling bi-annual failover test
Cyber Security Risk	Failover testing
Loss of Data	Providing cold and hot backup site for data backup

Service Failure	Providing multiple network connections to the DR Site to facilitate efficient backup
Compliance requirements failure	Documenting results of failover testing done

4.5 Steps to Failover to DR site

Step 1: Screenshot of Servers up and running at the Primary Site (23 Oba Akinjobi Street. Ikeja GRA. Lagos)

Servers are categorized in groups

Apps-Fullfailover2: Virtual machines on Nutanix HCI:

Name	vCPU	Memory	IP Addresses	Cluster	Hypervisor	OS	NGT	Project	Owner
VAccount	1	8 GiB	172.21.7.20	hqhci.xpresspay.c...	AHV	-	Not Installed	default	admin
AVS	1	8 GiB	172.21.7.63	hqhci.xpresspay.c...	AHV	-	Not Installed	default	admin
Xpress_View	2	6 GiB	172.21.7.60	hqhci.xpresspay.c...	AHV	-	Not Installed	default	admin
XPwallet	2	16 GiB	172.21.7.64	hqhci.xpresspay.c...	AHV	-	Not Installed	default	admin
XpressWalletFront	2	7 GiB	172.21.7.68	hqhci.xpresspay.c...	AHV	-	Not Installed	default	admin
XpressPayout	2	20 G...	172.21.7.36	hqhci.xpresspay.c...	AHV	Windows Serve...	Latest	default	admin
XpressGateway_	4	24 G...	172.21.9.9	hqhci.xpresspay.c...	AHV	Windows Serve...	Latest	default	admin
XpressDirect	4	64 G...	172.21.7.29	hqhci.xpresspay.c...	AHV	-	Not Installed	default	admin
XPAY	4	64 G...	172.21.9.13	hqhci.xpresspay.c...	AHV	Windows Serve...	Latest	default	admin

Step 2: Network

The Border Gateway Protocol (BGP) peering at the DR site is initially set to shutdown mode and must be turned on (Activated) before the primary site BGP peering is turned off for a planned failover process.

4.5.1 Border Gateway Protocol (BGP) before failover at the DR

```
XpressPayRTR#sh ip bg
XpressPayRTR#sh ip bgp nei
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 re
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 received-
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 received-routes

Total number of prefixes 0
XpressPayRTR#
XpressPayRTR#sh ip bgp neighbors 154.113.16.113ad
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 ad
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 advertised-routes

Total number of prefixes 0
```


BGP before failover at the head office

```
XpressPayR1#sh ip bgp neighbors 41.75.82.201 received-routes
BGP table version is 16928, local router ID is 80.88.8.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*    0.0.0.0          41.75.82.201             0  37282 3356 i

Total number of prefixes 1
XpressPayR1#
XpressPayR1#sh ip bgp neighbors 41.75.82.201 ad
XpressPayR1#sh ip bgp neighbors 41.75.82.201 advertised-routes
BGP table version is 16928, local router ID is 80.88.8.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*> 80.88.8.0/24      0.0.0.0              0         32768 i
```

Step 3: Login to the Nutanix Prism Element at the DR Site (KM 22 Ogombo Road, Ajah Lagos) and validate the Recovery plan.

Apps-FullFailover1 ▾ Validate Test Clean-up test VMs Update Failover ... More ▾ ? X

Summary Tasks Alerts

Name
Apps-FullFailover1

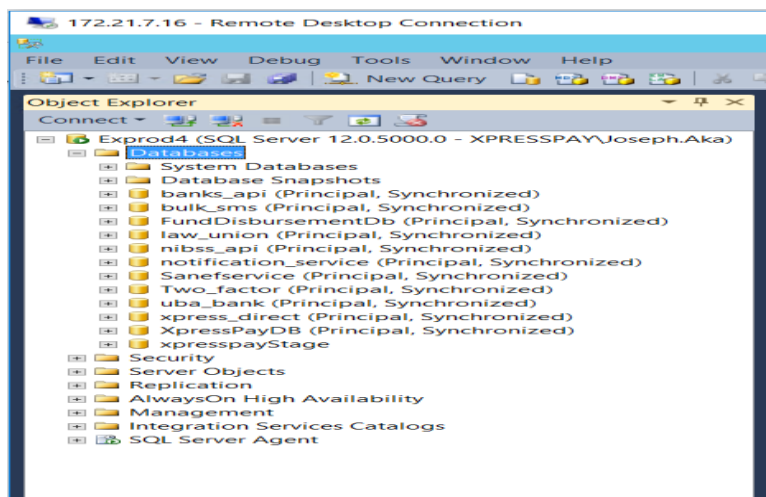
Validate Executed By
simisola.sah...@xpresspayments.com
Failover from
PC_172.22.51.38
Failover to
Local AZ
100%
1. Validating
Failed

Execution Reports 8 Recovery Plan Jobs search in table Q

Reports	Failover from	Failover to	Executed By	Generated On	Start Time	Duration	Status
Validate	PC_172.22.51.38	Local AZ	simisola.sah...	November 7, 2022 ...	November 7, 2022 ...	7 Seconds	Succeeded ...
Validate	PC_172.22.51.38	Local AZ	-	August 15, 2022 at ...	August 15, 2022 at ...	7 Seconds	Failed
Validate	PC_172.22.51.38	Local AZ	-	August 15, 2022 at ...	August 15, 2022 at ...	7 Seconds	Failed
Validate	PC_172.22.51.38	Local AZ	-	June 30, 2022 at 01...	June 30, 2022 at 01...	9 Seconds	Failed

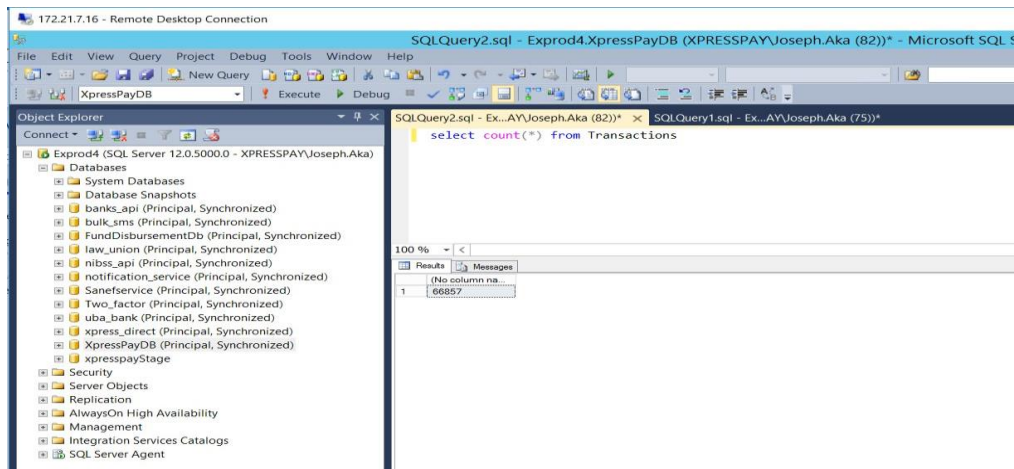
Database server before failover at the Head Office (23 Oba Akinjobi Street. Ikeja GRA. Lagos)

4.5.1.1 EXPROD1: For PSSP databases (PAYOUT, E-cashier, VAS and REPORT)



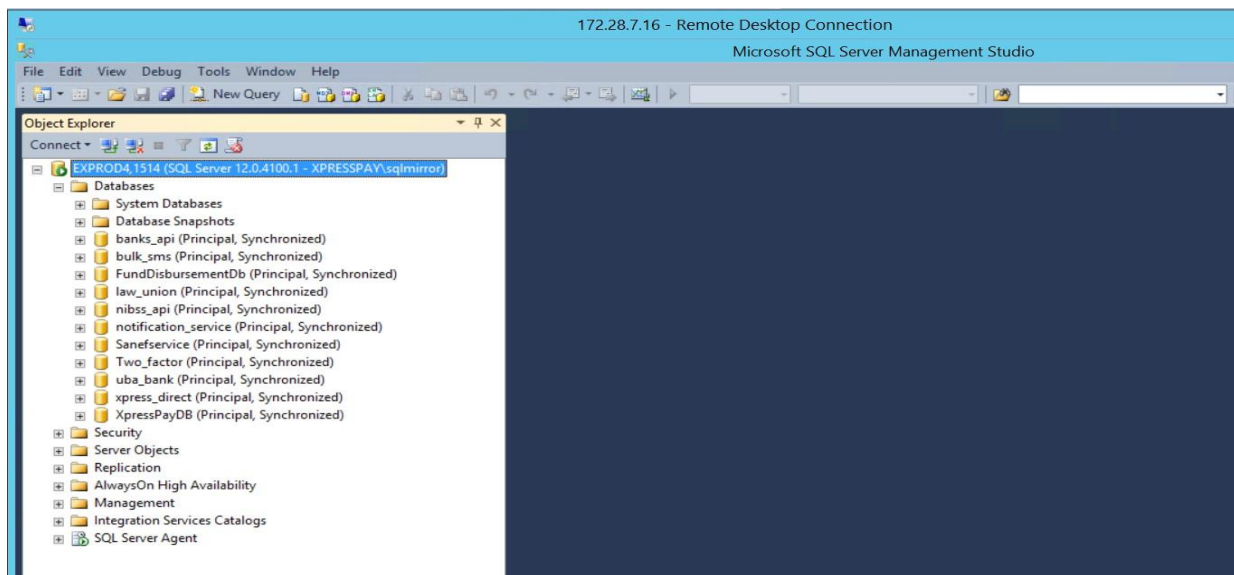
Transaction count on database before failover(e-cashier)

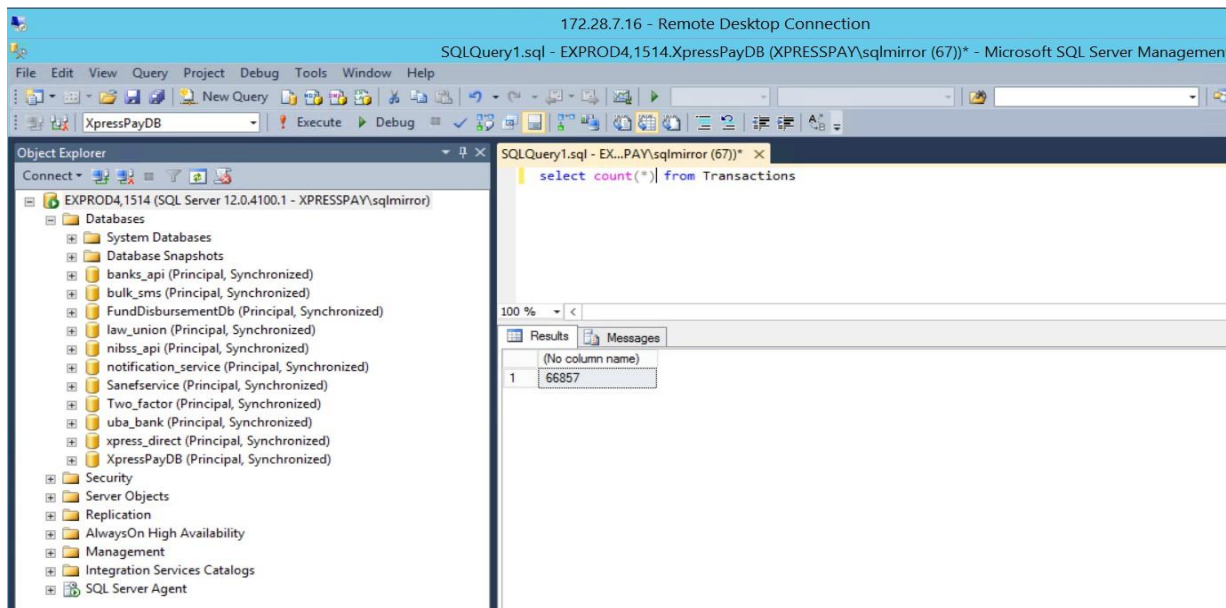
4.5.1.2 **EXPROD1:** Count of e-cashier transaction before failover



Database server before failover at the DR Site (KM 22 Ogombo rd, Ajah Lagos)

4.5.1.3 **EXPROD1:** Shows status of the database in DR site before failover





After the BGP peering is activated at the DR site, it is tuned off at the primary site so that the BGP peering IPs can failover to the DR site.

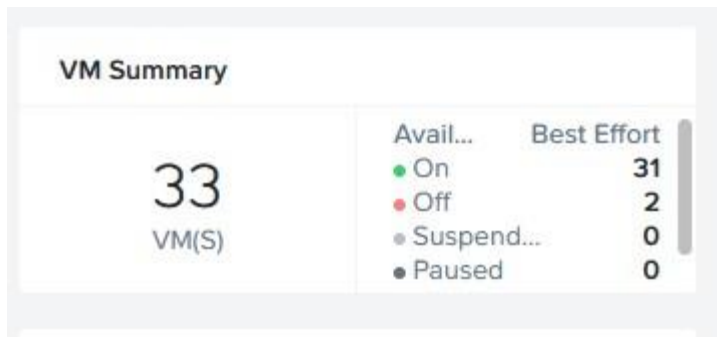
To failover to the DR site, all replicating servers had to be shut down and then failed over with some menu commands showing in the screenshots below.

Step4: Upon DBs Failover and Recovery plan Validations at the DR Prism element, Begin the failover of VMs to the DR site by clicking failover on the Recovery plans

Server Count Before Failover

VM Summary											
24 VM(S)	<table> <tr> <th>Avail...</th><th>Best Effort</th></tr> <tr> <td>● On</td><td>22</td></tr> <tr> <td>● Off</td><td>2</td></tr> <tr> <td>● Suspend...</td><td>0</td></tr> <tr> <td>● Paused</td><td>0</td></tr> </table>	Avail...	Best Effort	● On	22	● Off	2	● Suspend...	0	● Paused	0
Avail...	Best Effort										
● On	22										
● Off	2										
● Suspend...	0										
● Paused	0										

Server Count Before Failover

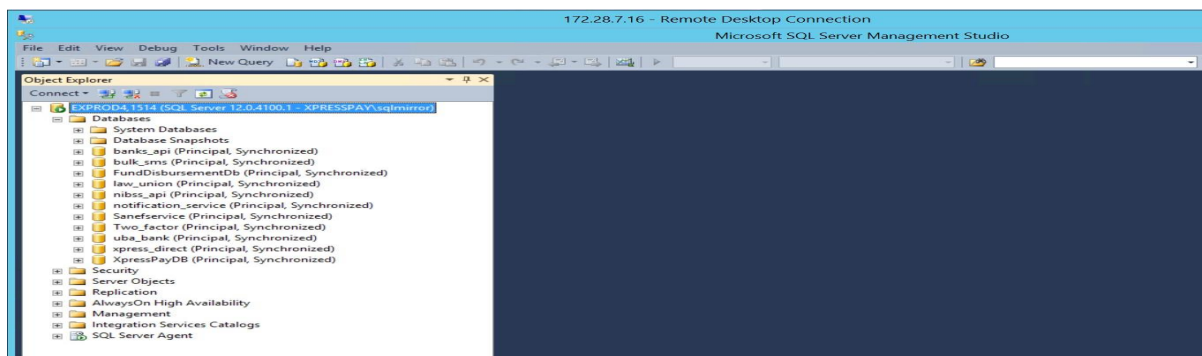


Servers in the DR

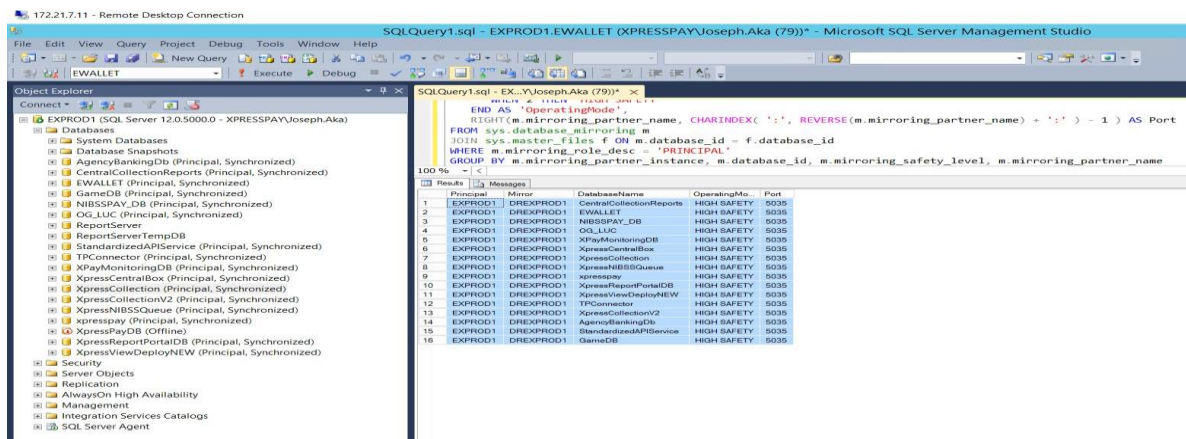
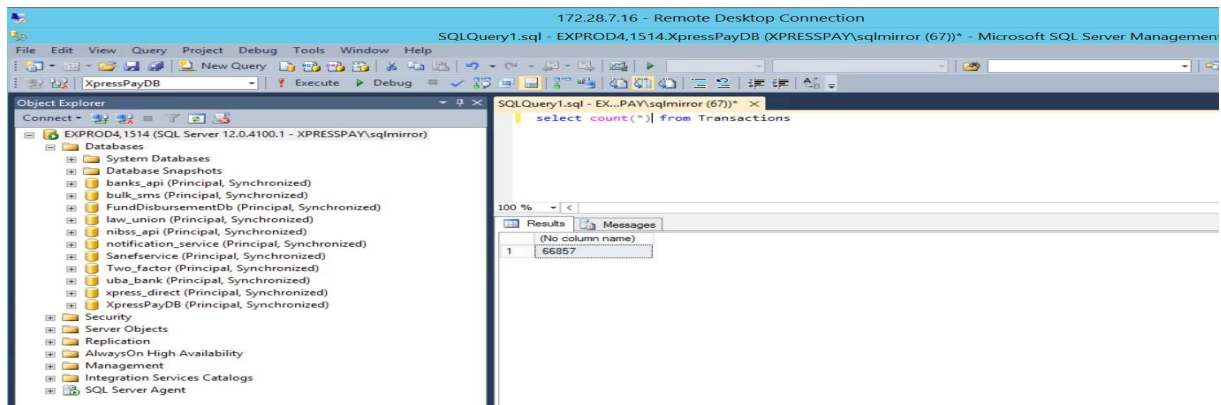
• Nutanix-Test-AppOthe...	XPRESSPAYN TNXDR03/AH V	1	9 GiB	69.1 GiB / 200 GiB	59.61 %	-	0	0	0 KBps	0 ms	Yes	No
• Nutanix-Test-Coperate...	XPRESSPAYN TNXDR01/AH V	1	4 GiB	11.02 GiB / 30 GiB	39.33 %	12.79 %	83	3	5.16 MBps	22.7 ms	Yes	No
• Nutanix-Test-ECashier	XPRESSPAYN TNXDR03/AH V	1	8 GiB	103.49 GiB / 500 GiB	62.76 %	-	-	-	-	-	Yes	No
• Nutanix-Test-EJournal	XPRESSPAYN TNXDR01/AH V	1	13 GiB	171.78 GiB / 300 GiB	60.09 %	-	26	0	2.76 MBps	35.44 ms	Yes	No
• Nutanix-Test-mCash A...	XPRESSPAYN TNXDR03/AH V	2	8 GiB	24.41 GiB / 300 GiB	29.68 %	-	0	0	0 KBps	0 ms	Yes	No
• Nutanix-Test-MiniTMS	XPRESSPAYN TNXDR03/AH V	2	5 GiB	51.51 GiB / 300 GiB	29.91 %	-	0	0	0 KBps	0 ms	Yes	No
• Nutanix-Test-MOFRep...	XPRESSPAYN TNXDR01/AH V	3	16 GiB	325.15 GiB / 350 GiB	19.57 %	0%	23	0	2.58 MBps	38.85 ms	Yes	No
• Nutanix-Test-Notificati...	XPRESSPAYN TNXDR03/AH V	1	16 GiB	21.5 GiB / 250.13 GiB	58.25 %	0%	-	-	-	-	Yes	No

Virtual Machine Servers in a running state at the DR Site (KM 22 Ogombo rd, Ajah Lagos)

Database server in DR now principal (KM 22 Ogombo Road, Ajah Lagos)

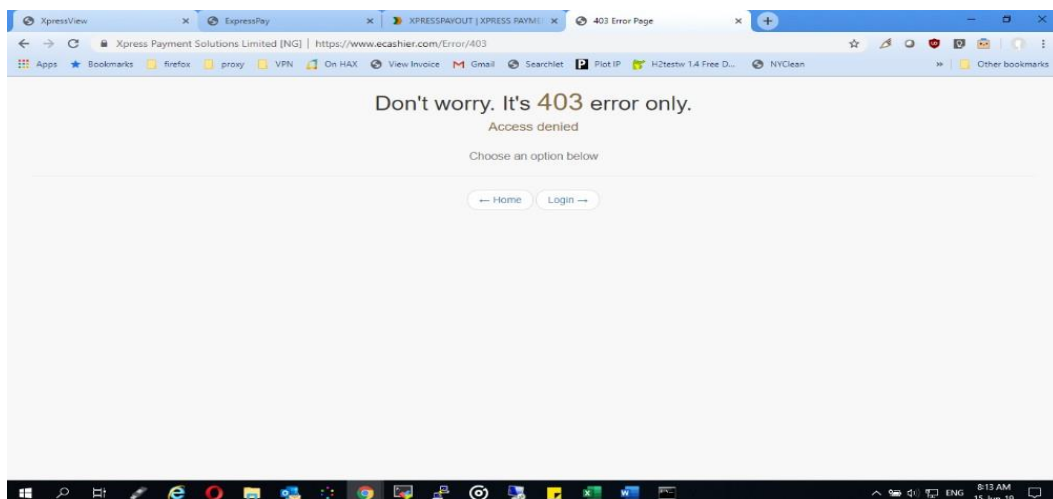


Database Transaction count in DR

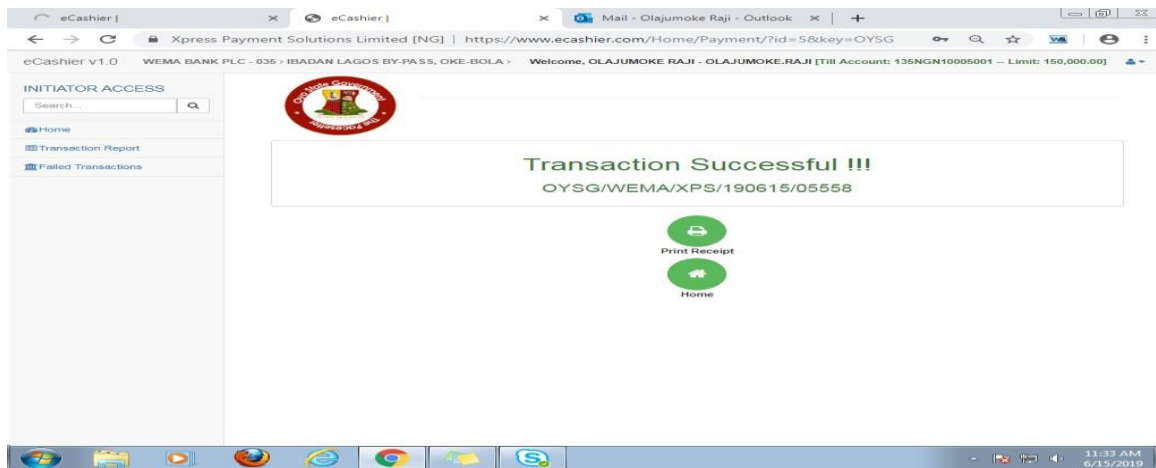


Step 6: Application Testing over the Internet to the DR Site during Failover

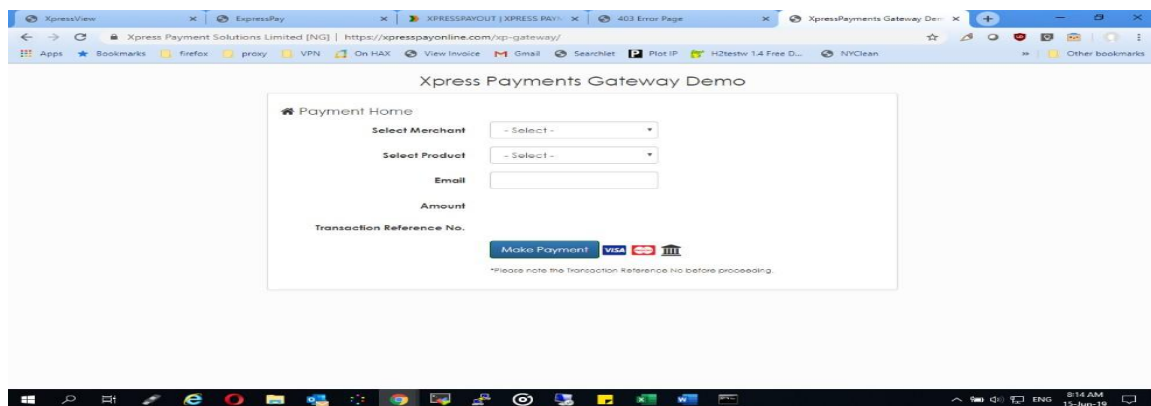
E-Cashier



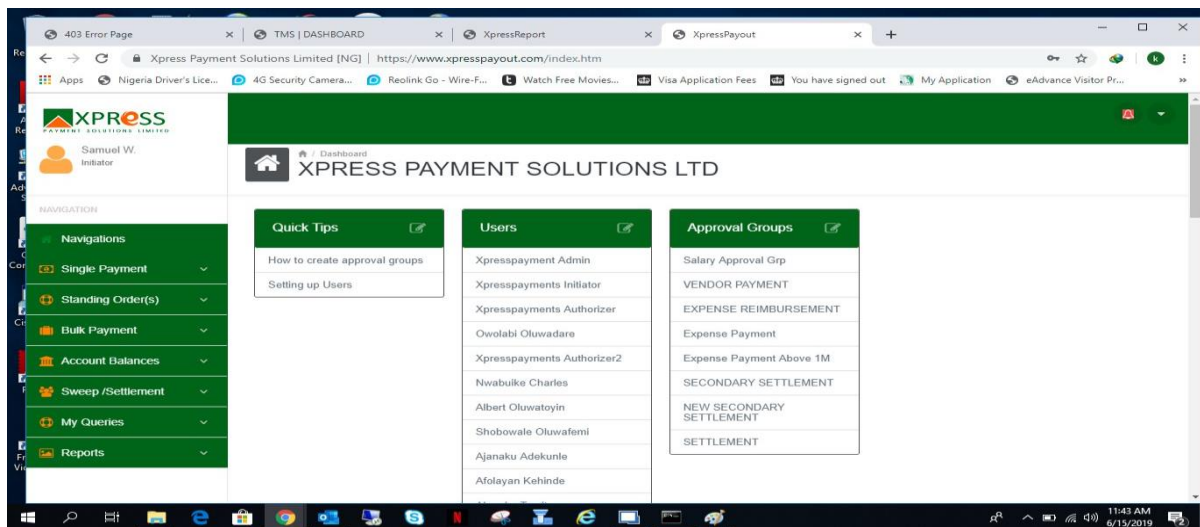
Ecashier Transactions from Wema Bank during Failover



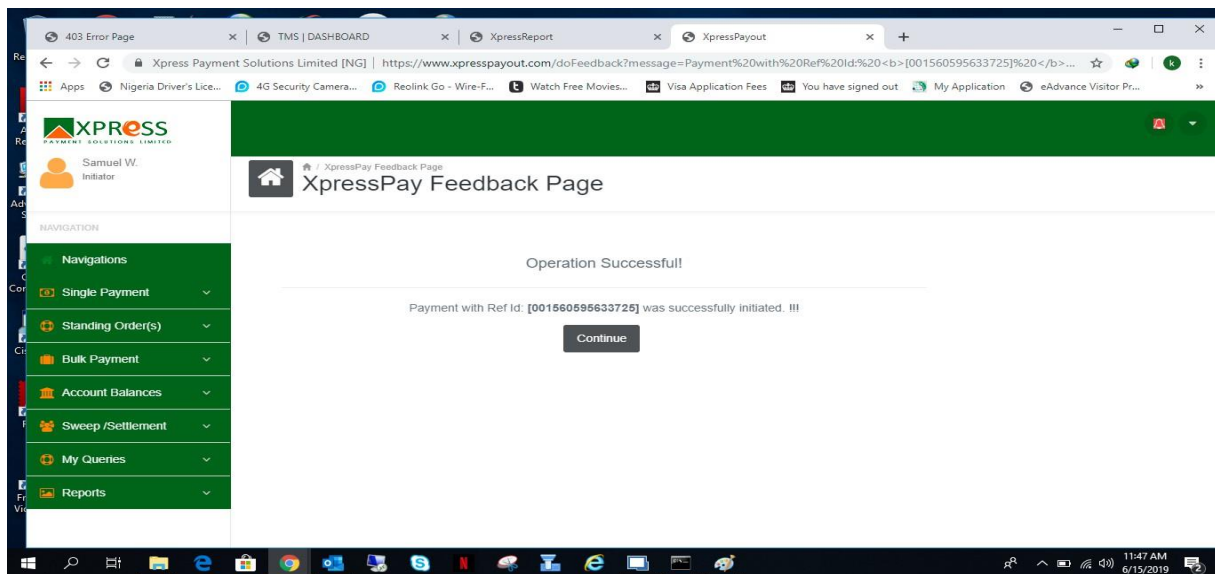
Payment Gateway



Xpress Payout Portal



Payment Upload on Xpress Payout



BGP after failover at the HQ (23 Oba Akinjobi Street. Ikeja GRA. Lagos)

```
XpressPayR1#sh ip bgp neighbors 41.75.82.201 ad
XpressPayR1#sh ip bgp neighbors 41.75.82.201 advertised-routes

Total number of prefixes 0
XpressPayR1#
XpressPayR1#sh ip bgp neighbors 41.75.82.201 rece
XpressPayR1#sh ip bgp neighbors 41.75.82.201 received-
XpressPayR1#sh ip bgp neighbors 41.75.82.201 received-routes

Total number of prefixes 0
XpressPayR1#
```

BGP after failover at the DR site (KM 22 Ogombo Road, Ajah Lagos)

```
XpressPayR1R#sh ip bgp neighbors 154.113.16.113 advertised-routes
BGP table version is 3, local router ID is 80.88.8.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
   *> 80.88.8.0/24  0.0.0.0                0         32768 i

Total number of prefixes 1
XpressPayR1R#
XpressPayR1R#
XpressPayR1R#sh ip bgp neighbors 154.113.16.113 received-routes
BGP table version is 3, local router ID is 80.88.8.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
   *> 0.0.0.0      154.113.16.113       0 37282 3356 i

Total number of prefixes 1
XpressPayR1R#
```

BGP before failover at the DR site (KM 22 Ogombo Road, Ajah Lagos)


```

XpressPayRTR#sh ip bg
XpressPayRTR#sh ip bgp nei
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 re
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 received-
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 received-routes

Total number of prefixes 0
XpressPayRTR#
XpressPayRTR#sh ip bgp neighbors 154.113.16.113ad
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 ad
XpressPayRTR#sh ip bgp neighbors 154.113.16.113 advertised-routes

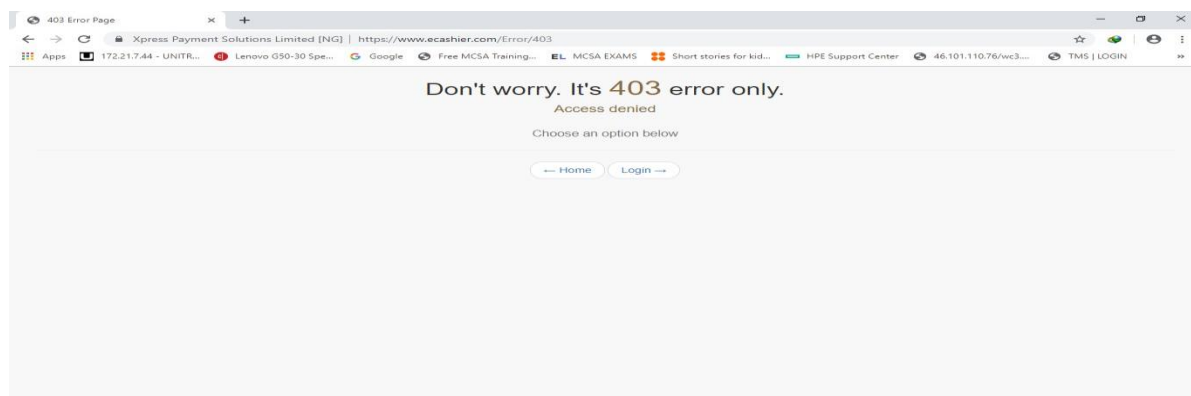
Total number of prefixes 0

```

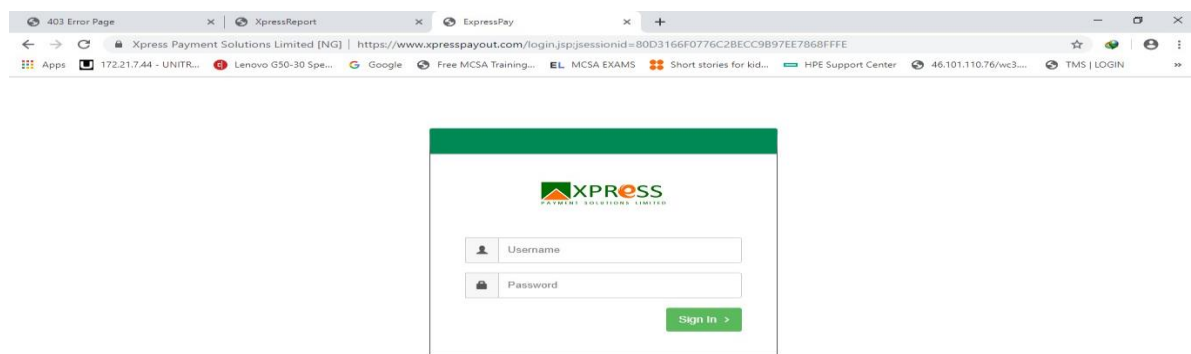
Step 7: Reverse failover back to Primary Site;

Initiate Step 1 to 4 at the head office

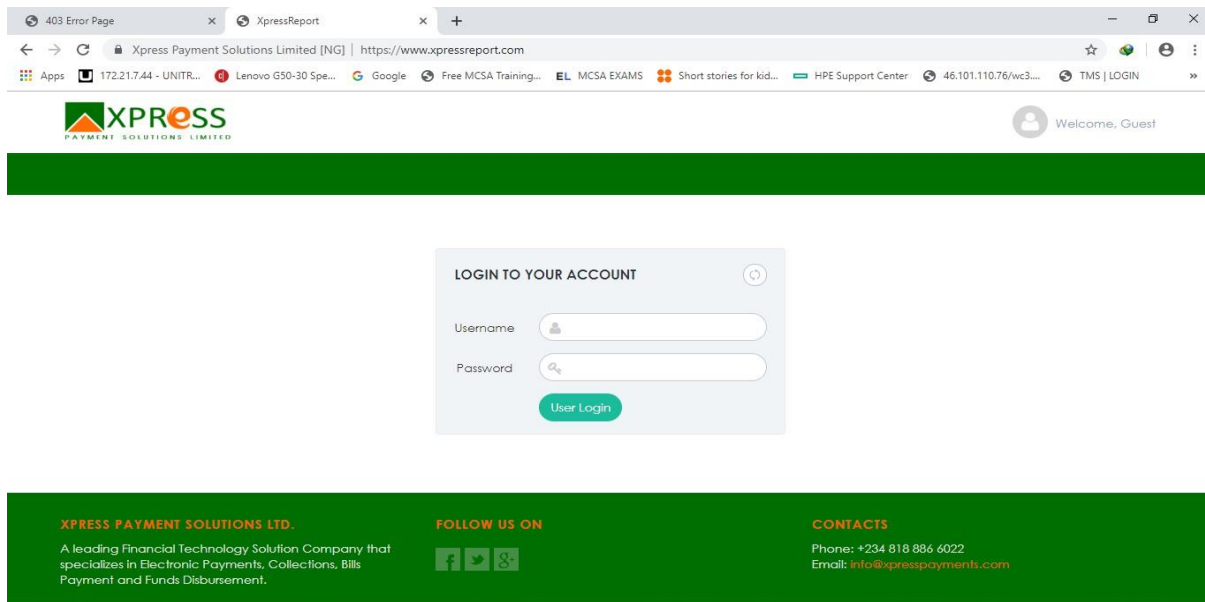
E-Cashier



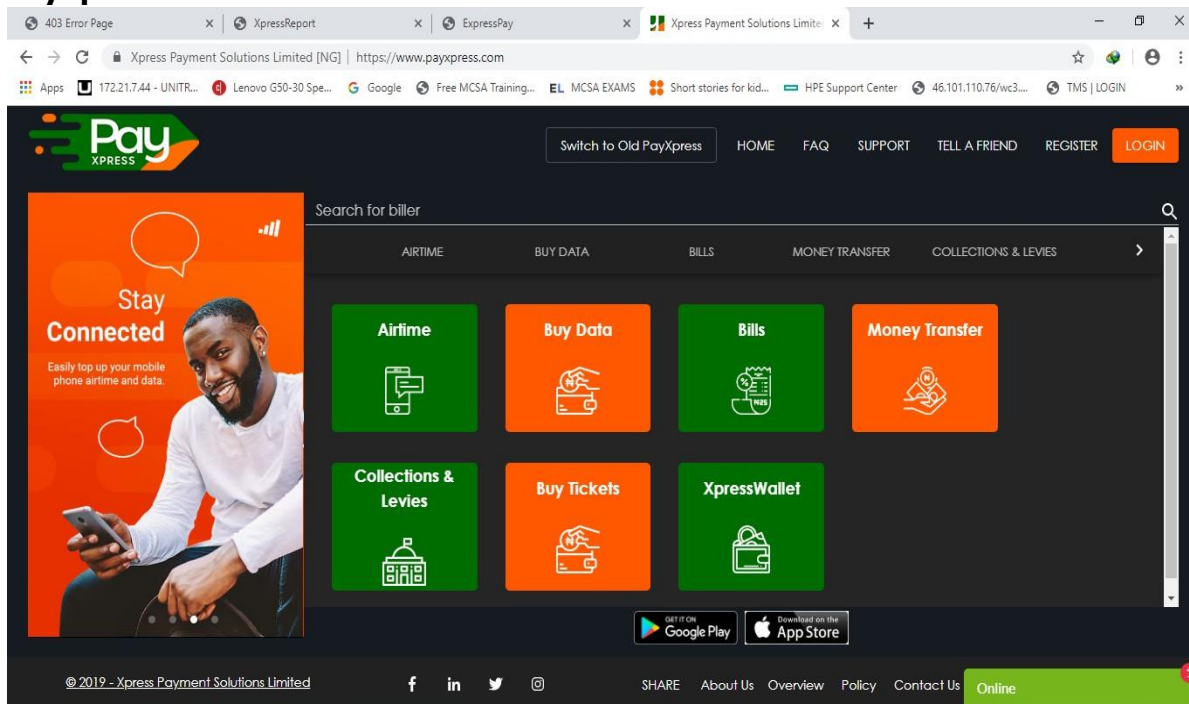
Xpress Payout



Xpress Report



PayXpress



5.0 How to deploy Wildfly on webserver

5.1 Background

This is a report dashboard designed to provide detailed transaction reports both in textual and graphical forms which can be used for various activities such as reconciliations, auditory activities, and treasury activities. This application run on Wildfly.

Wildfly is a flexible, lightweight, managed application runtime that helps you build amazing applications.

5.2 Process Objective (Purpose)

This process is a guideline for deploying Wildfly on **XpressReport** platform. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges

5.3 Process Owner

Group Head, IT

5.4 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	Wildfly deployment on xpressreport server	1.1 Login to the report server 1.2 Restart the server 1.3 Stop the wildfly_report service 1.4 Cut the deployment file to a different location 1.5 Start the wildfly_report 1.6 Copy the file back to deployment folder starting with "<i>CollectionReport.ear</i>" wait for it to finish deployed 1.7 Copy other file at once and paste it on deployment folder wait for it to finish deployed. 1.8 Launch a web browser and paste the url "https://www.xpressreport.com/" to confirm the site is up. 1.9 Sign out of the system.	30 Minutes

		2.0 Follow the screen shot below	
--	--	----------------------------------	--

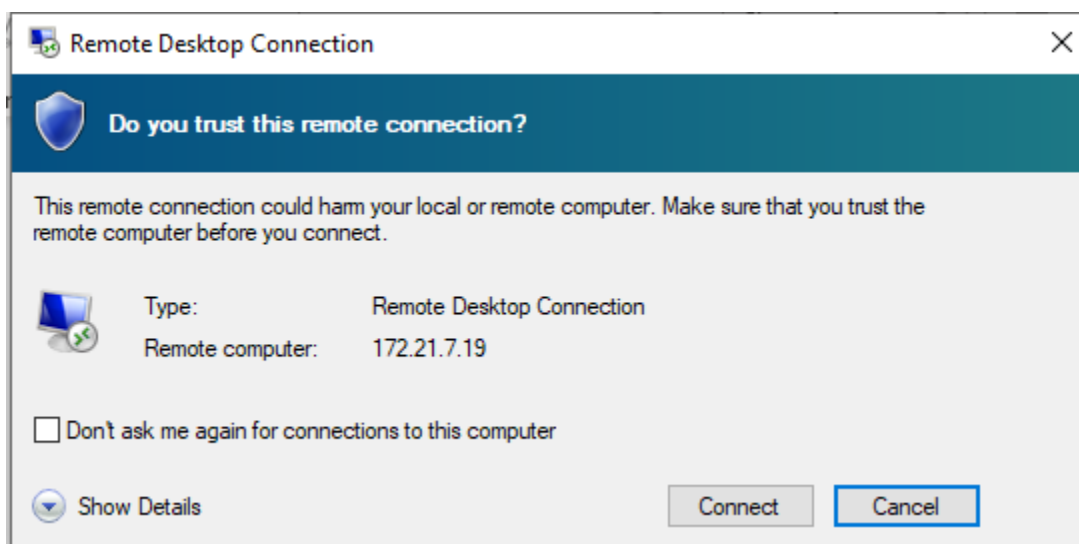
5.5 Risk/Controls

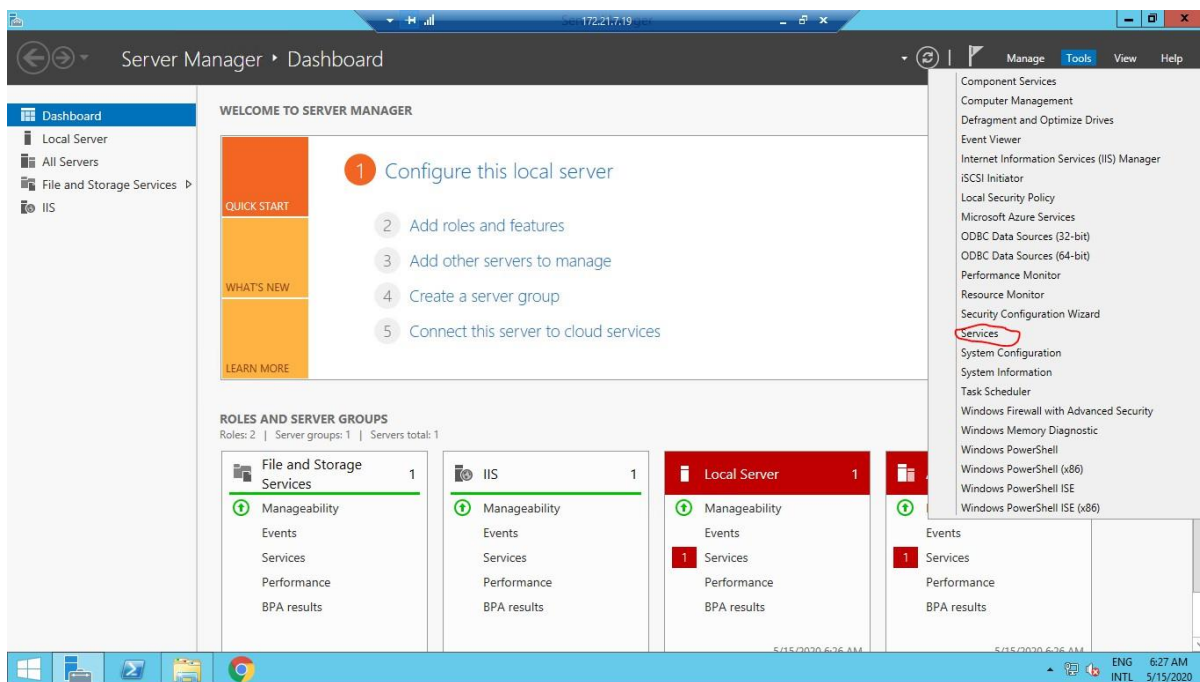
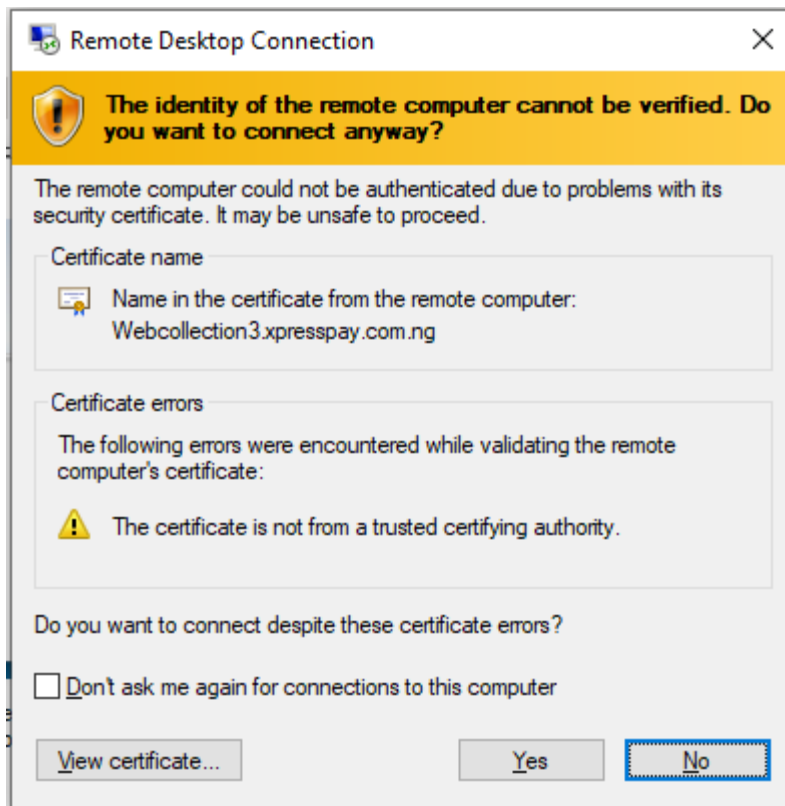
In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

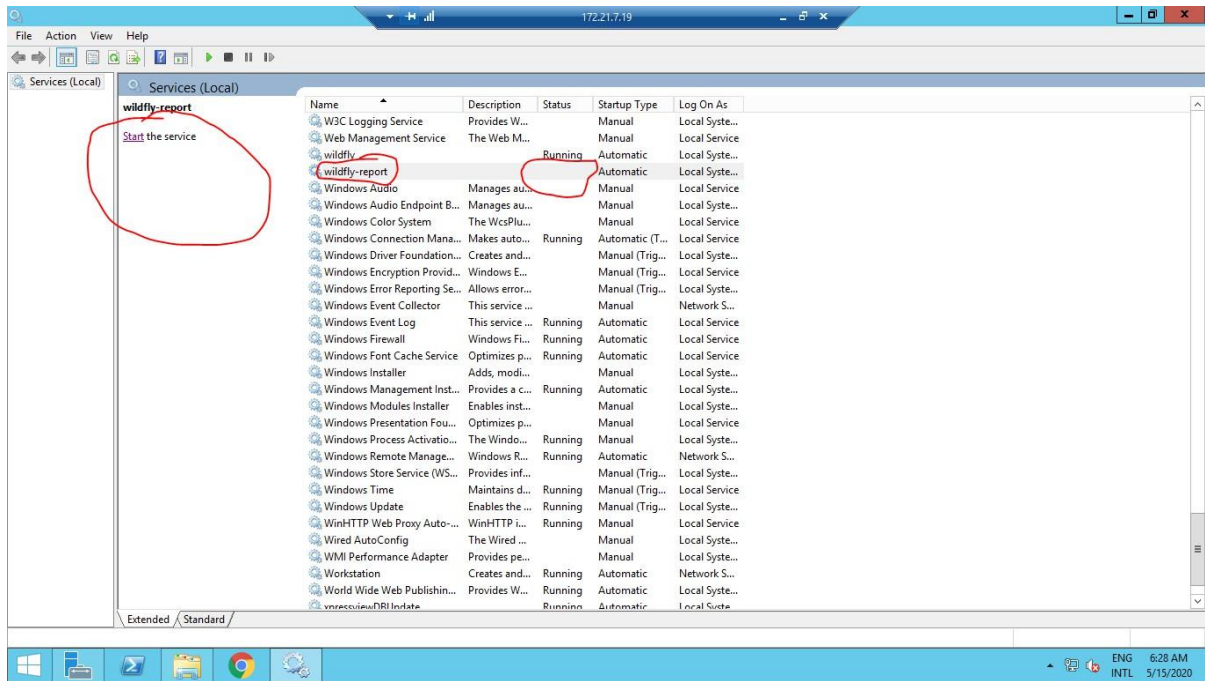
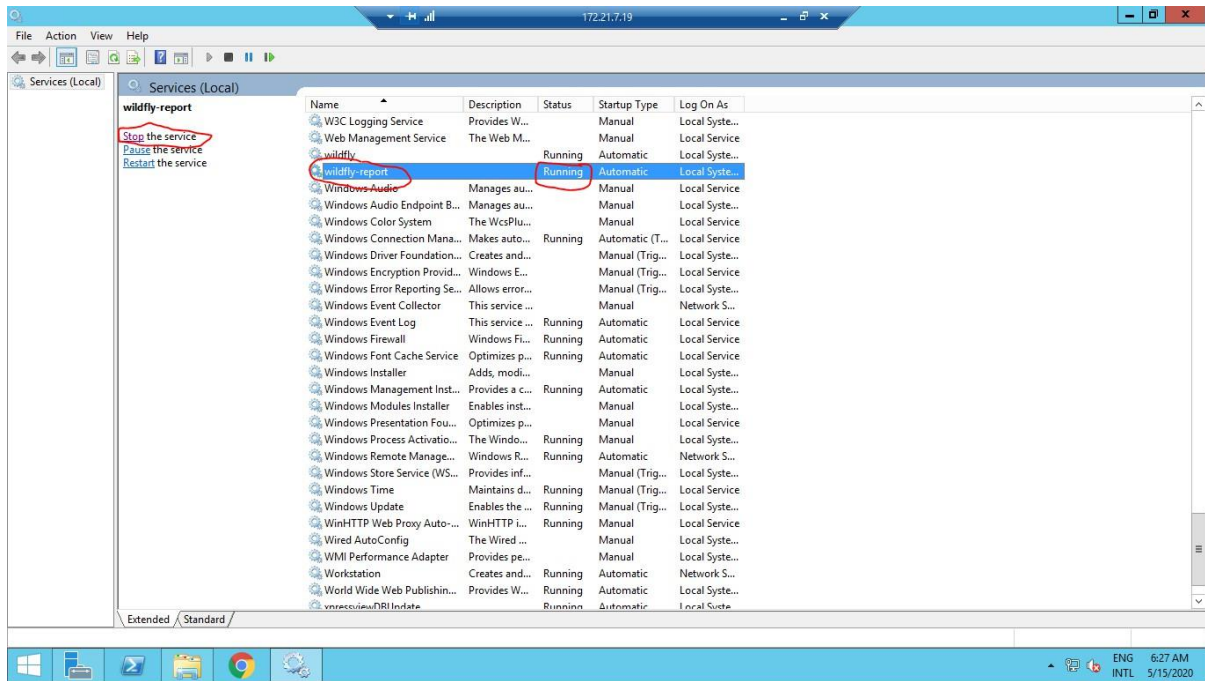
Risk	Controls
Application not following standards	Wildfly configuration audited to ensure consistent baseline configurations.

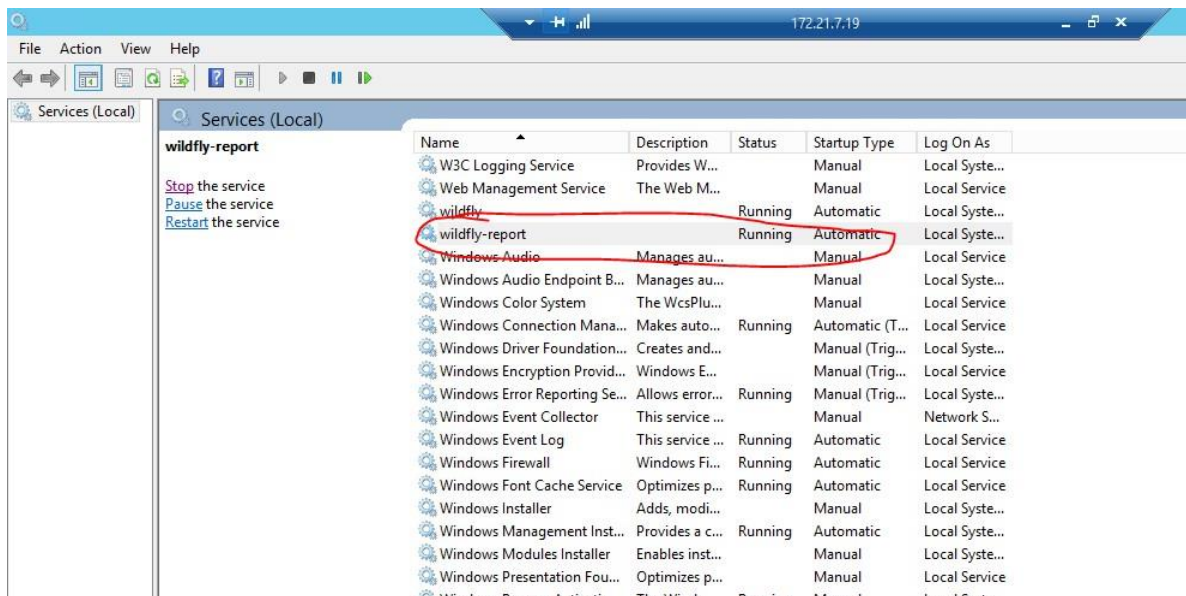
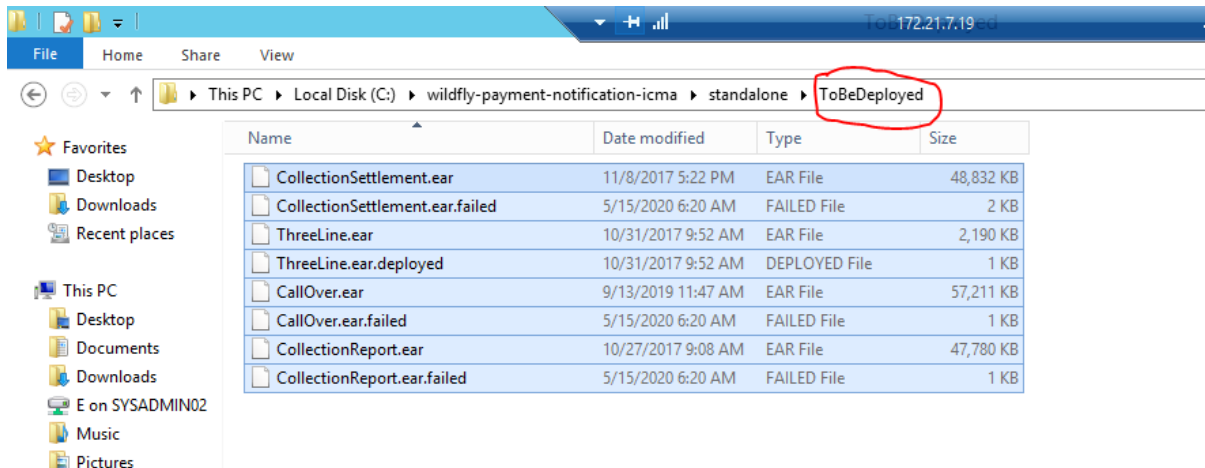
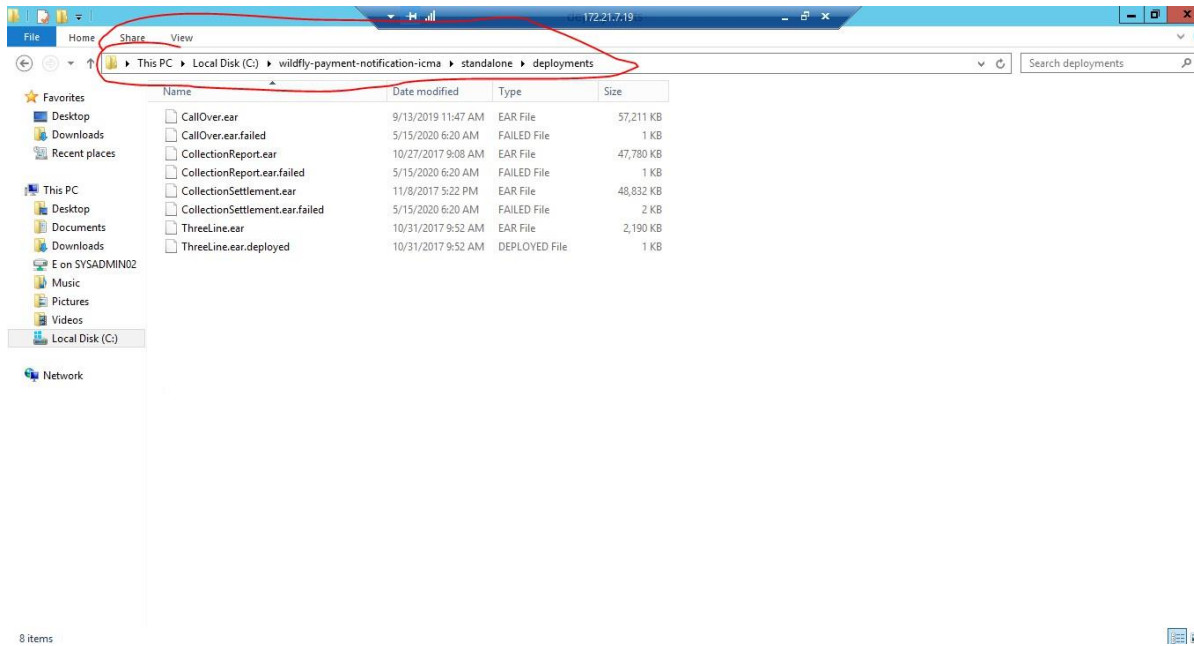
5.6 How to deploy wildfly on webserver

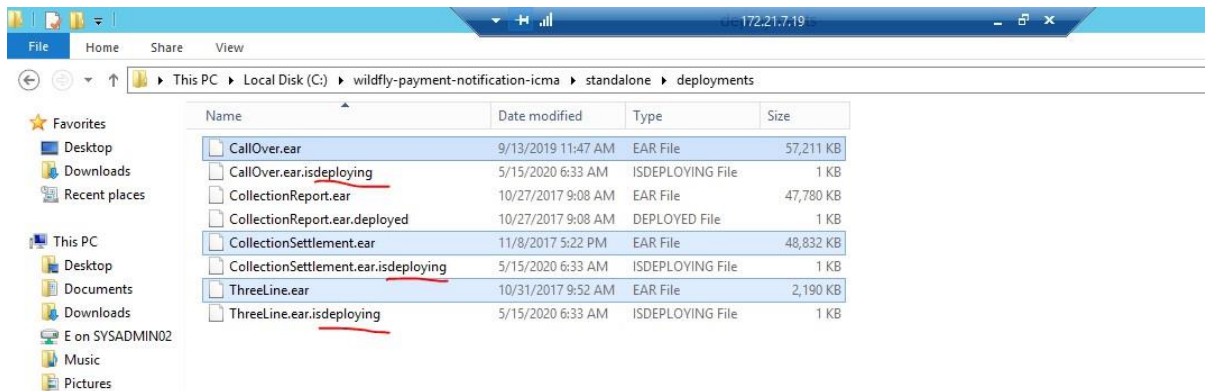
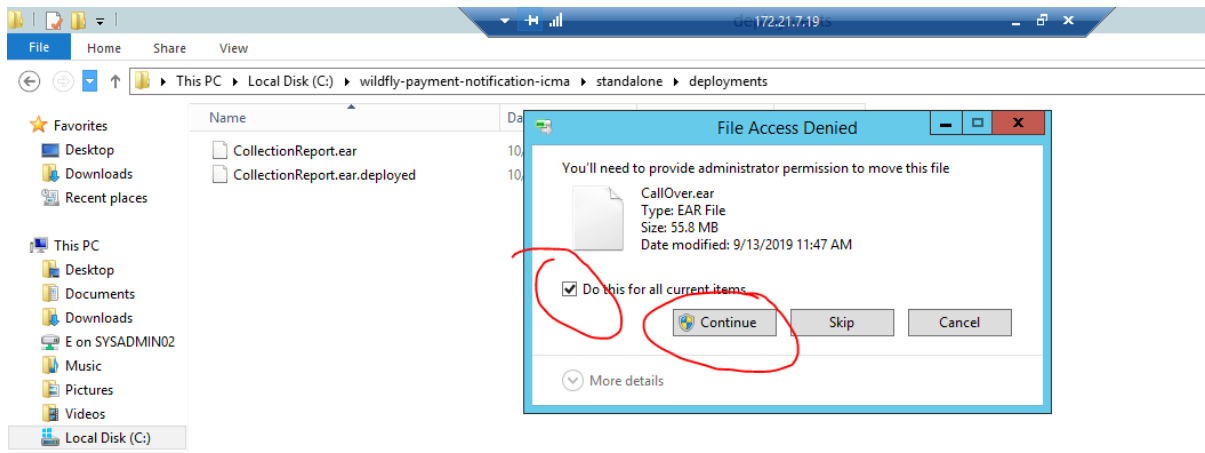
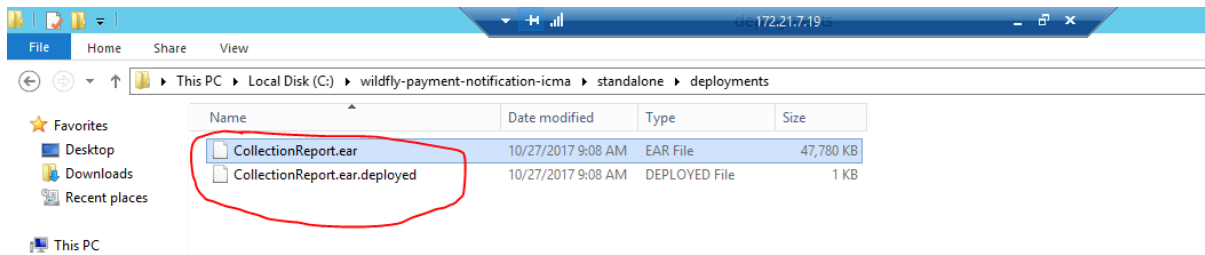
Login to the Report Server











LOGIN TO YOUR ACCOUNT

Username

Password

6.0 IIS Webserver Installation on Windows server

6.0 Background

Internet Information Services is an extensible web server software created by Microsoft for use with the Windows NT family. To be able to use IIS, you must install it on the server in question and then configure it.

6.1 Process Objective (Purpose)

To install IIS on a server that is running Microsoft Windows Server 2012 R2, Microsoft Windows Server 2016, and Microsoft Windows Server 2019. This process explains how IIS webserver installation is performed on Windows servers. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

6.2 Process Owner

Group Head, IT

6.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	IIS webserver Installation on Windows server	1.1 Press the [Windows] key and select Server Manager. 1.2 In the Server Manager Dashboard, click Manage > Add Roles and Features. 1.3 Click Installation Type. 1.4 Select the Role-based or feature-based installation option and click Next. 1.5 Select the server on which you want to install IIS and click Next. 1.6 Activate the Web Server (IIS) role. 1.7 To add the IIS Management Console, click Add Features. 1.8 Click Next. The Select Features window will open.	30 Minutes

		1.9 Click Next. The Web Server Role (IIS) window will open. 1.10 Click Next. The Select Role Services window will open. 1.11 Select the required role services and click Next. 1.12 To install the selected roles, role services, and features, click Install. 1.13 To complete the installation, click Close.	
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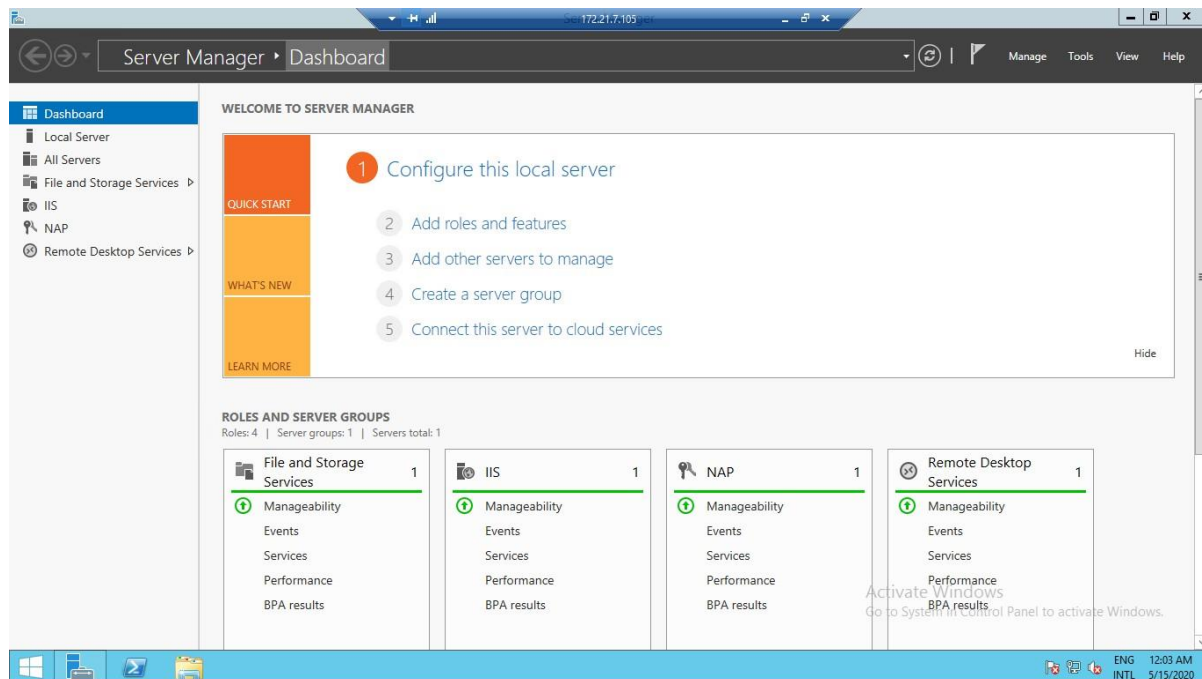
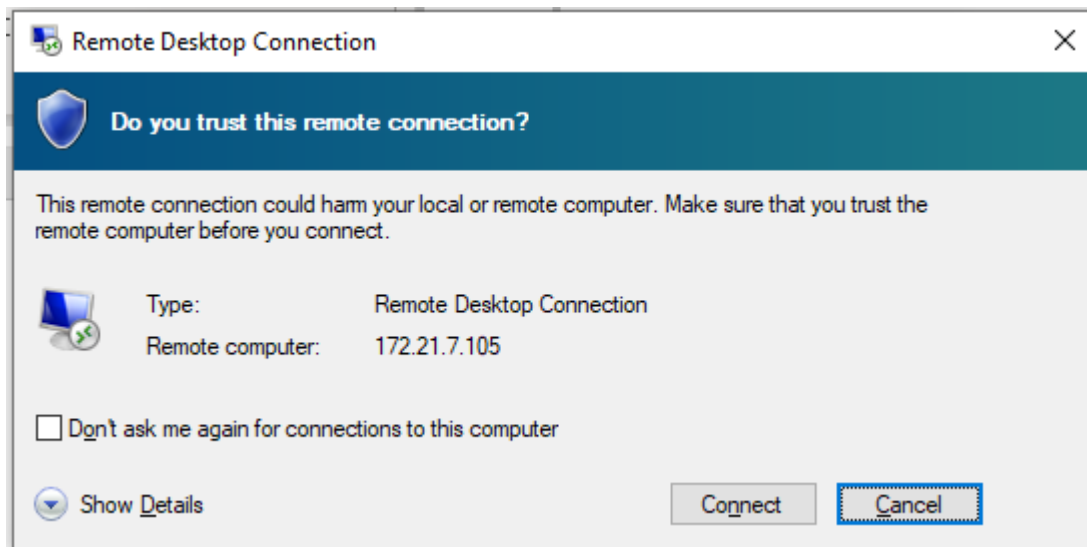
6.4 Key Risks/Control

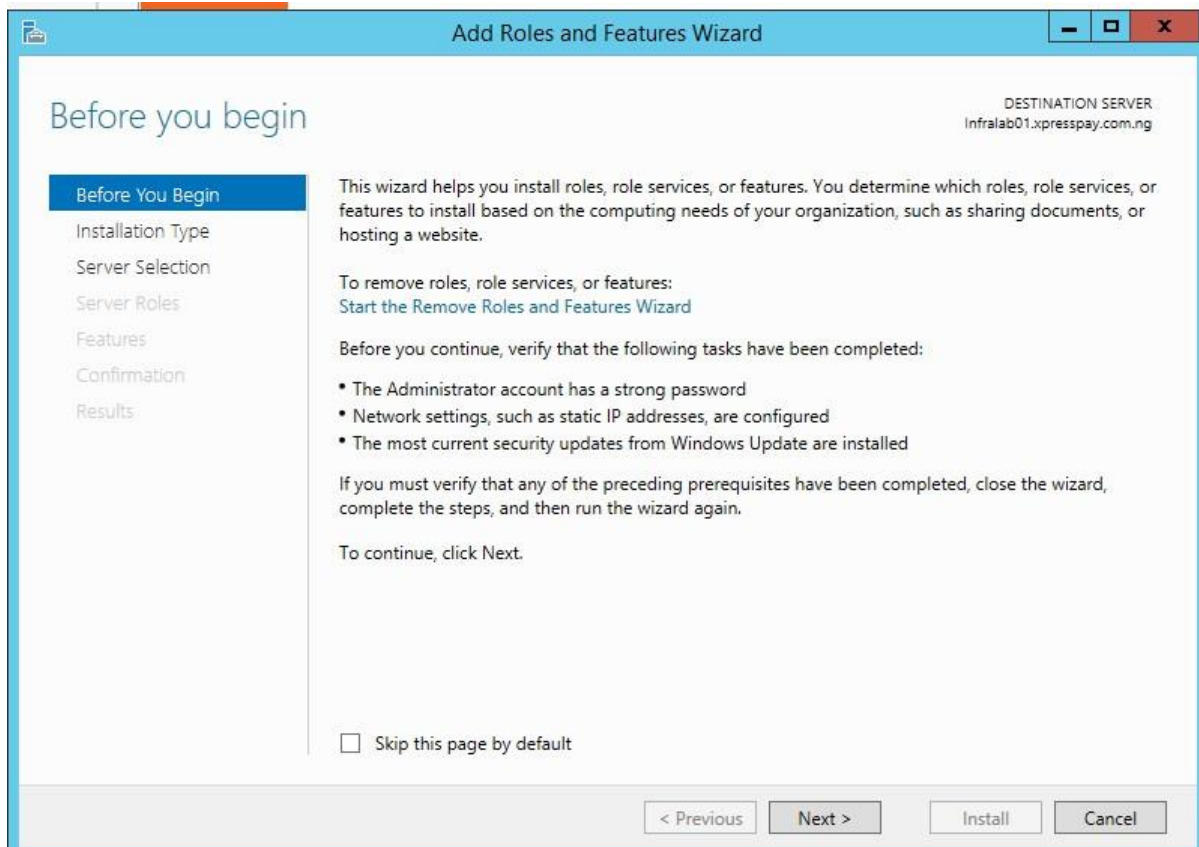
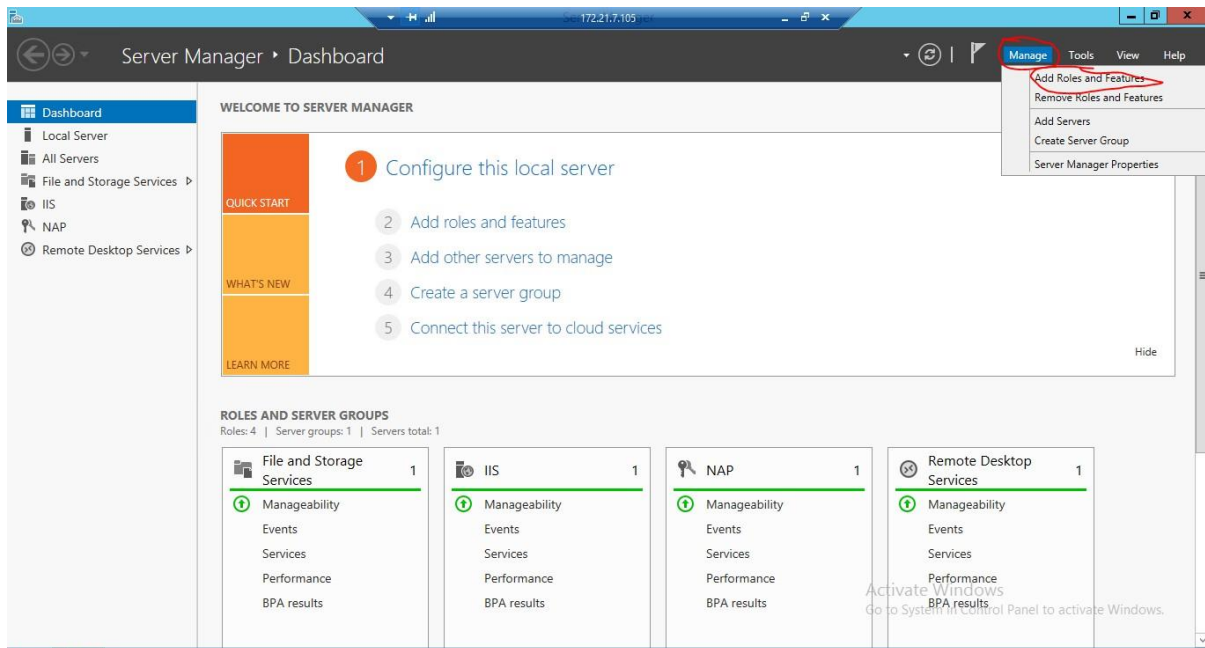
In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

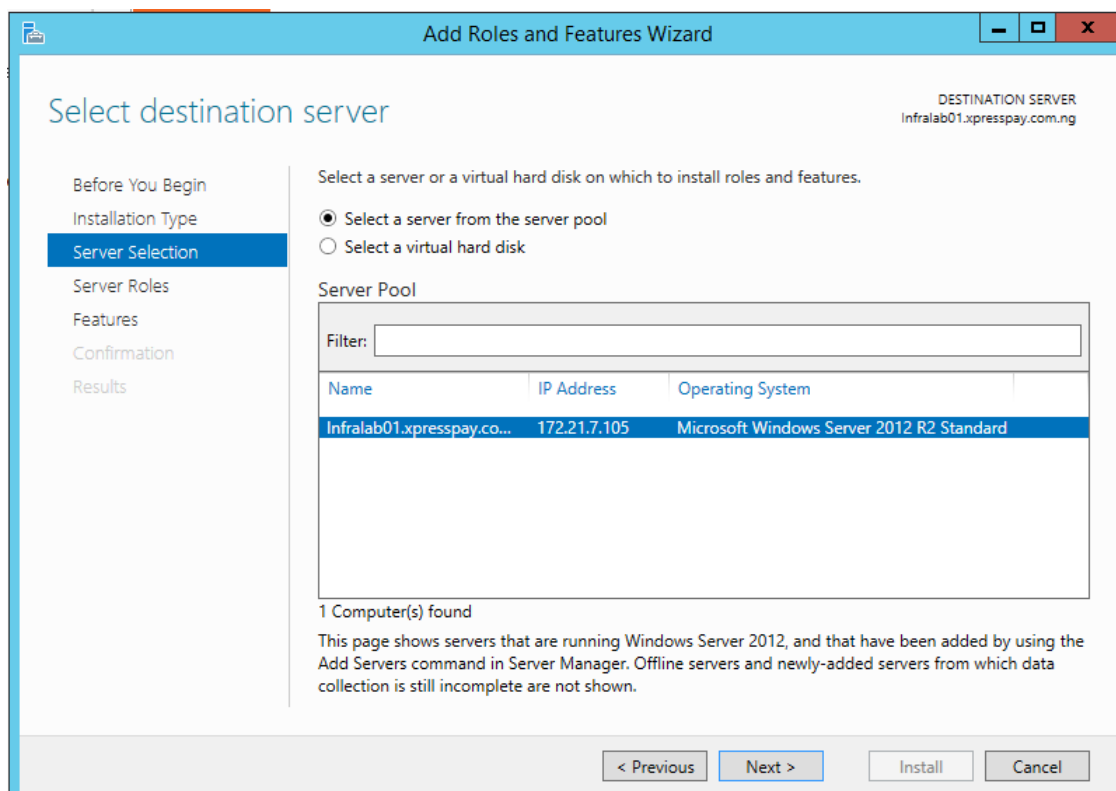
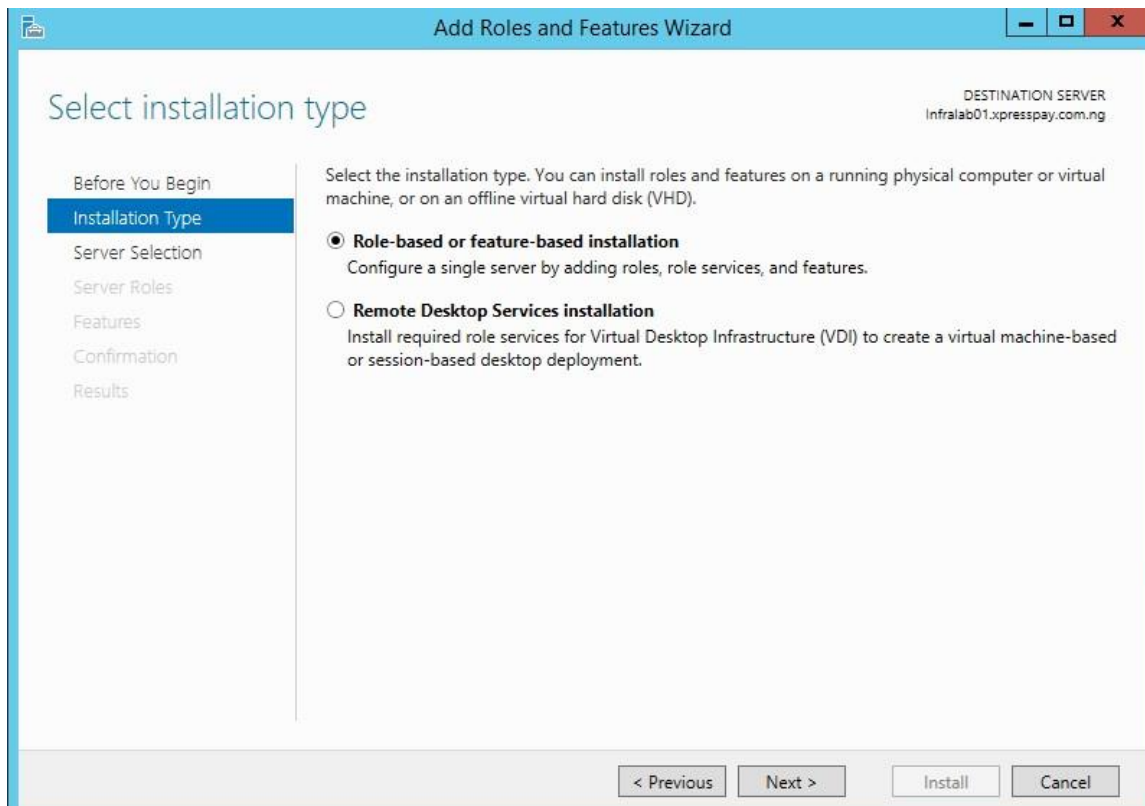
Risk	Control
Inability to run web applications leading to delayed projects	Ensure IIS are installed on web servers
Missing application dependencies	Liaise with the developers to know the application required dependencies and install them.

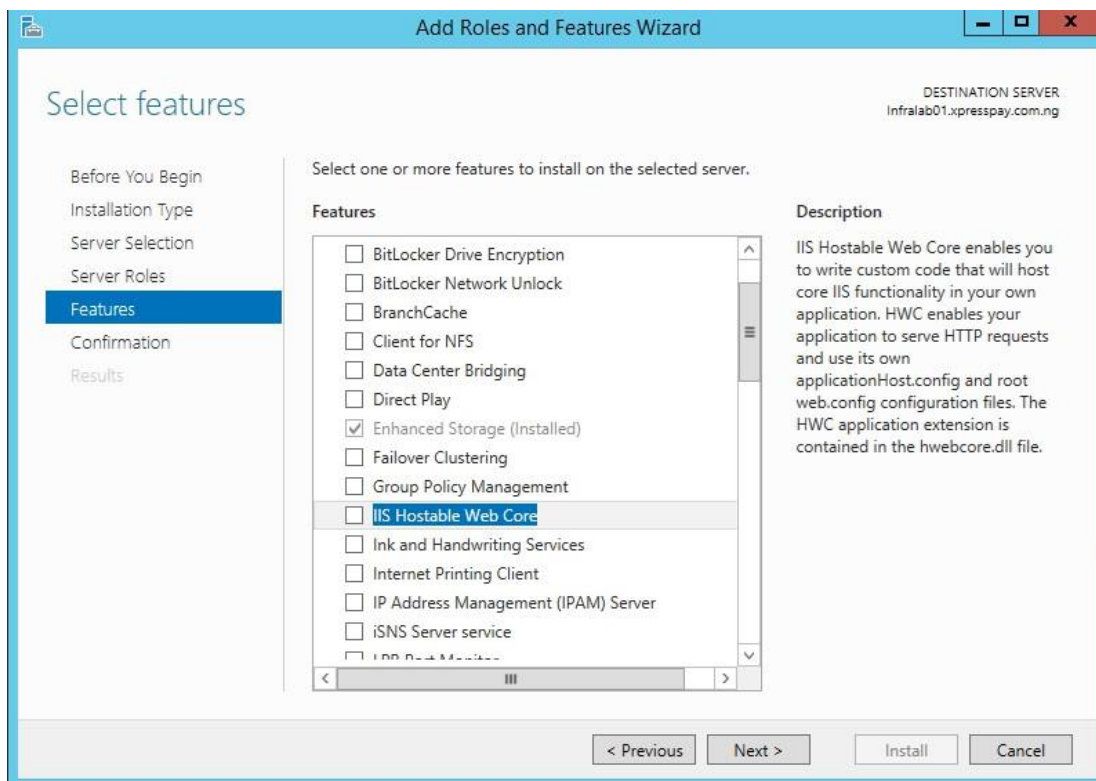
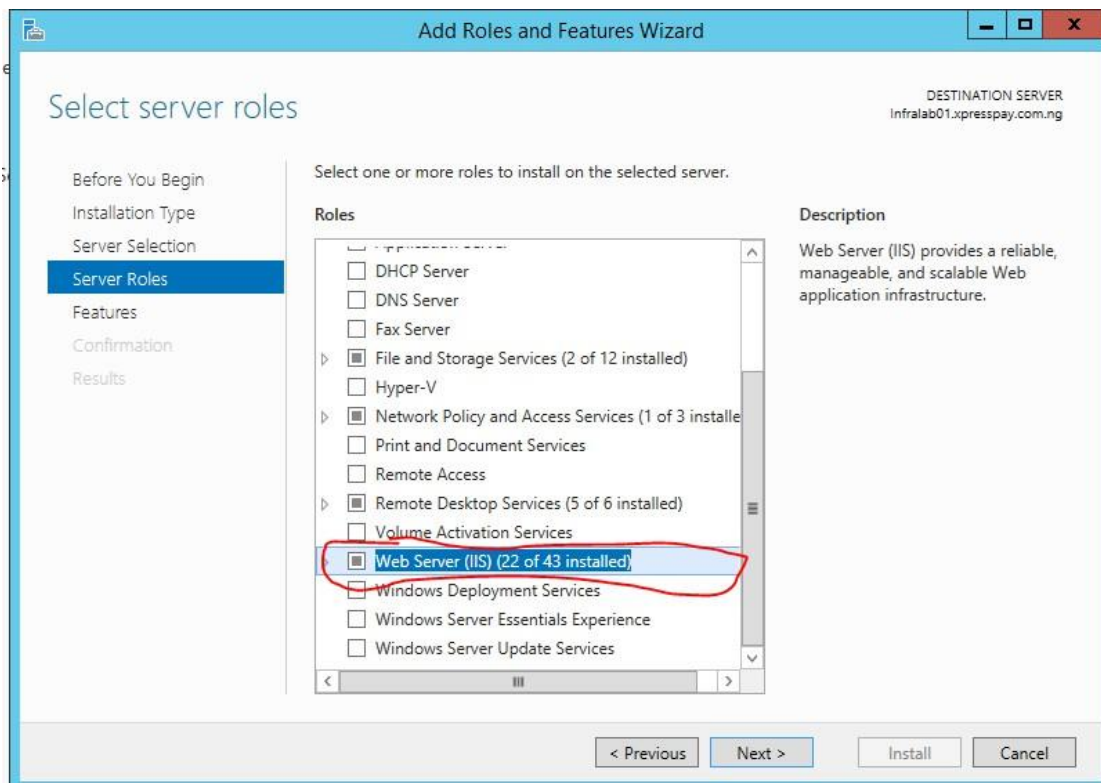
w8.6 IIS webserver Installation on Windows server

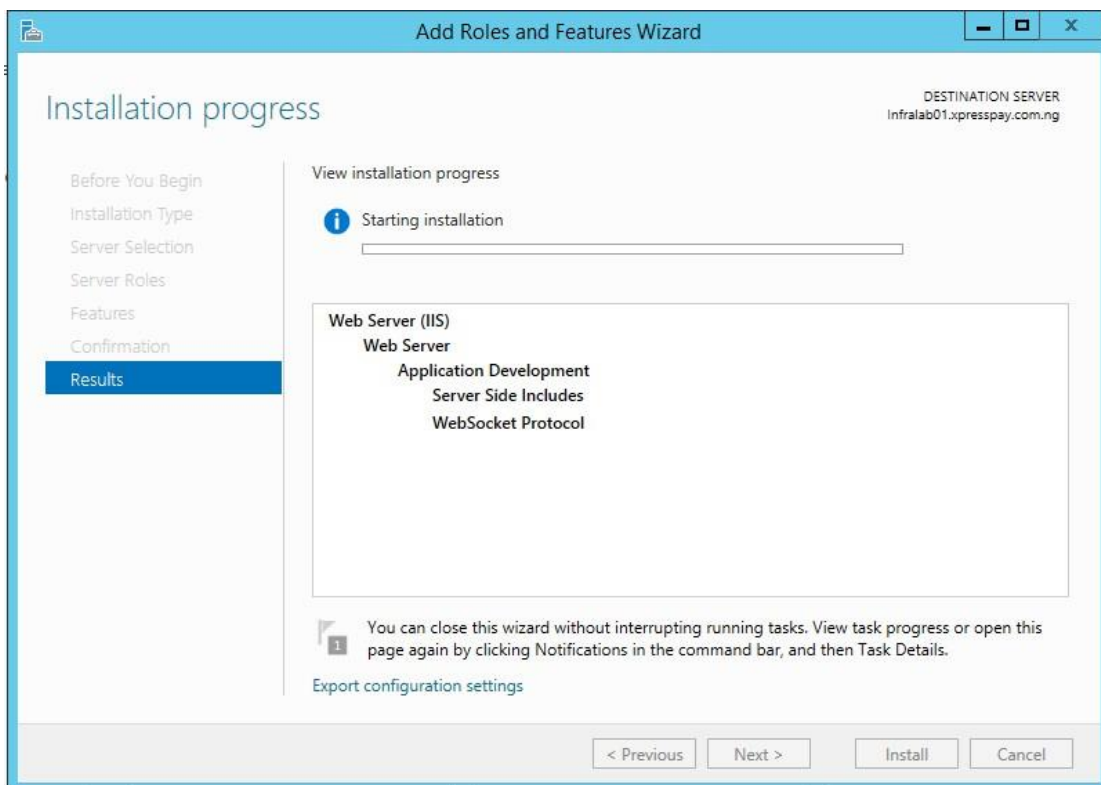
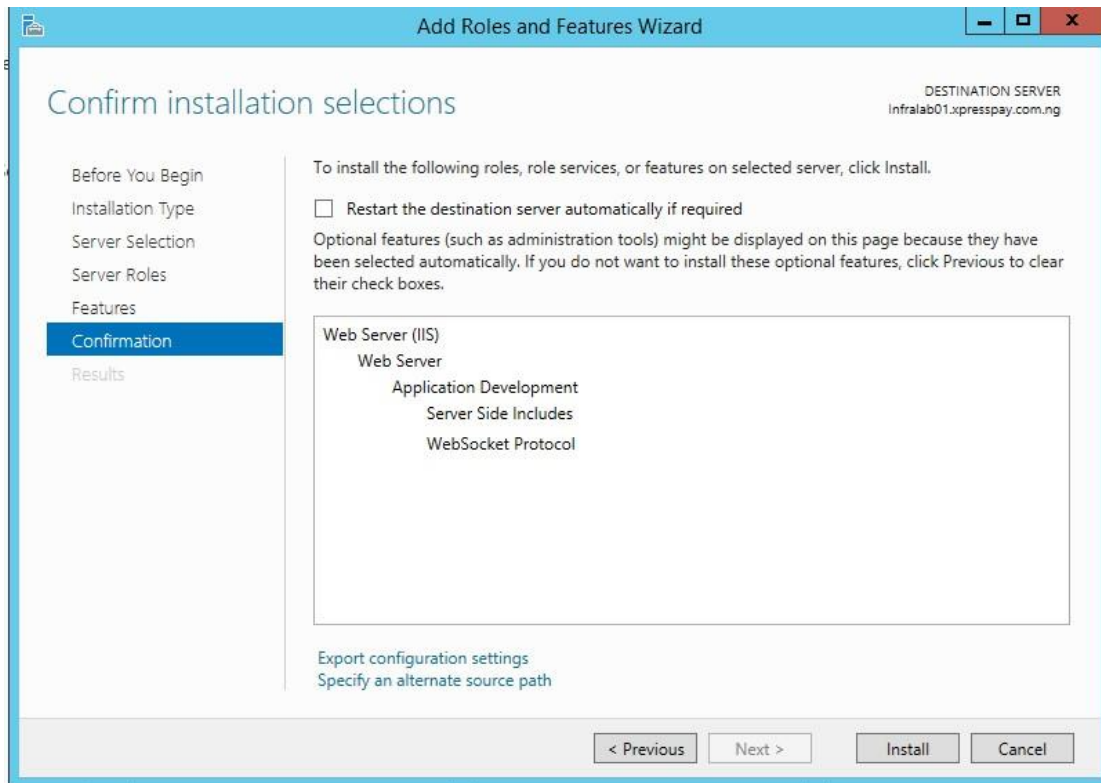
Login to the webserver

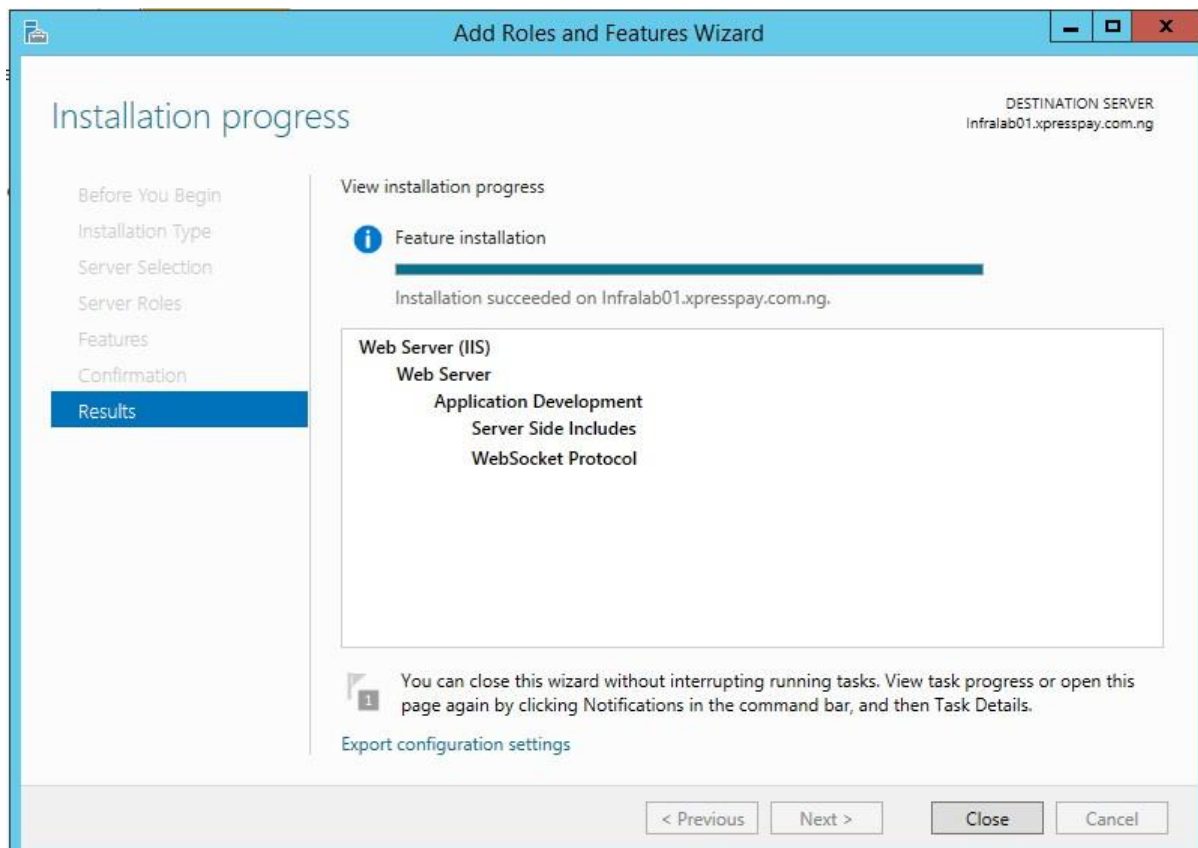












7.0 ILO configuration on servers

7.0 Background

Integrated Lights-Out (iLO) is a remote server management processor embedded on the system boards of HP ProLiant and Blade servers that allows controlling and monitoring of HP servers from a remote location. HP iLO management is a powerful tool that provides multiple ways to configure, update, monitor, and run servers remotely

7.1 Process Objective (Purpose)

This process is a guideline for accessing servers via Integrated Lights-Out (iLO). If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

7.2 Process Owner

Group Head, IT

7.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	ILO configuration on servers	1.1 Download the ilo setup from HP site 1.2 Copy the setup to the server 1.3 Install the setup on the server 1.4 Configure the ilo using the GUI 1.5 Access the server via browser	30 Minutes

7.4 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the

exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

Risk	Controls
Unable to troubleshoot hardware related issues with servers	Ensure iLO is installed on all the HP servers

7.5 ILO configuration on servers

Download the utility from HP site

← → ↻ 🏠 support.hpe.com/hpsc/swd/public/detail?swItemId=MTX_c2f8737292c74b3b852b063ff7

Menu **Hewlett Packard Enterprise Support Center** Search

[Printable version](#)

Drivers & software

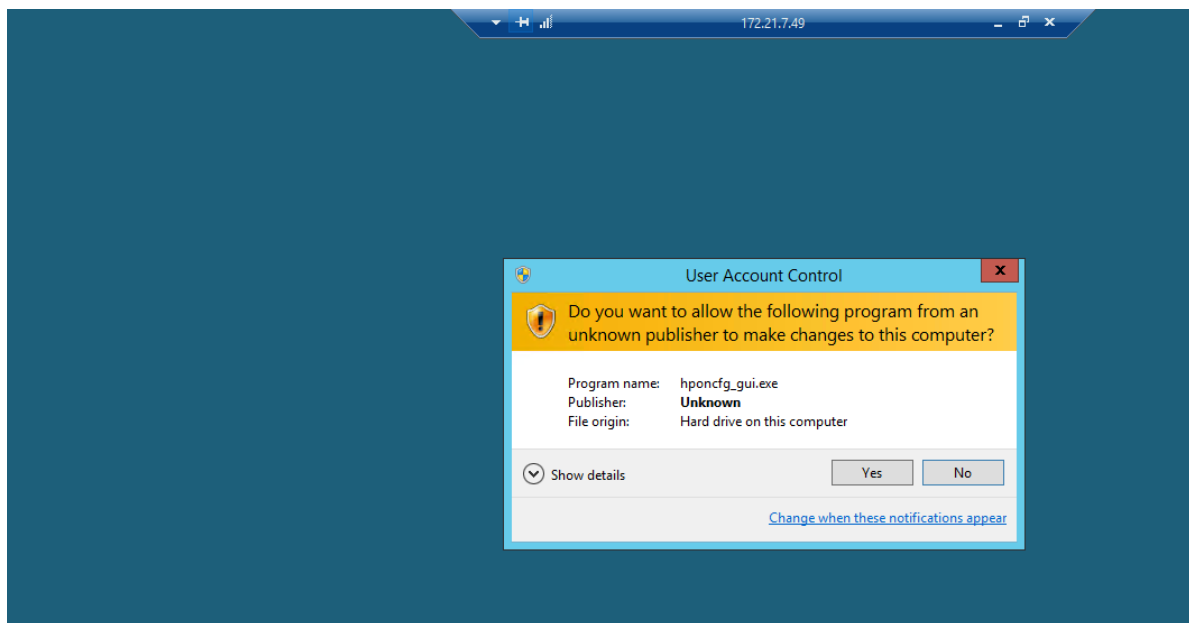
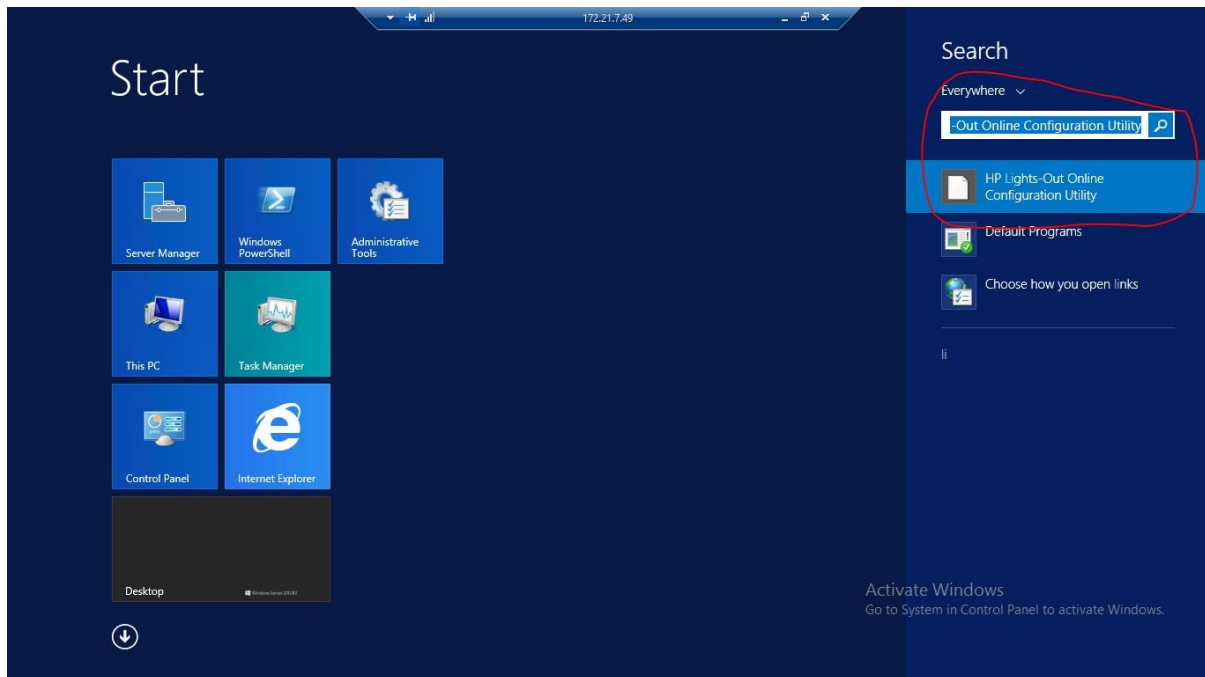
* RECOMMENDED * HP Lights-Out Configuration Utility

By downloading, you agree to the terms and conditions of the [Hewlett Packard Enterprise Software License Agreement](#).

Note: Some software requires a valid warranty, current Hewlett Packard Enterprise support contract, or a license fee.

Type:	Software - Lights-Out Management
Version:	1.5.0(24 Oct 2016)
Operating System(s):	Microsoft Windows 7 (32-bit) Microsoft Windows 7 (64-bit) Microsoft Windows 8 (32-bit) Microsoft Windows 8 (64-bit) Microsoft Windows 8.1 (32-bit) Microsoft Windows 8.1 (64-bit) Microsoft Windows Server 2008 R2 Microsoft Windows Server 2008 W32 Microsoft Windows Server 2008 x64

Login to the server



HP Lights-Out Online Configuration Utility

Summary **Network** User Settings About

Standard Network Settings

IPv4 IPv6

Enable DHCP ☐ Enable ☒ Disable

iLO IP Address 172.21.7.99

iLO Subnet Mask 255.255.255.0

iLO Gateway IP Address 172.21.7.1

iLO DNS Name ILOCZJ8030Q2M

iLO Mac Address 98f2:b3:32:4cfc

Network interface Adapter ☒ Enable ☐ Disable

Transceiver Speed Autoselect ☒ Yes ☐ No

HP Lights-Out Online Configuration Utility

Summary Network **User** Settings About

User Accounts

Select User
Administrator

Add
View/Modifv
Delete

User Name Administrator
User Login Administrator
☒ **Change Password**
Password
Confirm Password
Administer User Accounts ☒ Allow
Remote Console Access ☒ Allow
Virtual Power and Reset ☒ Allow
Virtual Media ☒ Allow
Configure iLO Settings ☒ Allow

Apply

HP Lights-Out Online Configuration Utility

Summary Network User Settings About

User Accounts

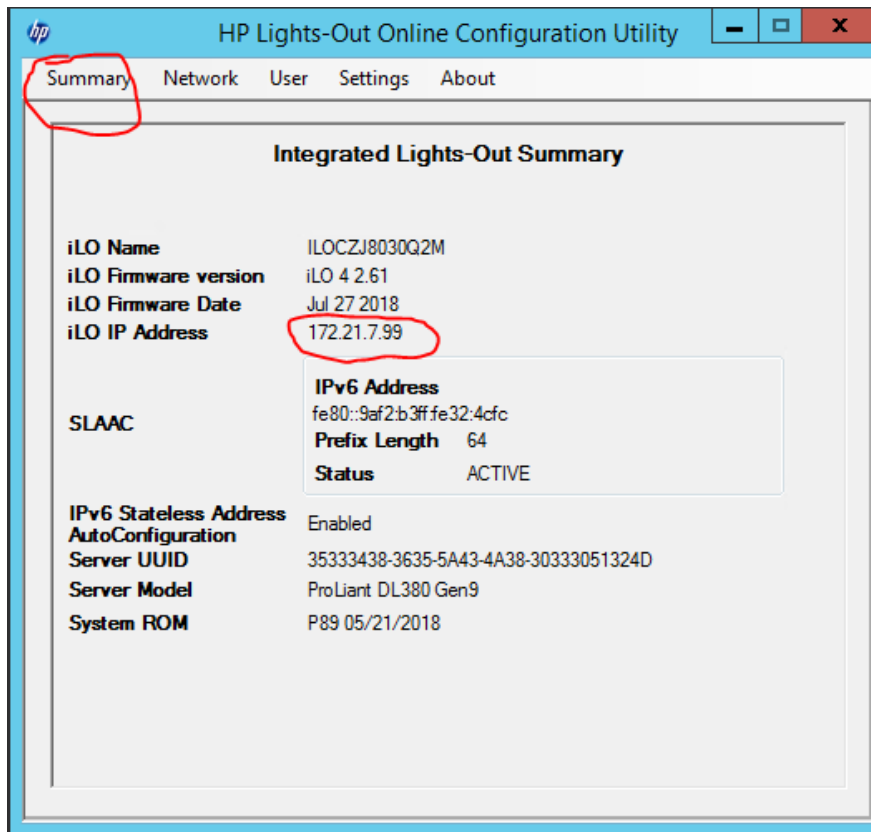
Select User
Administrator

Add
View/Modifv
Delete

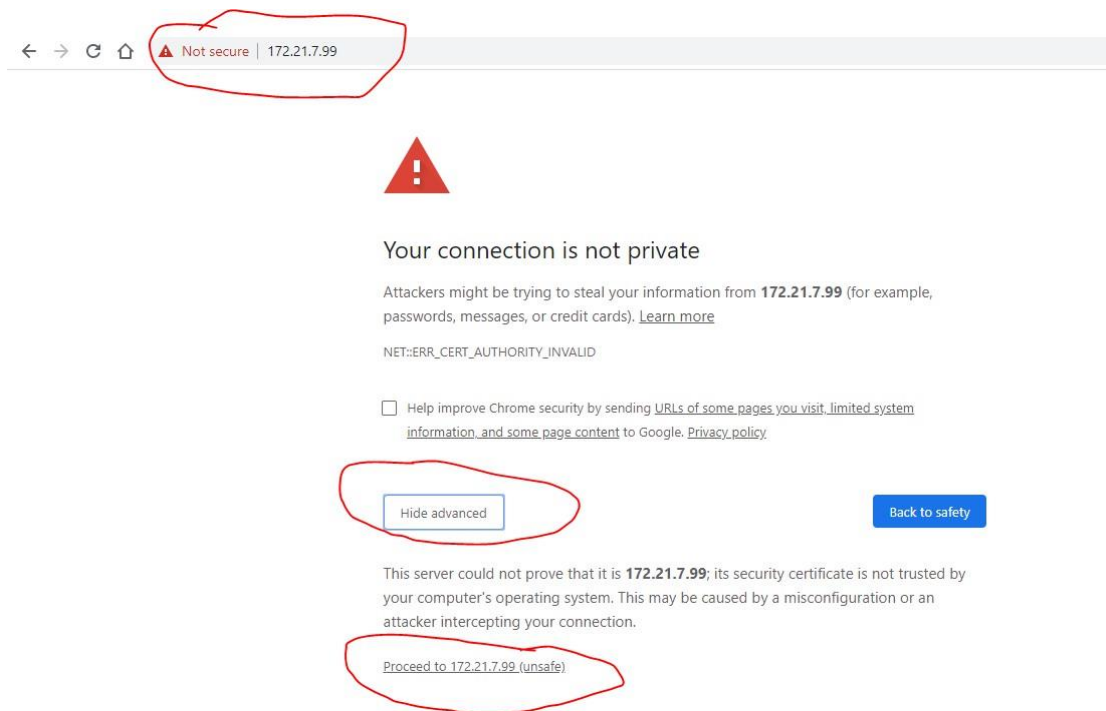
User Name Administrator
User Login Administrator
☒ **Change Password**
Password
Confirm Password
Administer User Accounts ☒ Allow
Remote Console Access ☒ Allow
Virtual Power and Reset ☒ Allow
Virtual Media ☒ Allow
Configure iLO Settings ☒ Allow

Apply

User Settings were modified Successfully.
OK



Login via web



The screenshot displays the iLO Overview page for a ProLiant DL380 Gen9 server. The interface includes a top navigation bar with user information (Local User: Administrator) and system status (POWER: ON). A left sidebar contains a menu with options like Information, Overview, System Information, iLO Event Log, Integrated Management Log, Active Health System Log, Diagnostics, Location Discovery Services, Insight Agent, iLO Federation, Remote Console, Virtual Media, Power Management, Network, Remote Support, and Administration. The main content area is divided into three sections: Information, Status, and Active Sessions.

Information		Status
Server Name	BACKUPHOST	System Health OK
Product Name	ProLiant DL380 Gen9	iLO Health OK
UUID	35333438-3635-5A43-4A38-30333051324D	Server Power ON
Server Serial Number	CZJ8030Q2M	UID Indicator UID ON
Product ID	843556-425	TPM Status Not Present
System ROM	P89 v2.60 (05/21/2018)	SD-Card Status Not Present
System ROM Date	05/21/2018	iLO Date/Time Sun May 24 09:06:21 2020
Backup System ROM	10/25/2017	
Integrated Remote Console	.NET Java Web Start Java Applet	Connection to HPE Not registered
License Type	iLO Standard	
iLO Firmware Version	2.61 Jul 27 2018	
IP Address	172.17.99	
Link-Local IPv6 Address	FE80::9AF2:B3FF:FE32:4CFC	
iLO Hostname	ILOCZJ8030Q2M.workgroup	

Active Sessions		
User	IP Address	Source
Local User: Administrator	172.17.51.49	HTTPS

8.0 Jboss administration

8.0 Background

This is a report dashboard designed to provide detailed transaction reports both in textual and graphical forms which can be used for various activities such as reconciliations, auditory activities, treasury activities. This application run on JBOSS

JBOSS is a flexible, lightweight, managed application runtime that helps you build amazing applications.

8.1 Process Objective (Purpose)

This process is a guideline for us in deploying Wildfly on **XpressReport** platform. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges

8.2 Process Owner

Group Head, IT

8.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	Wildfly deployment on xpressreport server	First Copy the Jboss into Sravan Home Directory <pre>cd sravan.bevara/ mkdir inst3 chown sravan.bevara:sravan.bevara inst3 ls cd inst3 cp -r ../EAP-7.2.0/ . or cp -R /EAP-7.2.0 .</pre> 1.2 Change ownership to sravan <pre>chown sravan.bevara:sravan.bevara EAP-7.2.0 -R or chown sravan.bevara:sravan.bevara EAP-7.2.0 -R</pre>	30 Minutes

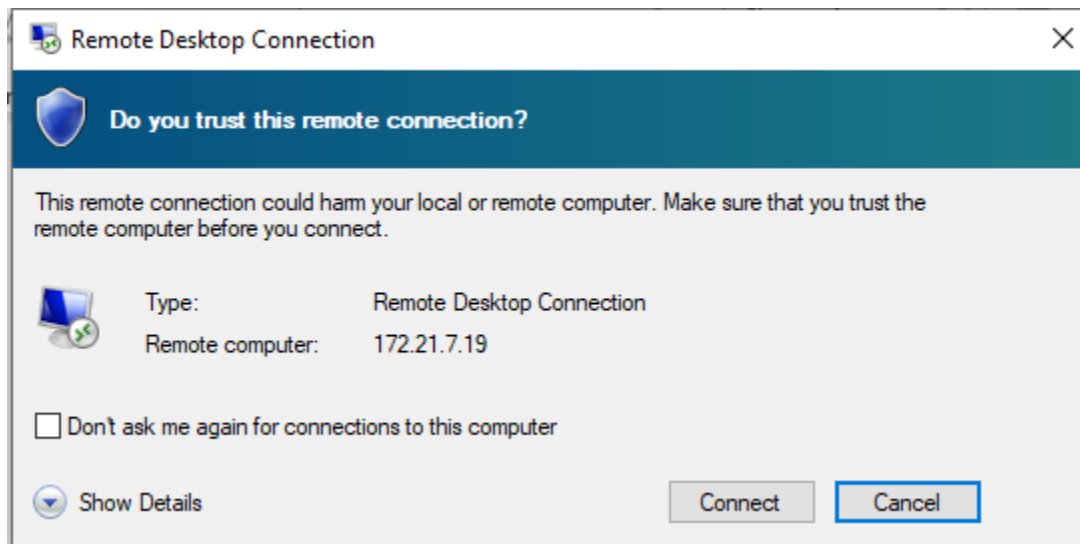
		cd EAP-7.2.0/ ll 1.3 Change Port in the standalone.xml cd /home/sravan.bevara/EAP-7.2.0/standalone/configuration vi standalone.xml -- Change all the port to a unique port leaving the ajp 1.4 Change the JBOSS_HOME, JBOSS_USER, JBOSS_CONSOLE_LOG cd /home/sravan.bevara/EAP-7.2.0/bin/init.d vi jboss-eap.conf 1.5 Add Port to Firewall firewall-cmd --add-port=9994/tcp --permanent firewall-cmd --add-port=8084/tcp --permanent firewall-cmd --reload 1.6 Start the new instance cd /home/sravan.bevara/EAP-7.2.0/bin nohup ./standalone.sh &	
--	--	--	--

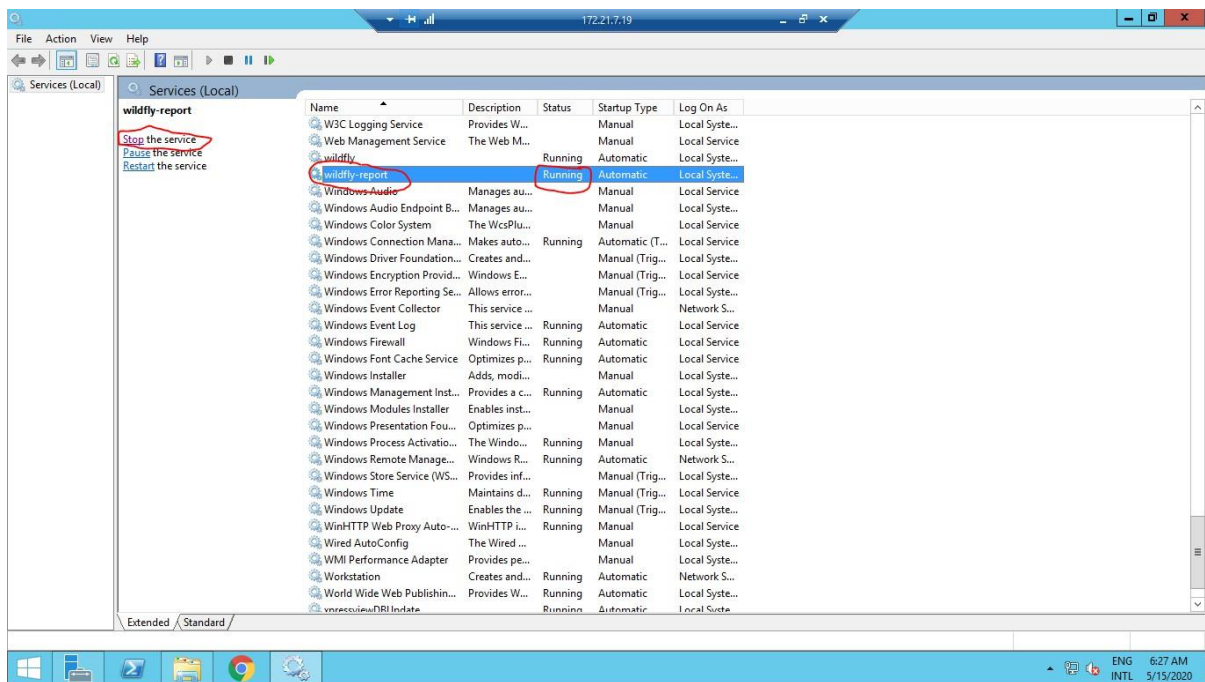
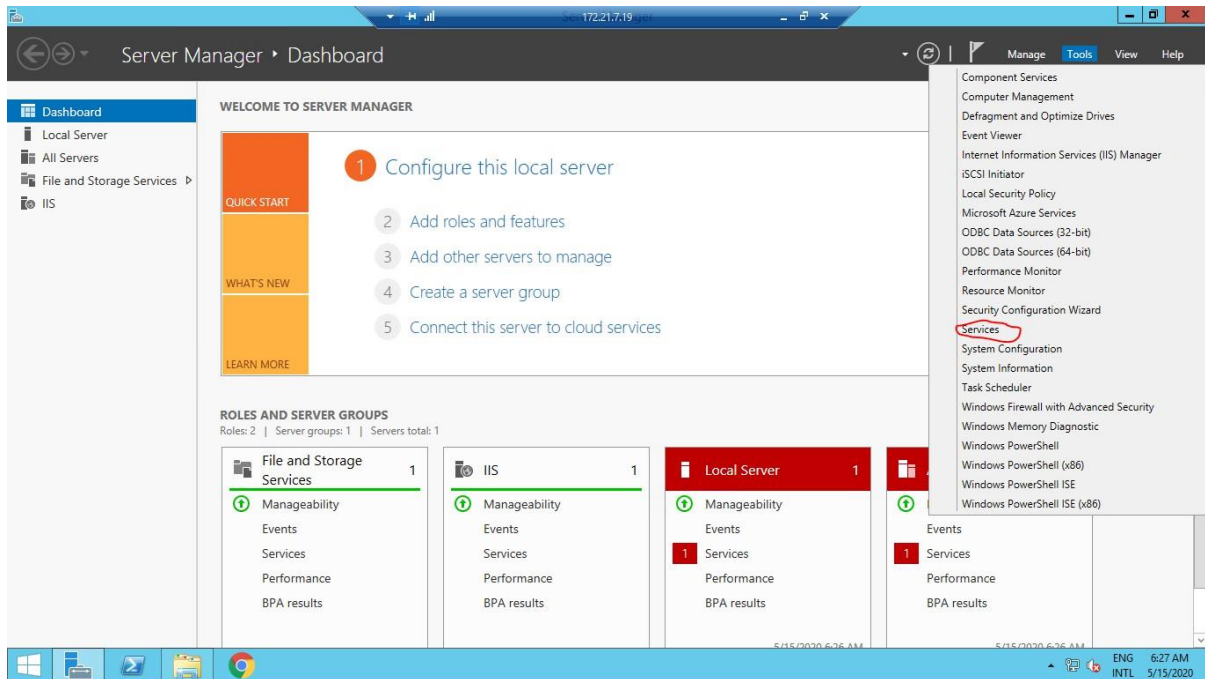
8.4 Key Risk/Control

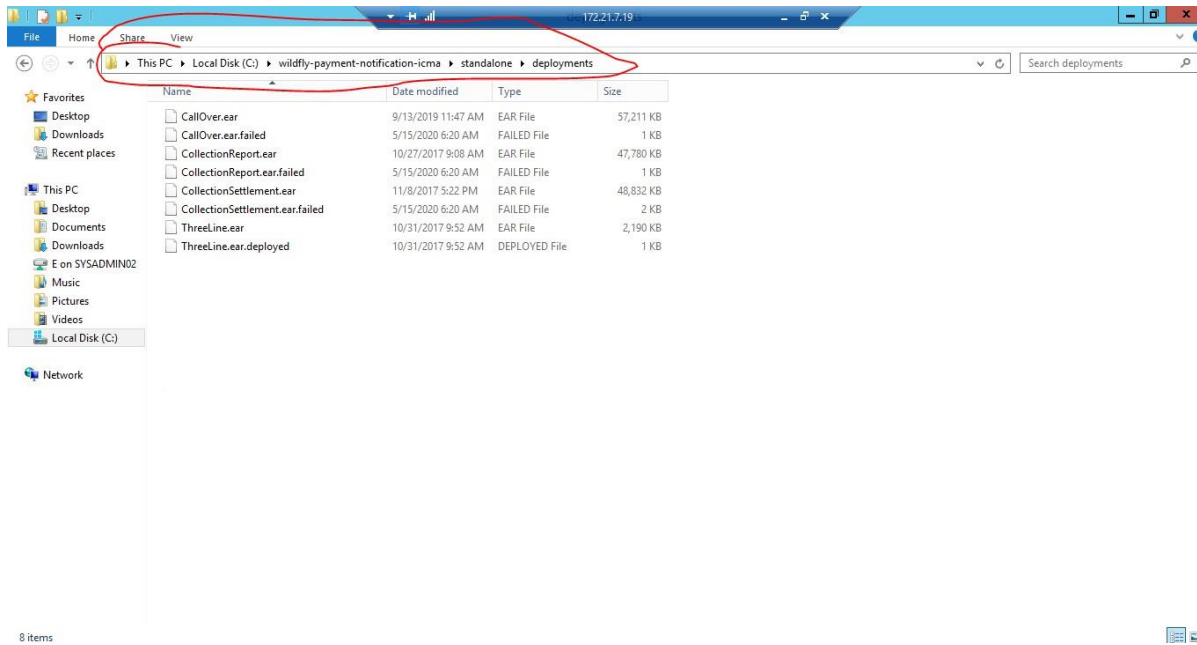
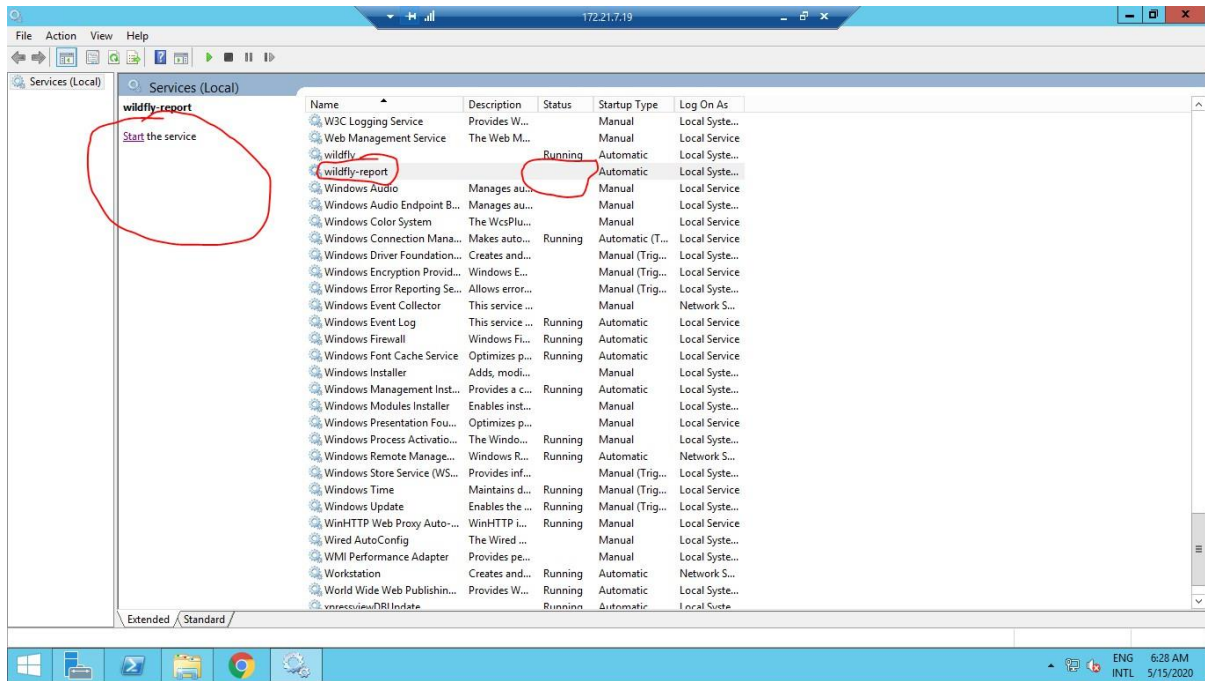
In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

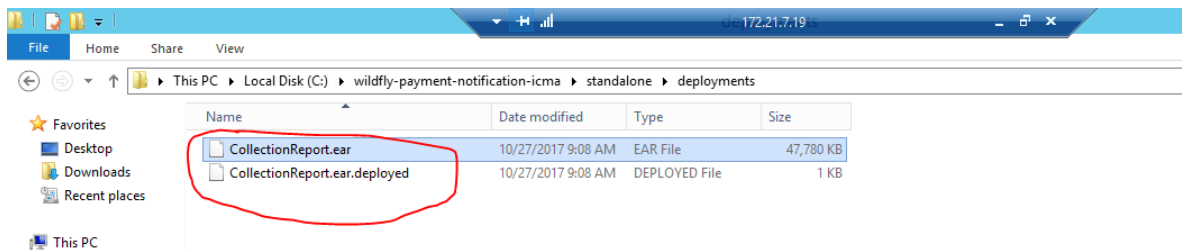
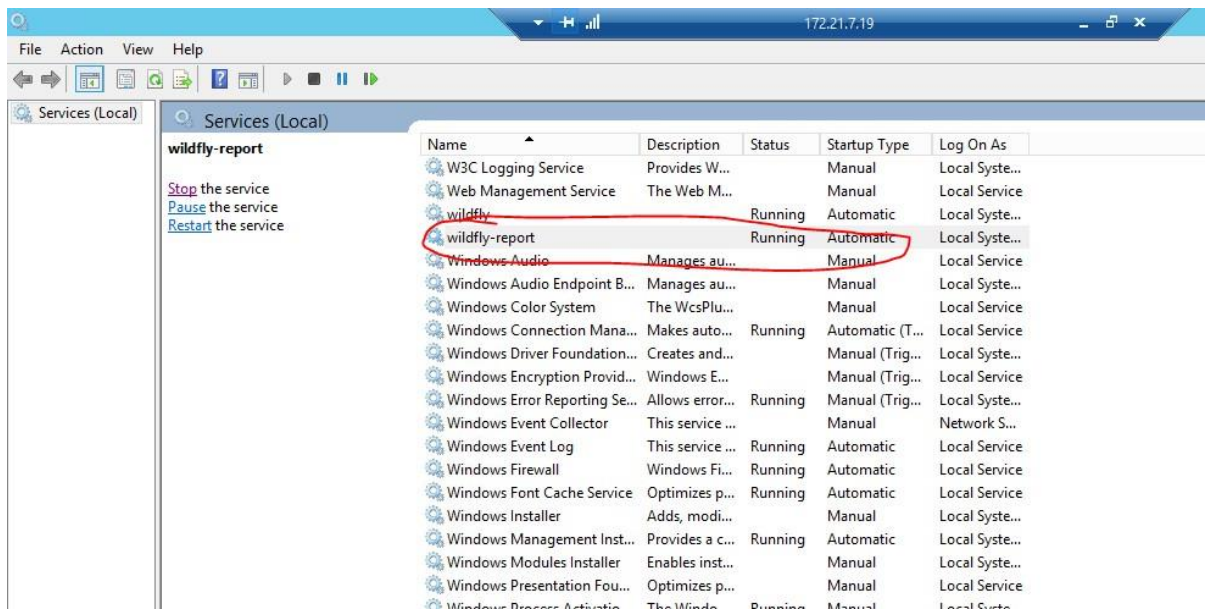
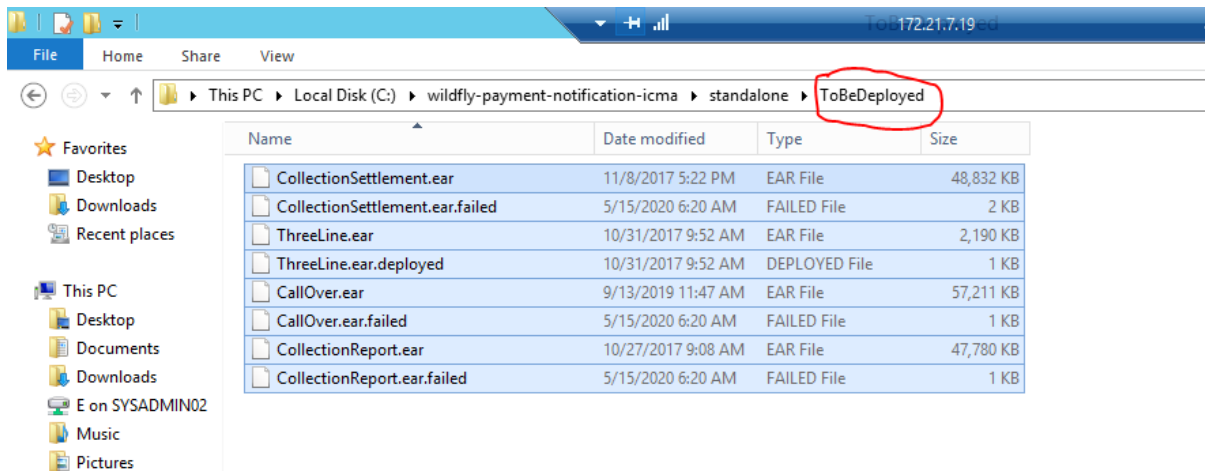
Risk	Controls
Unable to run JBOSS applications	Ensure the deployment steps are followed during JBOSS deployment

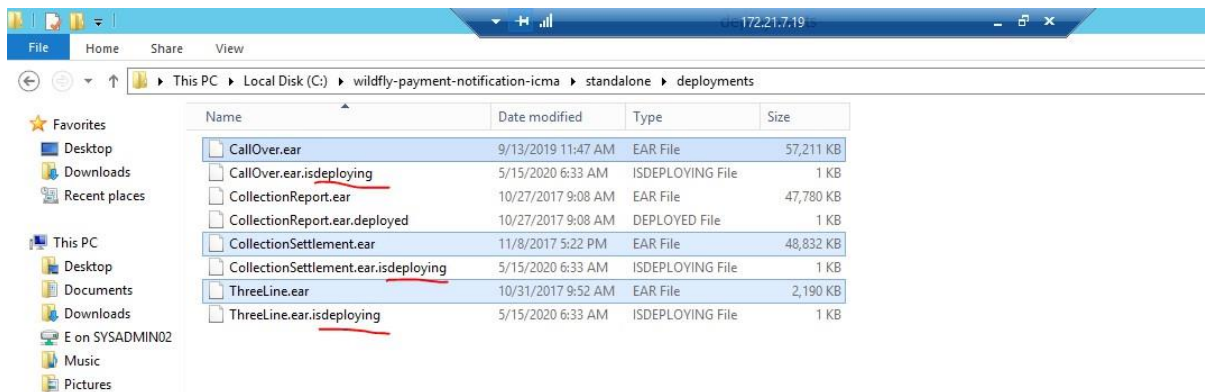
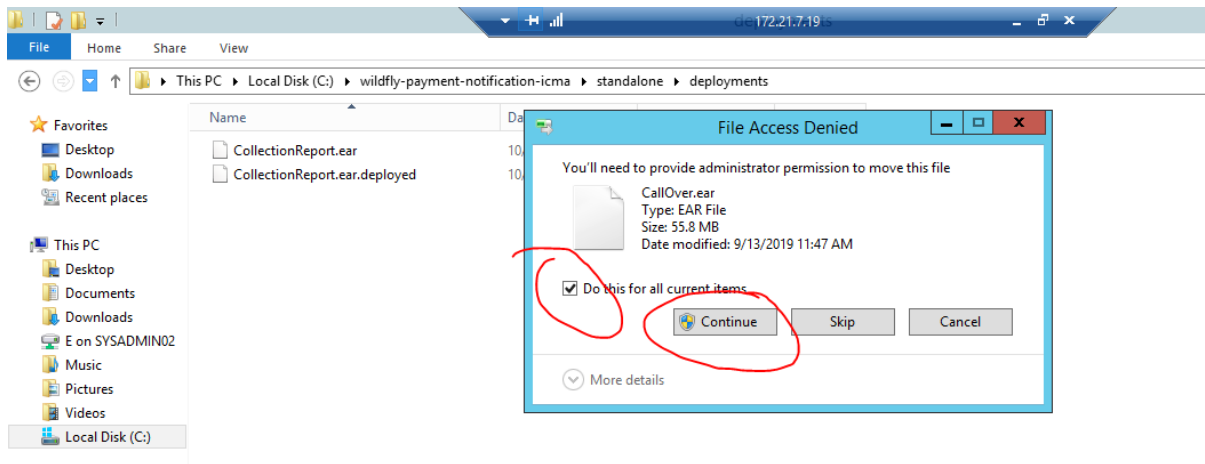
8.5 Login to the Report Server











LOGIN TO YOUR ACCOUNT

Username

Password

User Login

9.0 HP Server Setup, Installation and Configuration Process

9.0 Background

Server installation or setup is the act of making the system or program ready for execution. Without the OS installation and configuration users will not be able to access the servers to perform any task.

9.1 Process Objective (Purpose)

This process is a guideline to enable IT in Server setup, Installation, and configuration (Physical and Virtual). If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

9.2 Process Owner

Group Head, IT

9.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	To mount a new server on the rack	1.1 Remove the server, server frame and other accessories from the box 1.2 Mount the frame to the rack 1.3 Slot the server into the mounted frame. 1.4 plugin the power cable, network cable and other accessories.	30 minutes
	Server installation and configuration	2.1 Raid configuration (RAID 0, RAID 1, RAID 5, RAID 6). follow the screenshot below 2.2 Install the Operating system on the server (Windows or Linux). follow the screenshot below	1 hours

Support/Shift Team	Install approved default software's	3.1 Install the default approved software (TrendMicro, Windows update). follow the screen shot below	30 minutes

9.4 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

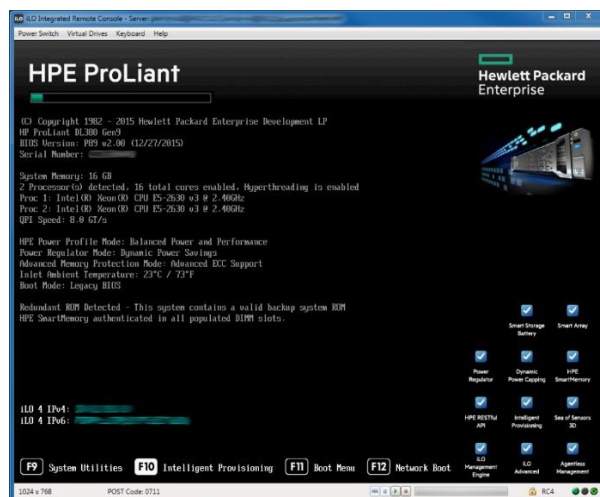
Risks	Control
Server failure leading to service failure	Confirmation of replication test from IS Control
Loss of Data	Failover test conducted bi-annually

9.5 SERVER RAID CONFIGURATION

1. Reboot or power on the server

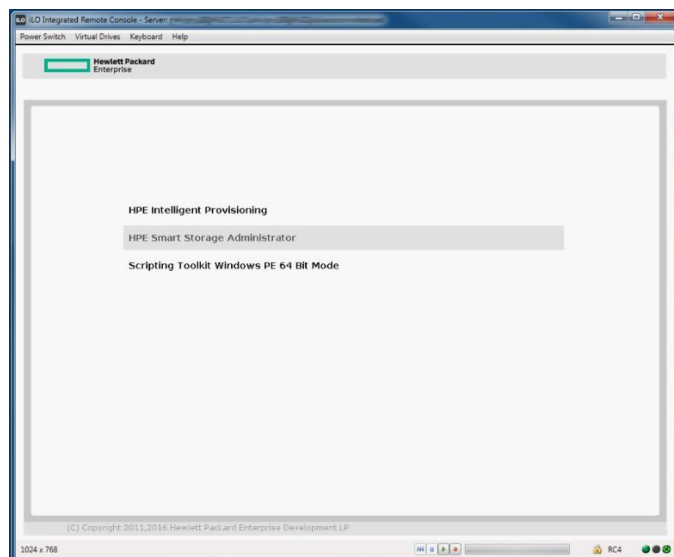
2. Launch Intelligent Provisioning

One of the fastest ways to get to HPE SSA leads through HPE Intelligent Provisioning menu. During system POST phase press **F10** to launch Intelligent Provisioning:

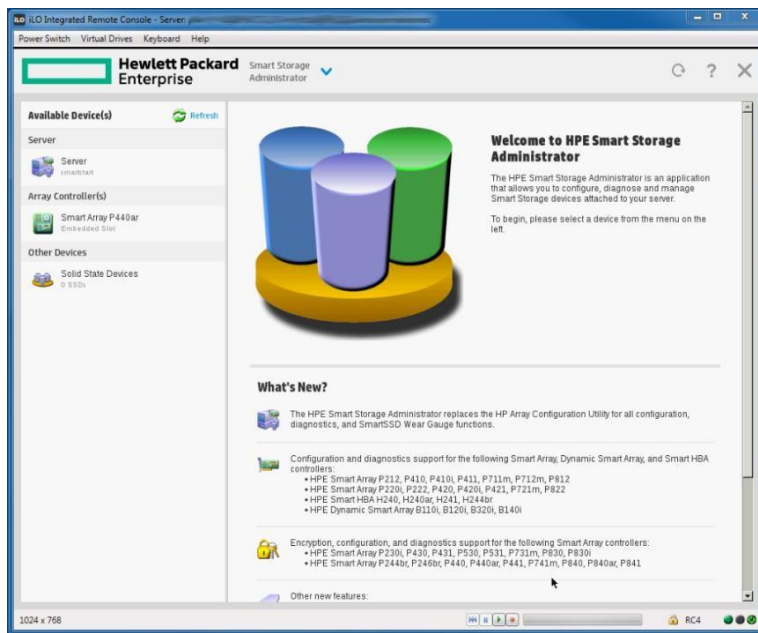


3. Access HPE Smart Storage Administrator

In the HPE Intelligent Provisioning menu select HPE Smart Storage Administrator – you have only **15 seconds**, so try to be quick, otherwise you will be redirected to the default option:

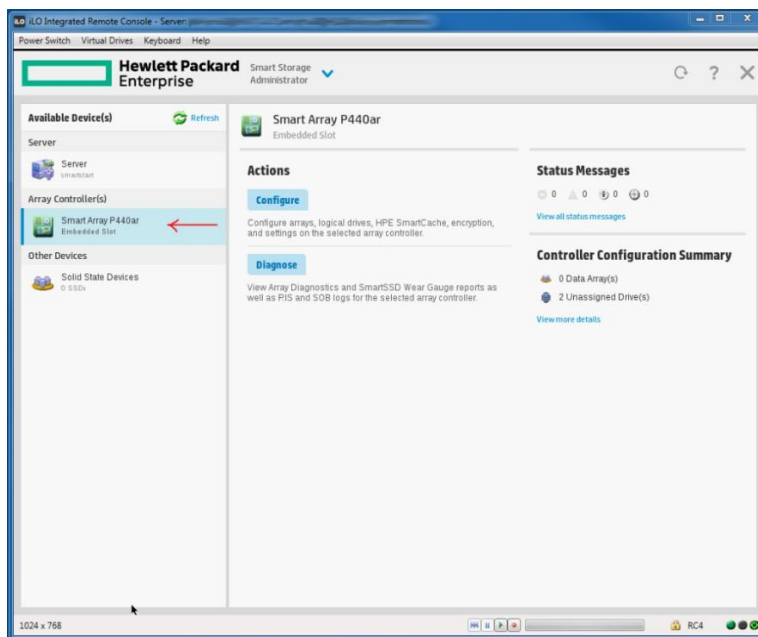


The below picture presents HPE Smart Storage Administrator GUI main screen:

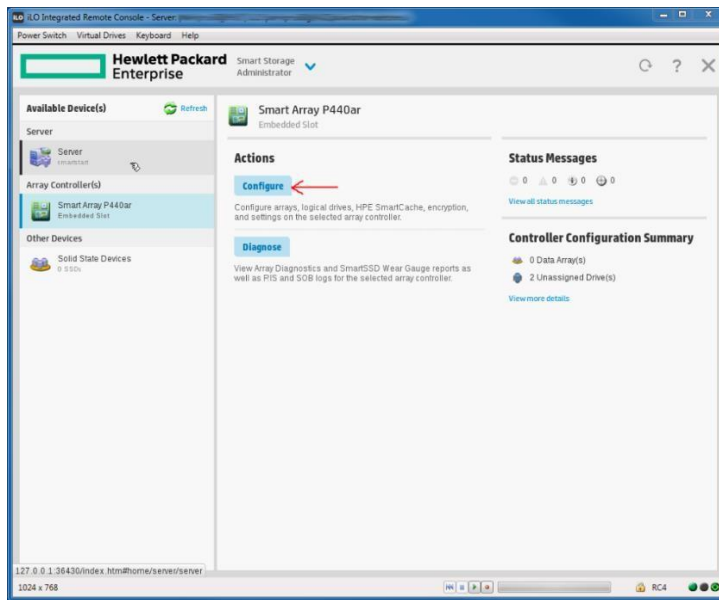


4. Create RAID 1 storage array in HPE SSA

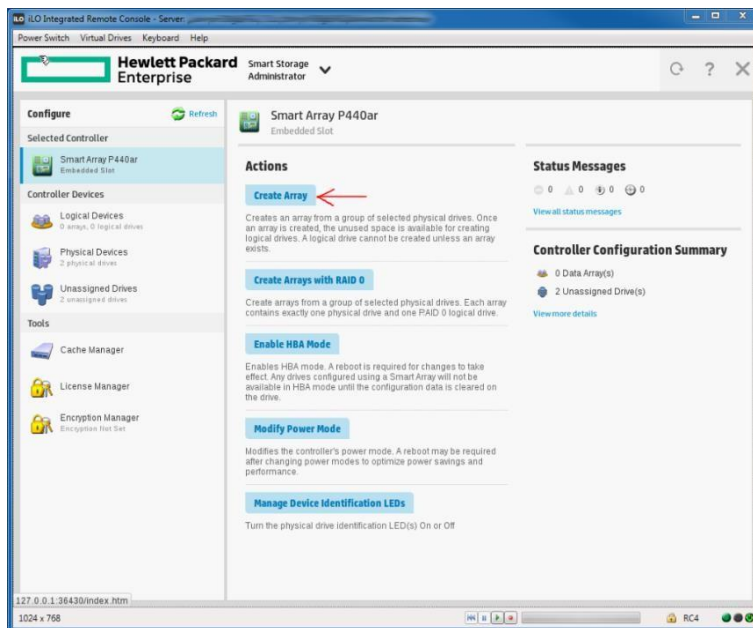
On HPE SSA main screen click on array controller:



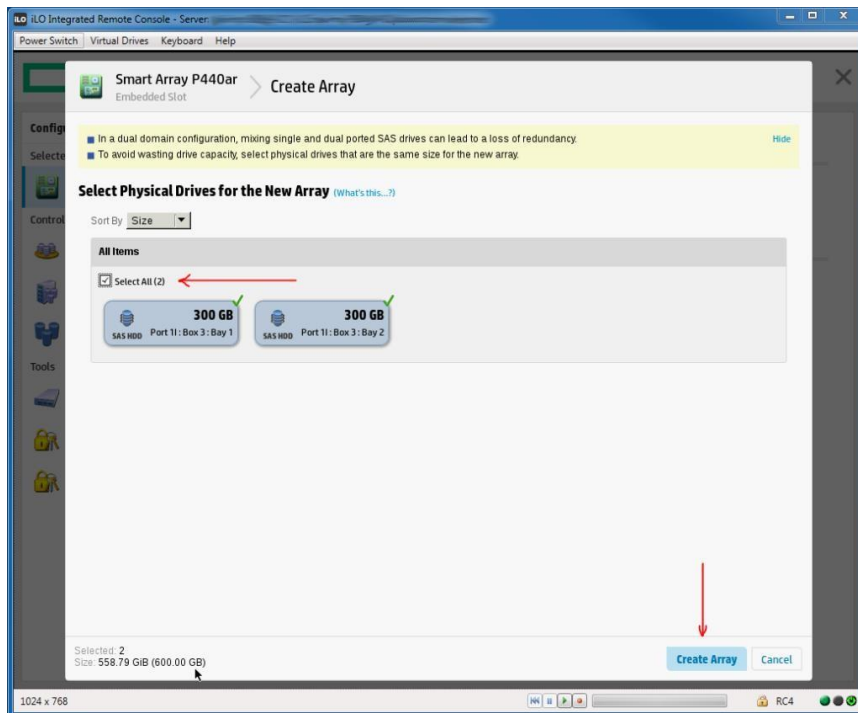
In the Smart Array **Actions** menu choose **Configure** option:



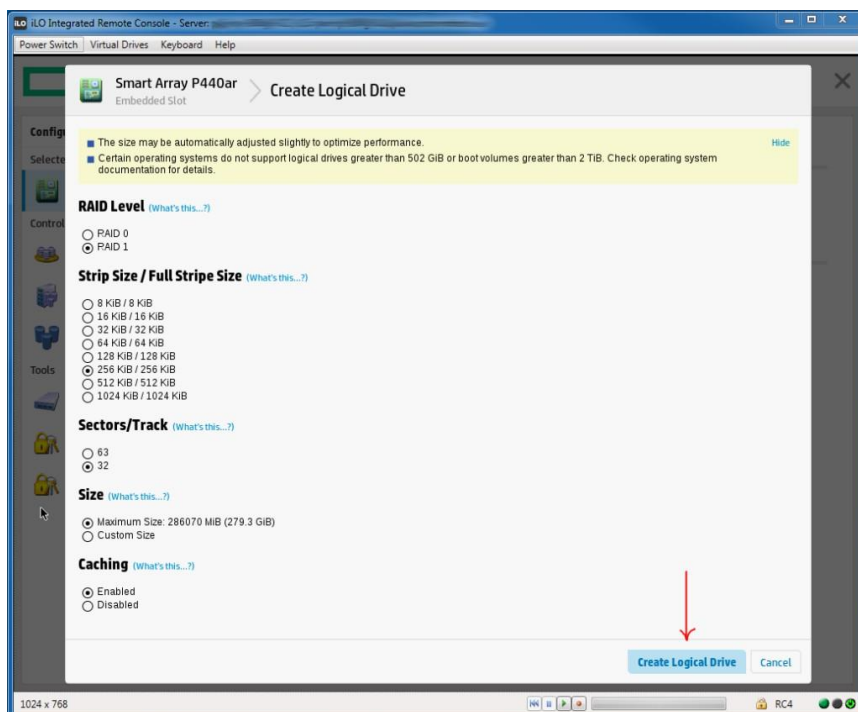
Click on **Create Array** button:



Since I have only two physical drives installed on the server, which is the minimum required for RAID 1 configuration, I must select all the disks available to create RAID 1 array:

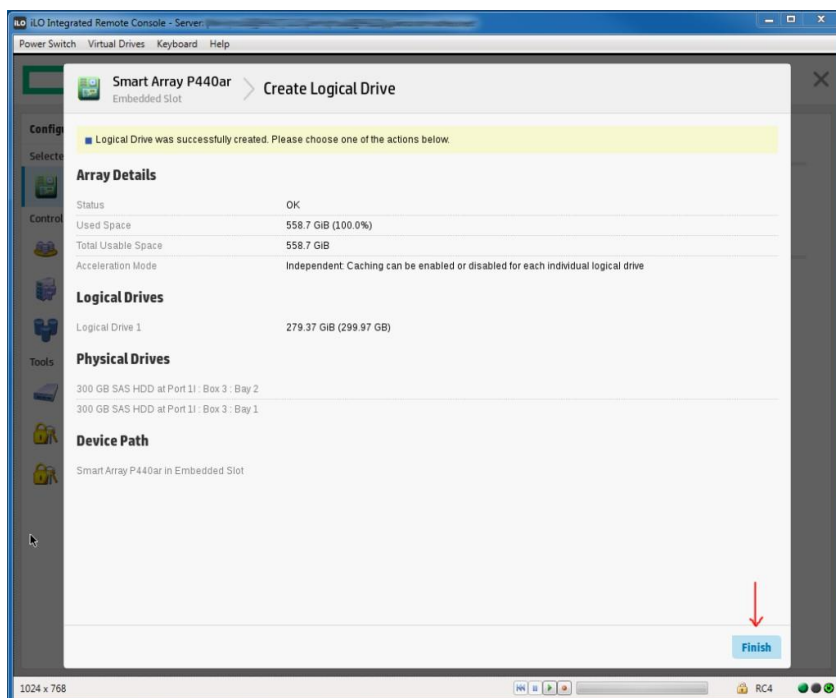


In **RAID level** section ensure that **RAID 1** has been chosen, you can leave the remaining parameters in their default values:

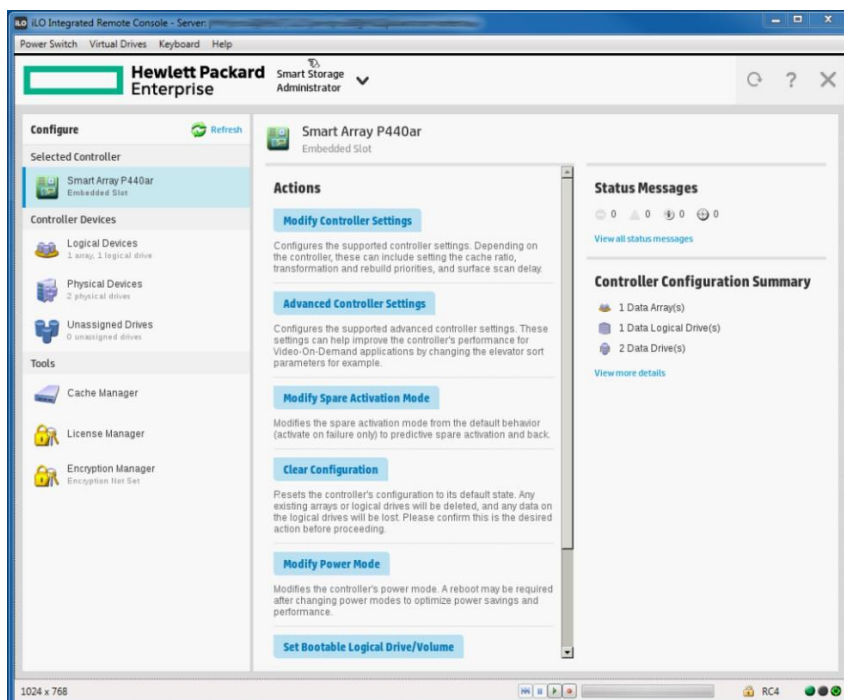


Click on **Create Logical Drive** button to create RAID 1 array.

Check summary screen to verify newly created array and click on **Finish** button:

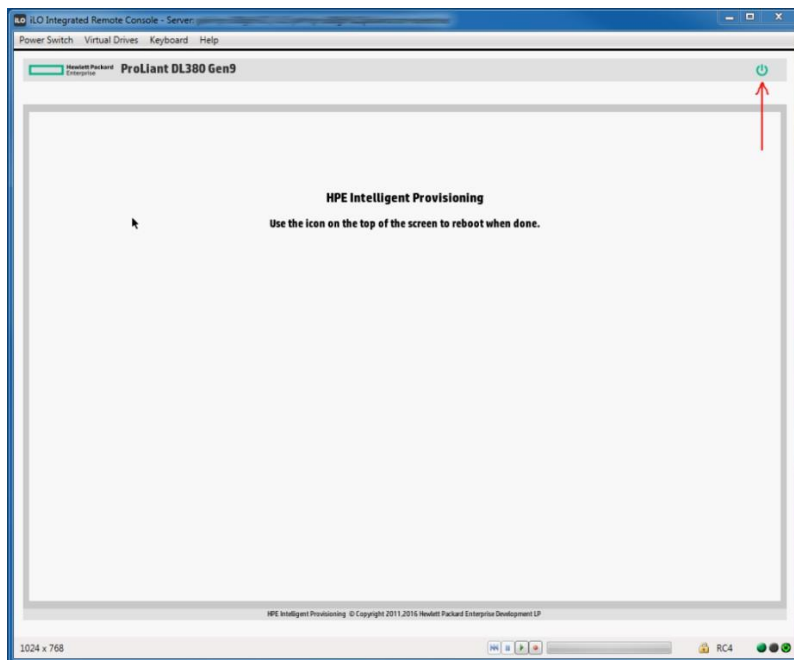


The new array should now appear in the left pane of the main HPE SSA window:



5. Exit HPE Smart Storage Administrator

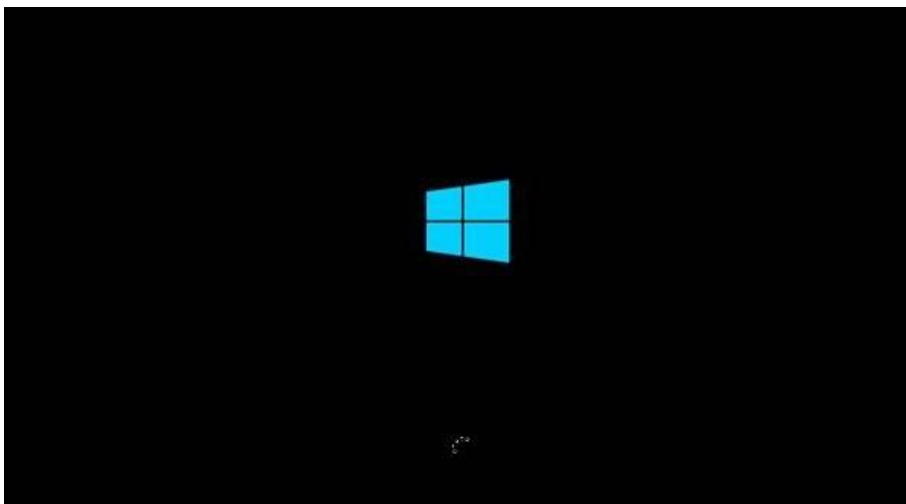
Click on the **X** button to exit the application, you will be redirected to Intelligent Provisioning screen. On Intelligent Provisioning screen click on the **Power Button** in the upper right corner to reboot the server:



OPERATING SYSTEM INSTALLATION

Step 1 – Create a boot USB driver which can be created with the Rufus App

Step 2 – After completing the above given steps, plug-in the USB to the server and wait for a while till it loads the files. This will continue for about a couple of minutes as shown in the following screenshot.



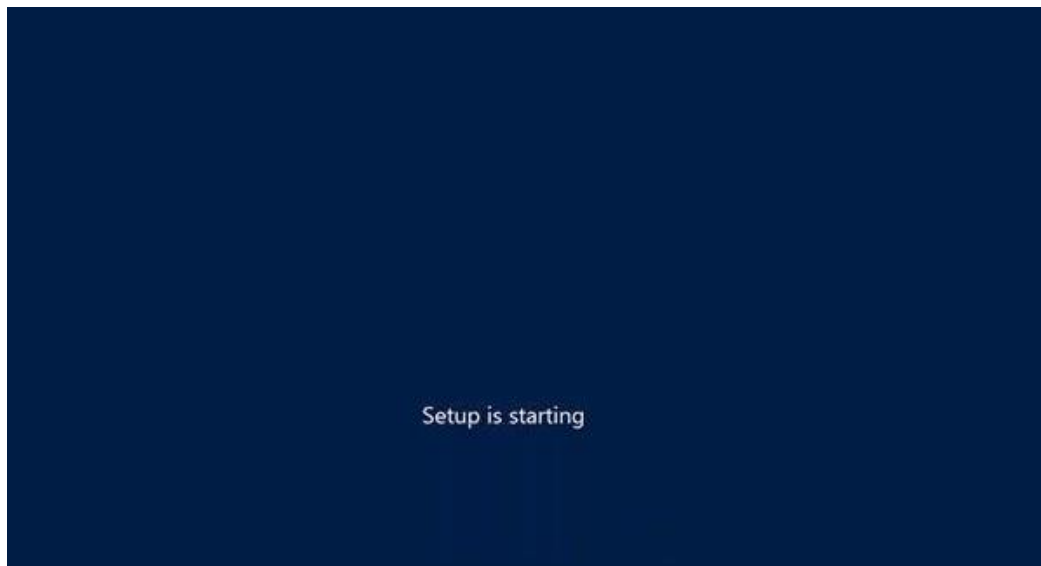
Step 3 – After the files are loaded, you will see the screen of language settings of installation, keyboard, time and currency format. Generally, all the default ones are also good enough to start with. Click Next.



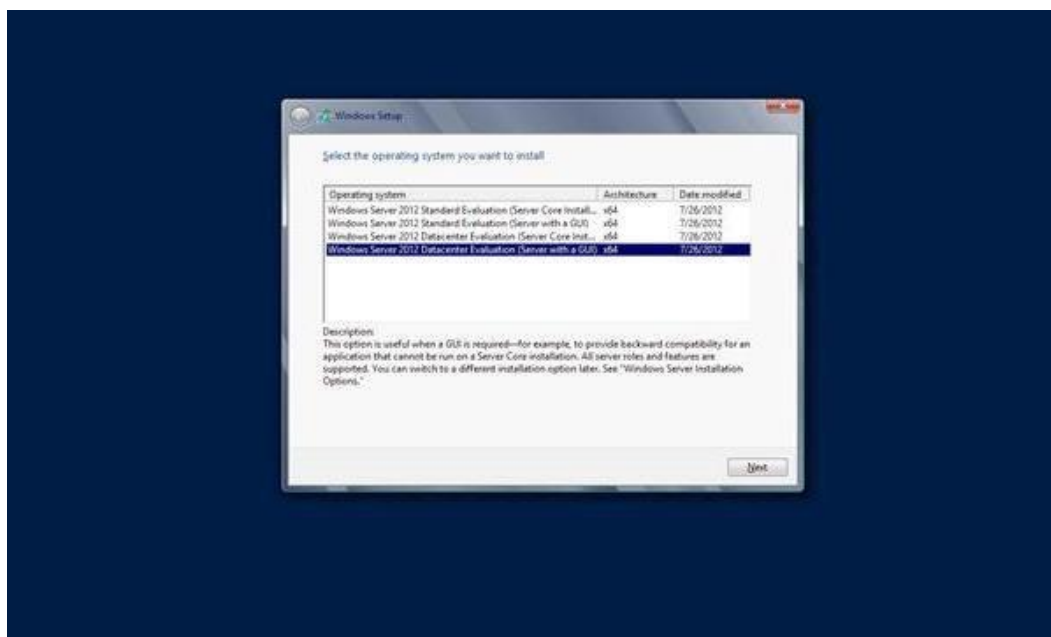
Step 4 – Click “Install now”.



Step 5 – Once you have clicked on Install Now, the setup will start, and it will load all the files and the screen will look as shown in the following screenshot.



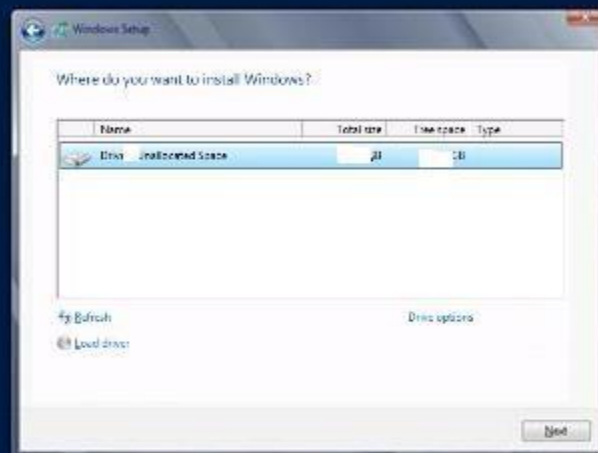
Step 6 – Wait until the files are loaded and then you will see the following screen. select Windows Server 2012 Standard Edition (Server with GUI) and click Next.



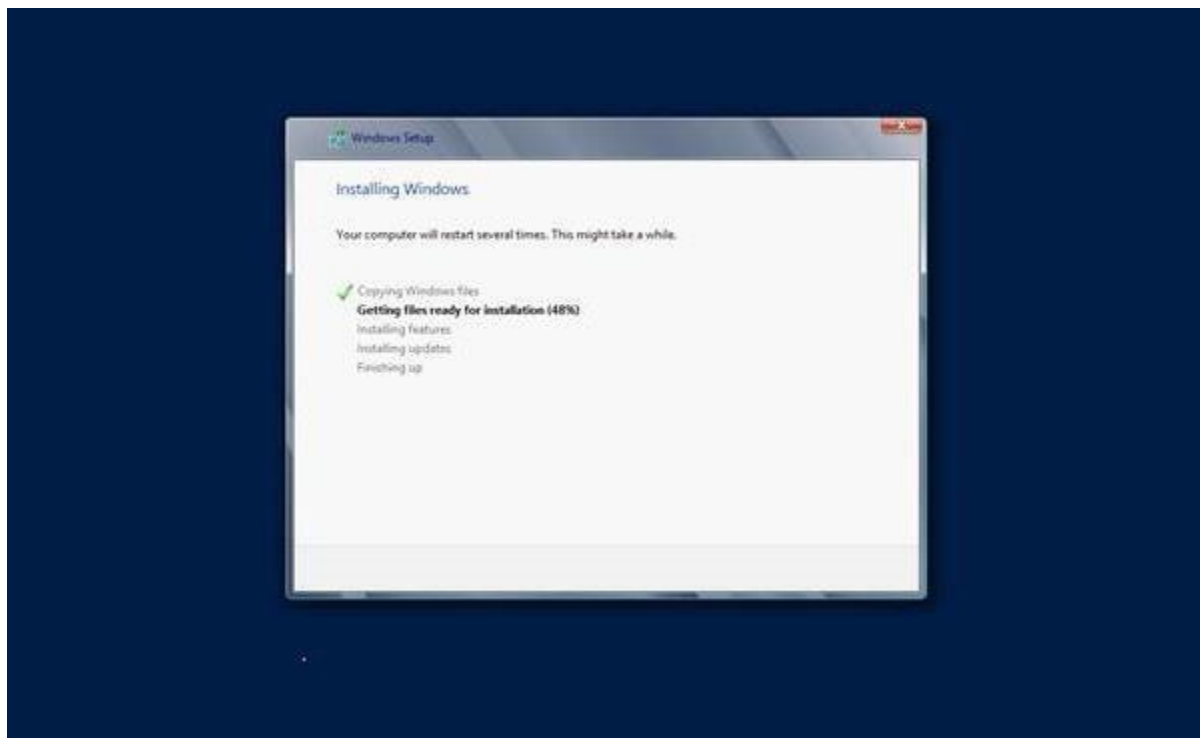
Step 7 – Click "I accept the license terms" and then click on the Next button as shown in the following screenshot.



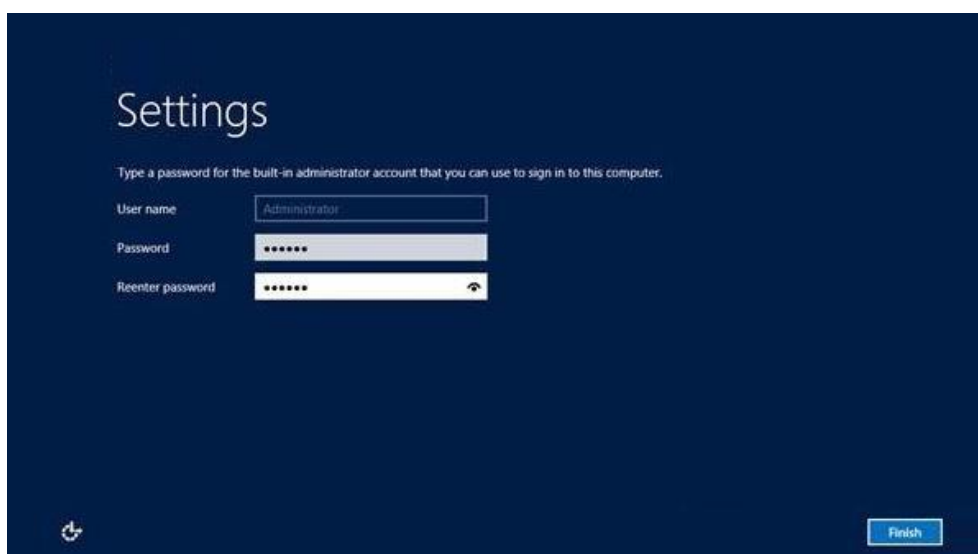
Step 8 – The following screen will appear. In “Driver Options” you can create a new partition, delete, or format the Hard disk. After you have done this process then you can select the partition where the Windows server 2012 R2 will be installed in our case we have one partition. Once all this is done, then click on Next.



Step 9 –Wait until this process finishes during this time and then the server will reboot.



Step 10 – Once the reboot is done the following screen will appear. Set the password for the server and then click on “Finish”.



Step 11 – It will take some minutes until the setup finishes completely.



Step 12 – Once all this is done, you have completed the installation process and the following screen will appear.



10.0 SSL Management (Certificate generation and renewal)

10.0 Background

SSL Certificates are small data files that digitally bind a cryptographic key to an organization's details. When installed on a web server, it activates the padlock and the https protocol and allows secure connections from a web server to a browser. Typically, SSL is used to secure credit card transactions, data transfer and logins, and more recently is becoming the norm when securing browsing of social media sites.

10.1 Process Objective (Purpose)

This process is a guideline when renewing SSL certificate.

If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

10.2 Process Owner

Group Head, IT

10.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

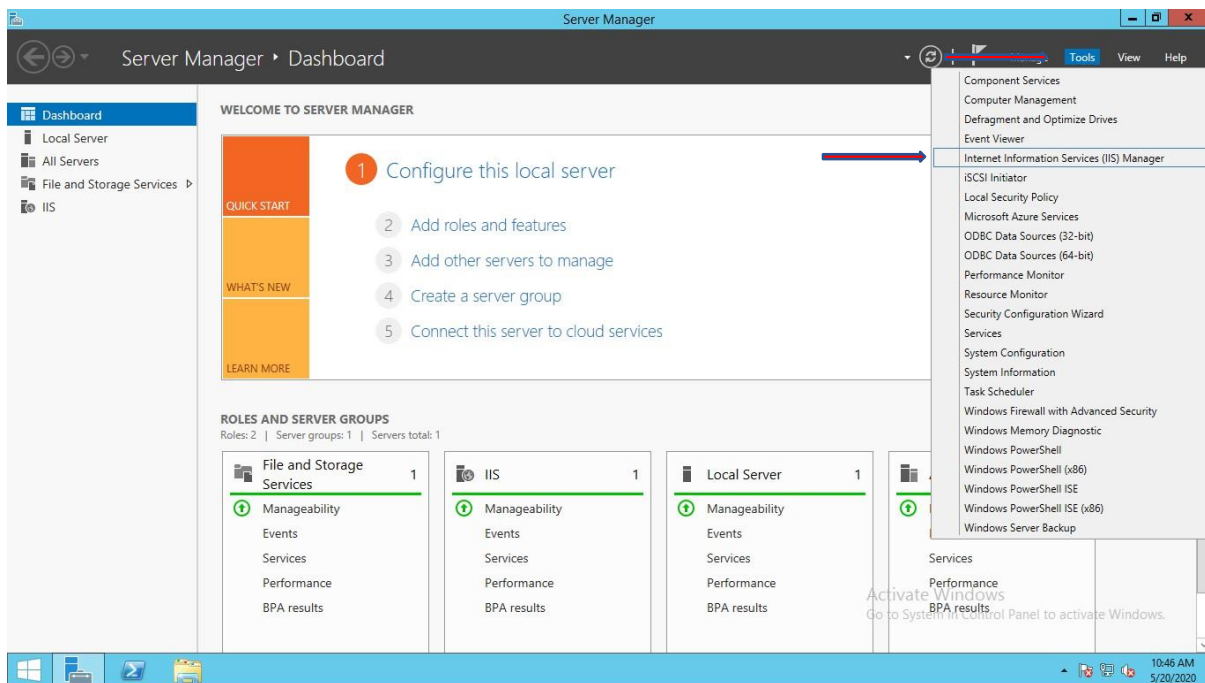
Role	Task	Process Steps	Estimated TAT
Server Administrator	SSL Management	1.1 Login to the server in which the web application resides 1.2 Follow the screen shot below to generate CSR, and attached a new certificate	30 Minutes

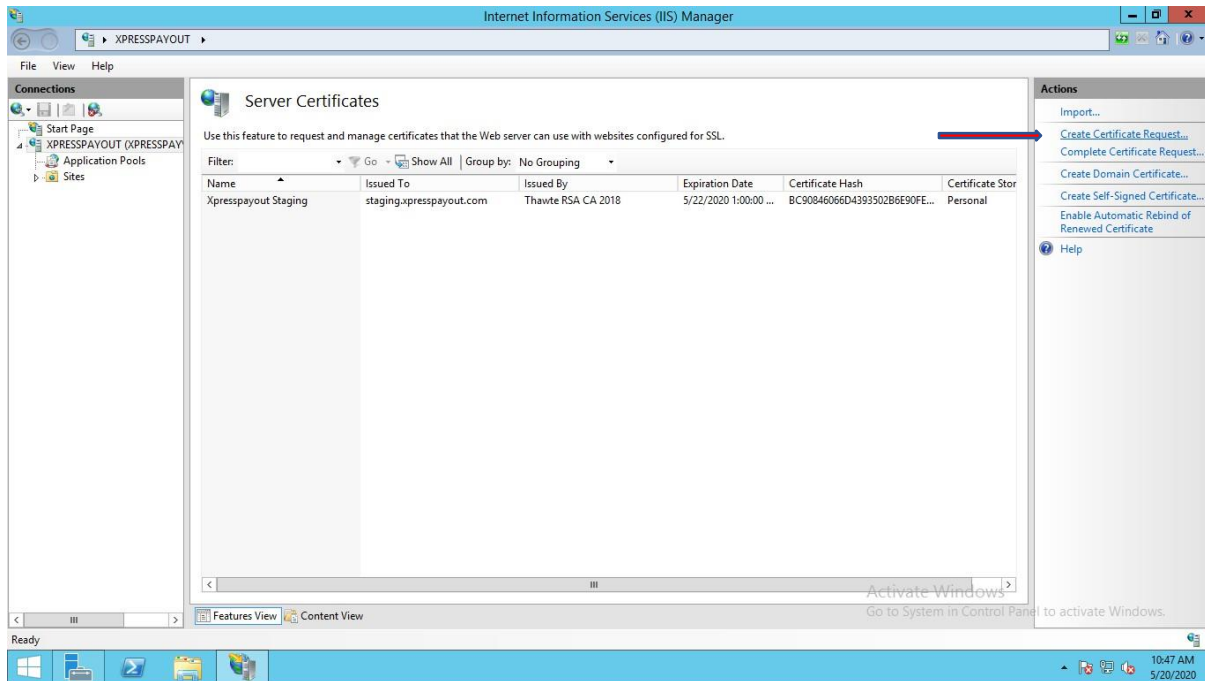
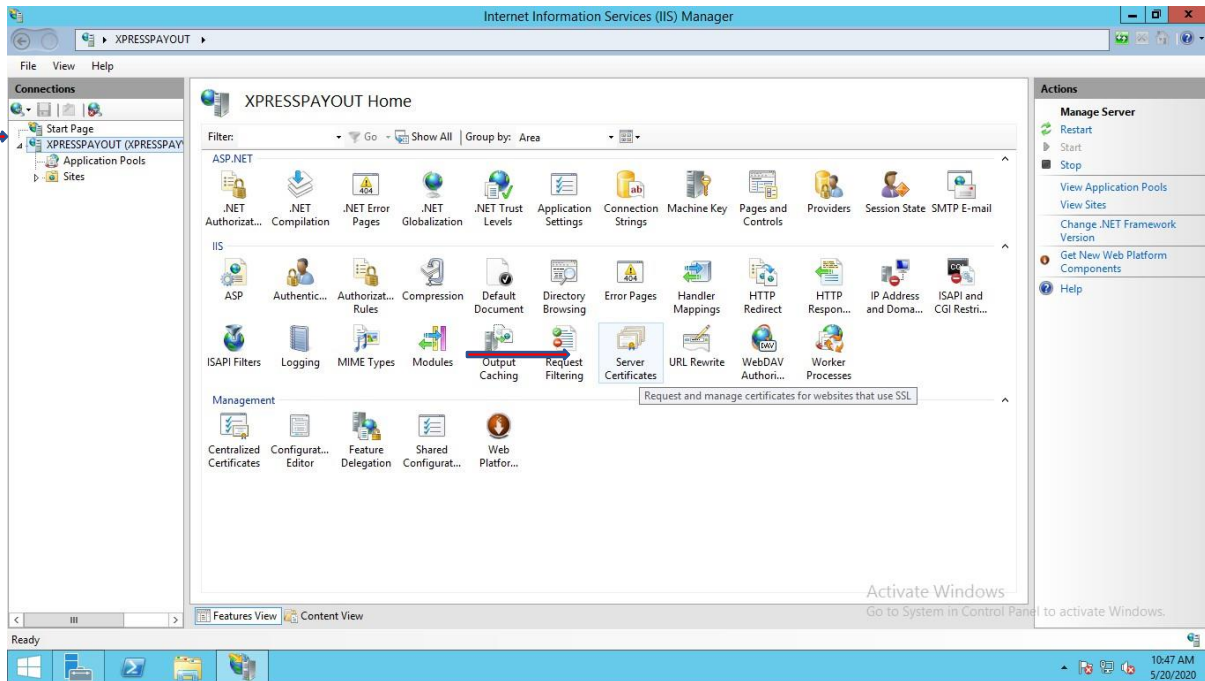
10.4 Key Risks/Controls

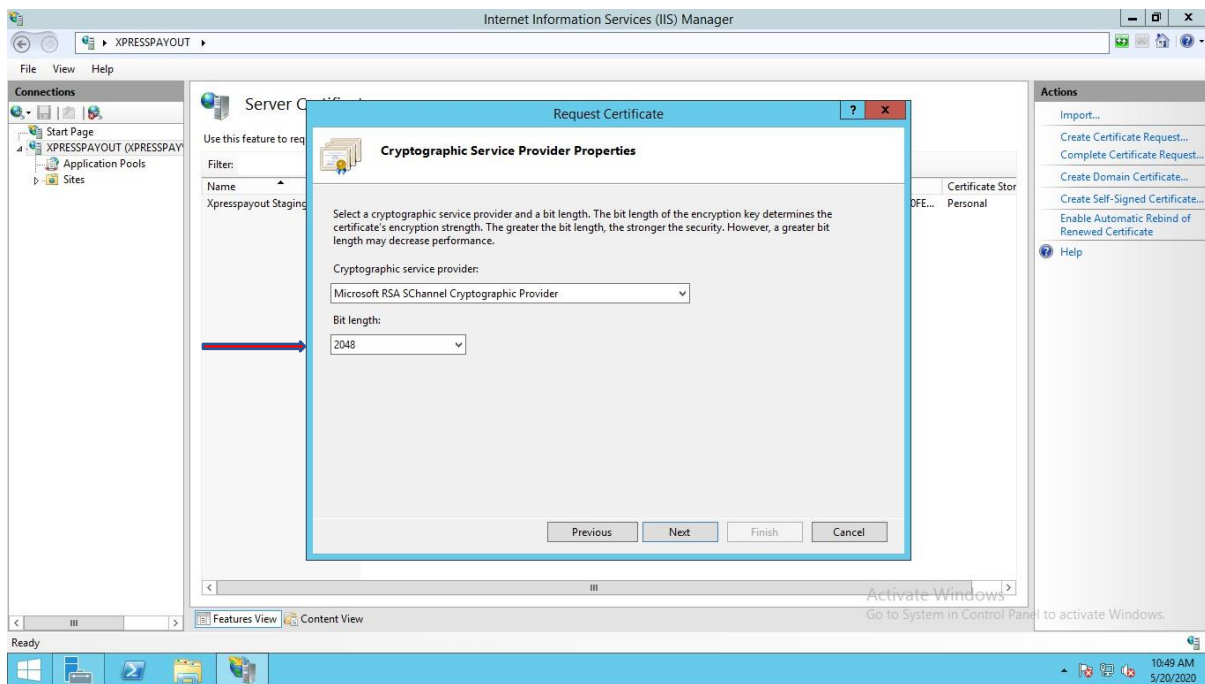
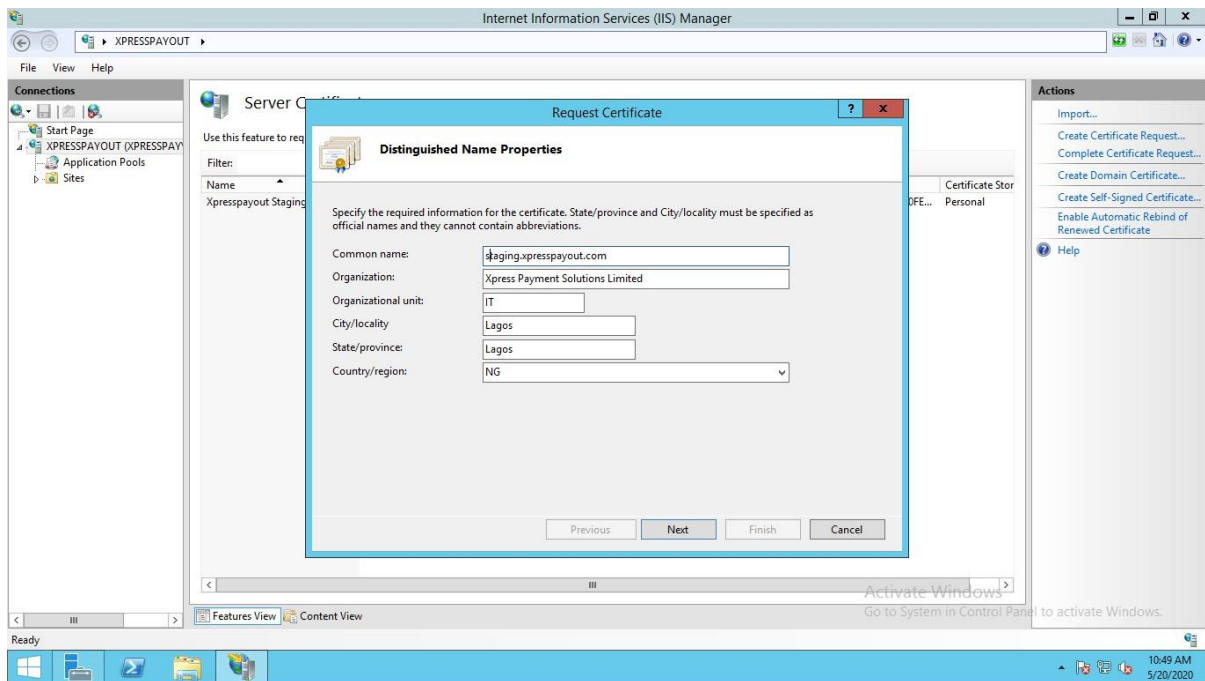
In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

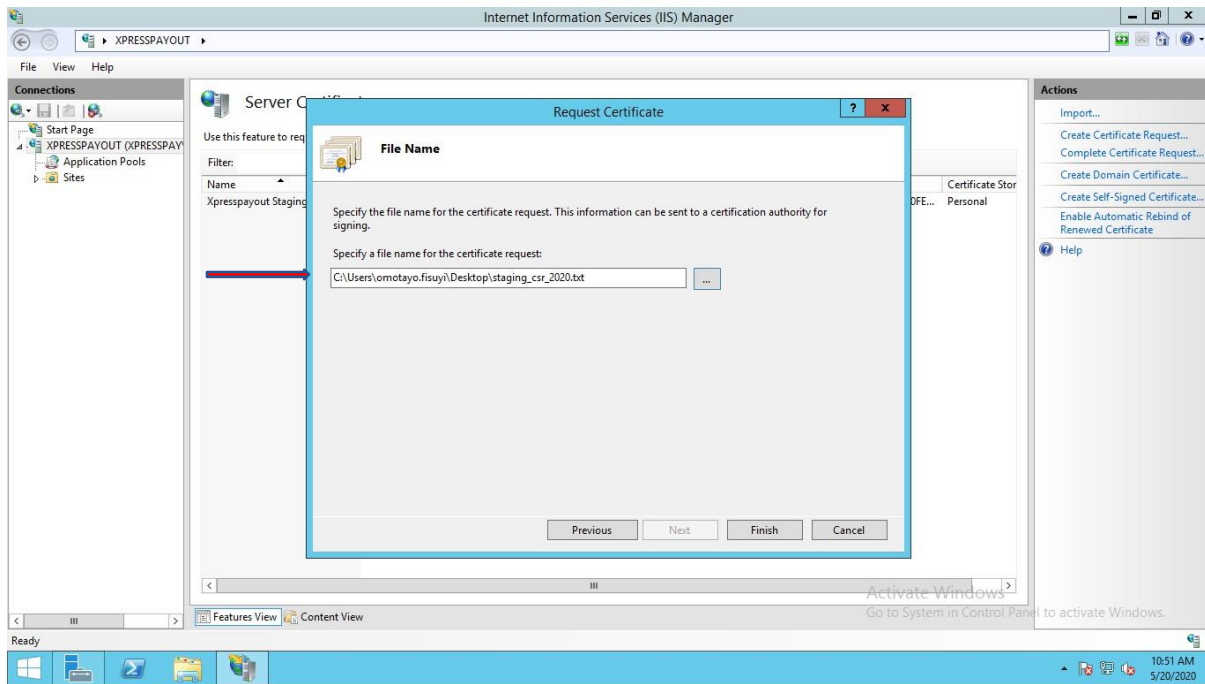
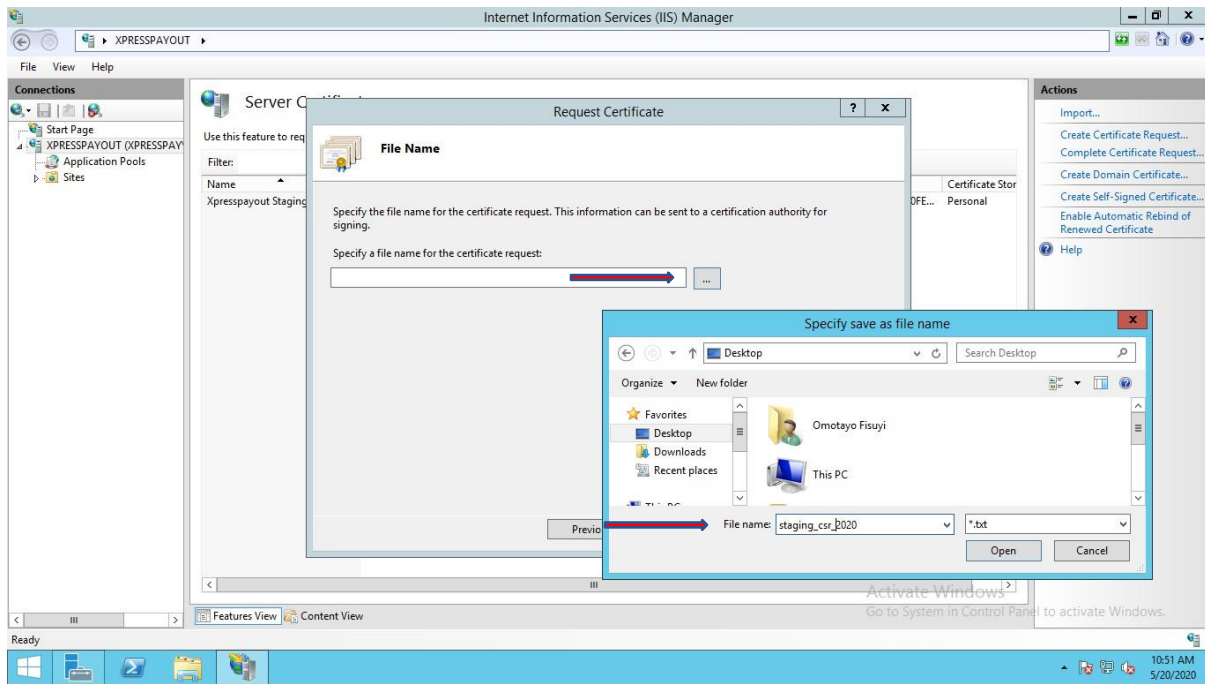
Risk	Controls
Lack of security on our web servers leading to fraud	Ensure SSL certificates are renewed as at when due

10.5 Login to the Server in which the web application resides.





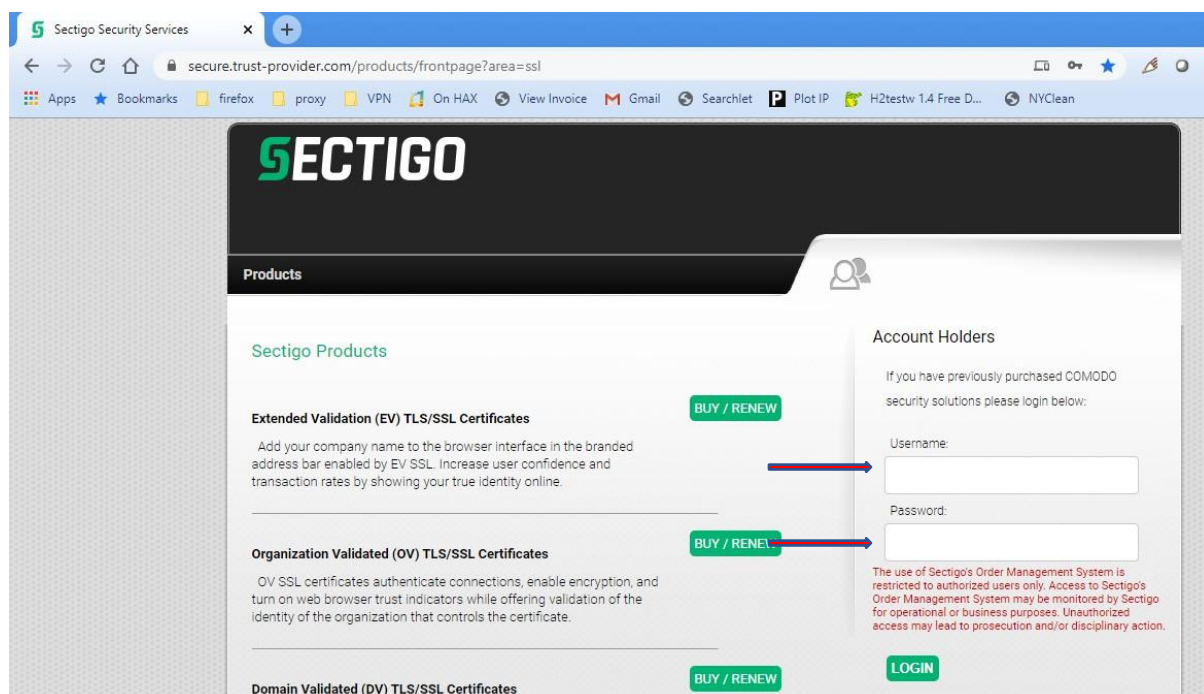






Login to the following url

<https://secure.trust-provider.com/products/frontpage?area=ssl>



Sectigo Security Services

secure.trust-provider.com/products/Management

Apps Bookmarks firefox proxy VPN On HAX View Invoice Gmail Searchlet Plot IP H2testw 1.4 Free D... NYClean

SECTIGO

Logout

Account Options: Management

Welcome: Omotayo Fisuyi
Xpress Payment Solutions Limited
IT

My Account Areas:

- E-PKI Manager**
Place orders through your E-PKI Manager
- IdAuthority**
Add / Update details of your website(s) in the IdAuthority

My Account Summary:

Last Login Time
28-APR-2020 09:38:27 (UTC)

Status
Active

Verification Level
Class 3

Options
[Change Password](#)
Current password expires in 43 days

Support

Thank you for being a Sectigo customer. We are here to provide you with the best experience

Sectigo Security Services

secure.trust-provider.com/products/EPKIOptions

Apps Bookmarks firefox proxy VPN On HAX View Invoice Gmail Searchlet P instan 1/1

(New - Click "Buy") (Replacement - First run a "Report on your orders")	
Enterprise SSL Pro with EV	BUY
Enterprise SSL Pro with EV Multi-Domain (New - Click "Buy") (Replacement - First run a "Report on your orders")	BUY
Optimum SSL with EV	BUY
COMODO SSL Certificate	BUY
COMODO SSL Wildcard Certificate	BUY
COMODO SSL Unified Communications Certificate (New - Click "Buy") (Replacement - First run a "Report on your orders")	BUY
InstantSSL Pro Certificate	BUY
PremiumSSL Certificate	BUY
PremiumSSL Wildcard Certificate	BUY
PremiumSSL Legacy Certificate	BUY
PremiumSSL Legacy Wildcard Certificate	BUY
EliteSSL Certificate	BUY
Enterprise SSL Certificate	BUY
Enterprise SSL Pro Certificate	BUY
Enterprise SSL Pro Wildcard Certificate	BUY
Unified Communications Certificate (New - Click "Buy")	BUY

Paste the CSR generated

- ☒ CNAME CSR Hash
- ☐ HTTP CSR Hash
- ☐ HTTPS CSR Hash
- ☐ None of the above. <— Choose this one if you don't know!!

CONTINUE >

Management area>>>>>E-KPI manager



E-PKI Manager: Account Options: Management

Welcome to E-PKI Manager. Use the E-PKI Manager to securely manage your account and your digital certificates.

Account Actions:

-  **Deposit additional funds**
Deposit additional funds
-  **Buy Prices**
View your current Buy Prices

Using your E-PKI Manager:

-  **E-PKI Manager pages**

Management Facilities:

-  **User Management**
User Management

Reporting Facilities:

-  **Report on Your Orders**
Run report on your Orders



Report on Your Orders:

Please enter the order number:

Order Number

Alternatively please select a date range. You can also search for orders for specific products or enter product specific search criteria.

Date

Range

From

1

JAN

2001

To

20

MAY

2020

Further options

- ☒ SSL Certificate
- ☒ Corporate Secure Email Certificate
- ☒ Deposit Funds
- ☒ Personal Authentication Certificate

RUN REPORT

WRITE REPORT TO CSV

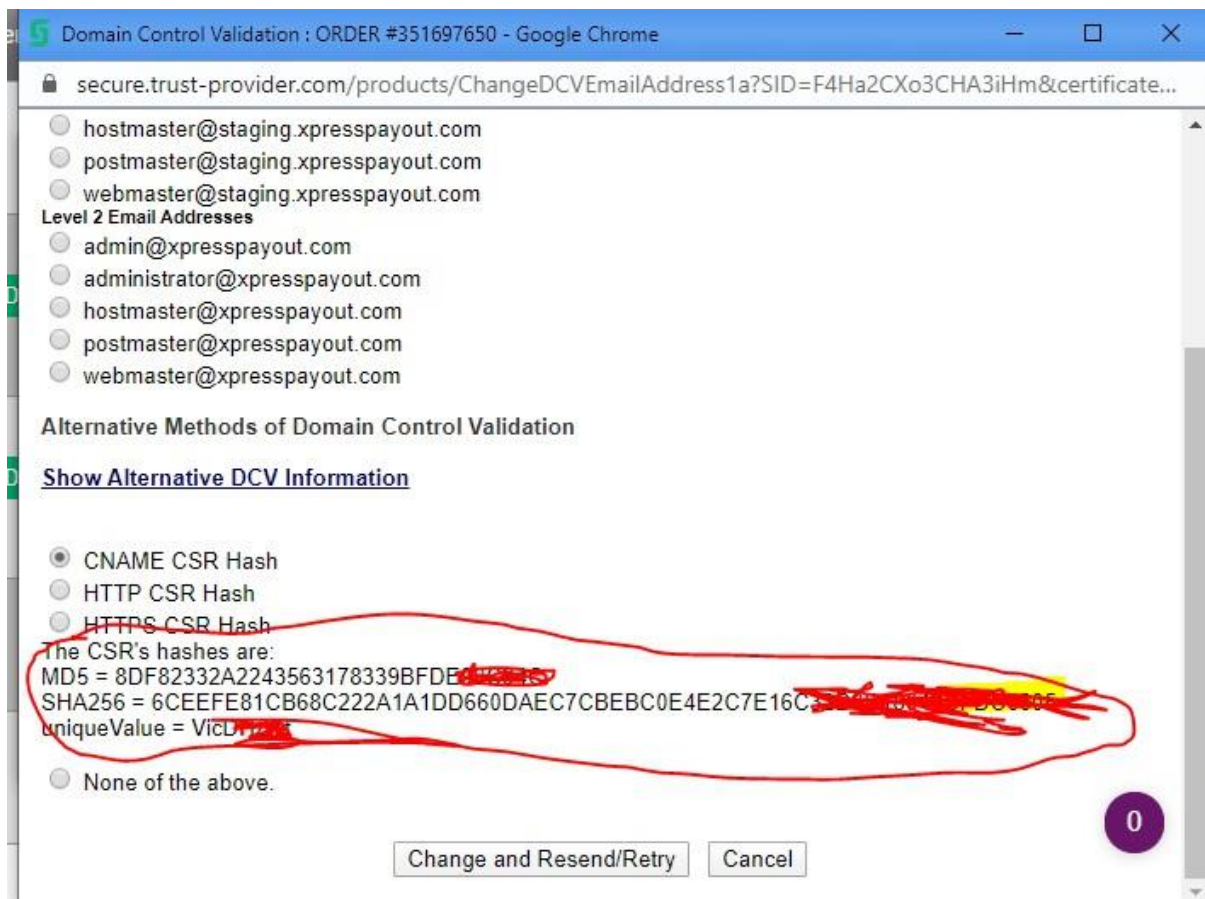
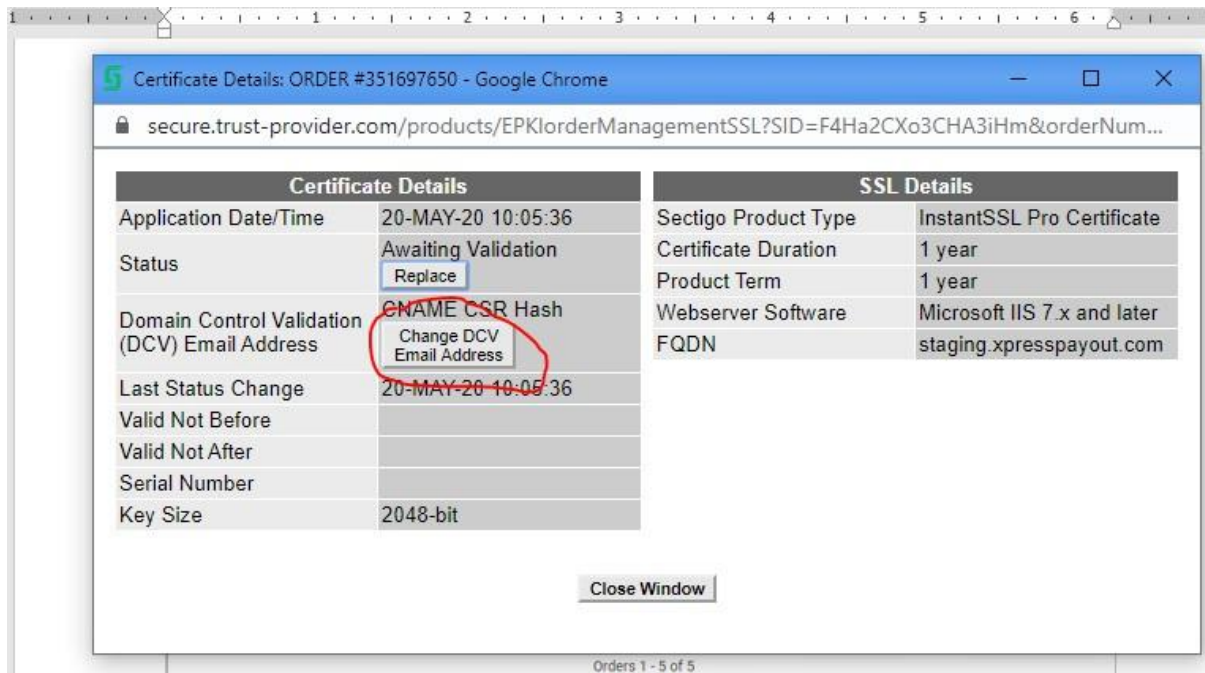
RETURN TO E-PKI MANAGER

Orders 1 - 5 of 5

\$9,967.01

ORDER # Date (GMT) Time (GMT)	Item(s)	COST
351697650 20-MAY-20 10:05am	DETAILS... Awaiting Validation InstantSSL Pro Certificate (1 year) for staging.xpresspayout.com RESEND CONFIRMATION EMAIL	\$97.97
343510289 27-APR-20 9:10am	DETAILS... Issued COMODO SSL Wildcard Certificate (731 days) for *.xpresspayonline.com CERTIFICATE DISCOVERY RESEND CONFIRMATION EMAIL DOWNLOAD AS .ZIP	\$684.52
339340598 09-APR-20 3:56pm	DETAILS... Issued COMODO SSL Wildcard Certificate (731 days) for *.xpresspayments.com CERTIFICATE DISCOVERY RESEND CONFIRMATION EMAIL DOWNLOAD AS .ZIP	\$684.52
338494570 06-APR-20 4:07pm	Topup Funds RESEND CONFIRMATION EMAIL SEND INVOICE EMAIL	\$3,500.00
337882825 03-APR-20 2:29pm	Signup Funds RESEND CONFIRMATION EMAIL SEND INVOICE EMAIL	\$5,000.00

Orders 1 - 5 of 5



Copy it to note pad and edit as follows

Login to the dns manager page of the domain (Godaddy, Upperlink or 1and1)

domains.upperlink.ng/clients/index.php

Facebook TravelStart eBay AliExpress

Credit Balance

You have a credit balance of N0.00 and this will be automatically applied to any new invoices.

DNS Manager

Portal Home / Client Area / My Domains / xpresspayout.com / BusyRack DNS Manager

BusyRack DNS Manager

manage DNS records for domain name xpresspayout.com

Host Name	TTL	Type	Value	Delete
www.staging	14440	A	Address: 80.88.8.32	[x] ☐
cp calendars	14400	A	Address: 127.0.0.1	[x] ☐
staging	14440	A	Address: 80.88.8.32	[x] ☐
cpanel	14400	A	Address: 127.0.0.1	[x] ☐
cpcontacts	14400	A	Address: 127.0.0.1	[x] ☐
whm	14400	A	Address: 127.0.0.1	[x] ☐

Add a new record from the edited above

Add a new DNS record

Host Name	TTL	Type	Value
_8DF82332A2243563178 .xpresspayout.com.	14440	CNAME	Address/Value: 6CEEFE81CB68C222A1A

Save Changes

Go to Sectigo report section

Orders 1 - 5 of 5 \$9,967.01

ORDER # Date (GMT) Time (GMT)	Item(s)	COST
351697650 20-MAY-20 10:05am	DETAILS... Awaiting Validation InstantSSL Pro Certificate (1 year) for staging.xpresspayout.com RESEND CONFIRMATION EMAIL	\$97.97
343510289 27-APR-20 9:10am	DETAILS... Issued COMODO SSL Wildcard Certificate (731 days) for *.xpresspayonline.com CERTIFICATE DISCOVERY RESEND CONFIRMATION EMAIL DOWNLOAD AS .ZIP	\$684.52
339340598 09-APR-20 3:56pm	DETAILS... Issued COMODO SSL Wildcard Certificate (731 days) for *.xpresspayments.com CERTIFICATE DISCOVERY RESEND CONFIRMATION EMAIL DOWNLOAD AS .ZIP	\$684.52
338494570 06-APR-20 4:07pm	Topup Funds RESEND CONFIRMATION EMAIL SEND INVOICE EMAIL	\$3,500.00
337882825 03-APR-20 2:29pm	Signup Funds RESEND CONFIRMATION EMAIL SEND INVOICE EMAIL	\$5,000.00

Orders 1 - 5 of 5

Certificate Details: ORDER #351697650 - Google Chrome

secure.trust-provider.com/products/EPKlorderManagementSSL?SID=7O1aXO6tFMbPQVh&orderNum...

Certificate Details		SSL Details	
Application Date/Time	20-MAY-20 10:05:36	Sectigo Product Type	InstantSSL Pro Certificate
Status	Awaiting Validation Replace	Certificate Duration	1 year
Domain Control Validation (DCV) Email Address	CNAME CSR Hash Change DCV Email Address	Product Term	1 year
Last Status Change	20-MAY-20 10:05:36	Webserver Software	Microsoft IIS 7.x and later
Valid Not Before		FQDN	staging.xpresspayout.com
Valid Not After			
Serial Number			
Key Size	2048-bit		

Close Window

0

Validated page

ORDER # Date (GMT) Time (GMT)	Item(s)
351697650 20-MAY-20 10:05am	InstantSSL Pro Certificate (366 days) for staging.xpresspayout.com DETAILS... Issued CERTIFICATE DISCOVERY RESEND CONFIRMATION EMAIL DOWNLOAD AS .ZIP

Download the new certificate

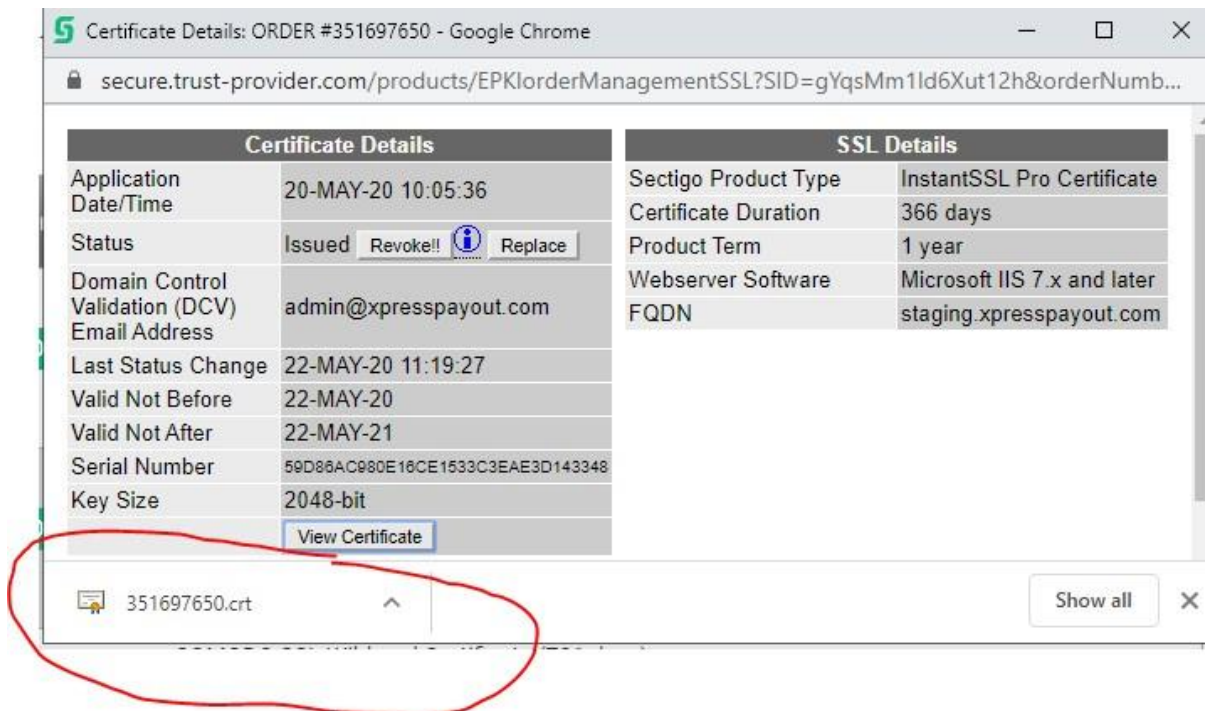
Certificate Details: ORDER #351697650 - Google Chrome

secure.trust-provider.com/products/EPKlorderManagementSSL?SID=gYqsMm1ld6Xut12h&orderNumb...

Certificate Details		SSL Details	
Application Date/Time	20-MAY-20 10:05:36	Sectigo Product Type	InstantSSL Pro Certificate
Status	Issued Revoke!! Replace	Certificate Duration	366 days
Domain Control Validation (DCV) Email Address	admin@xpresspayout.com	Product Term	1 year
Last Status Change	22-MAY-20 11:19:27	Webserver Software	Microsoft IIS 7.x and later
Valid Not Before	22-MAY-20	FQDN	staging.xpresspayout.com
Valid Not After	22-MAY-21		
Serial Number	59D86AC980E16CE1533C3EAE3D143348		
Key Size	2048-bit		

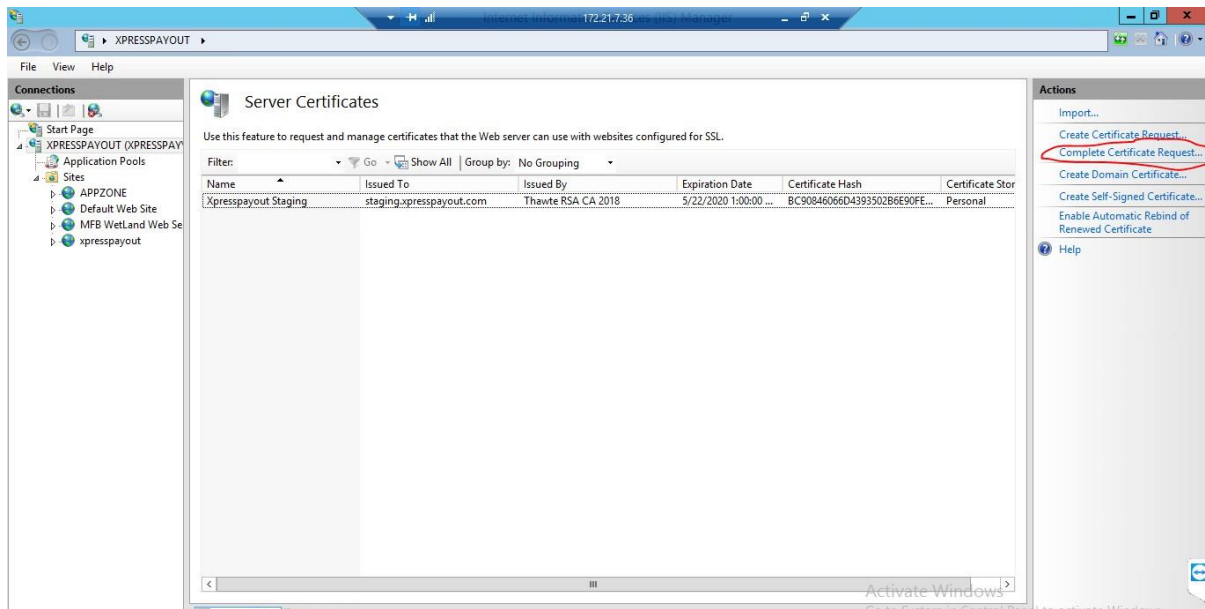
View Certificate

Close Window

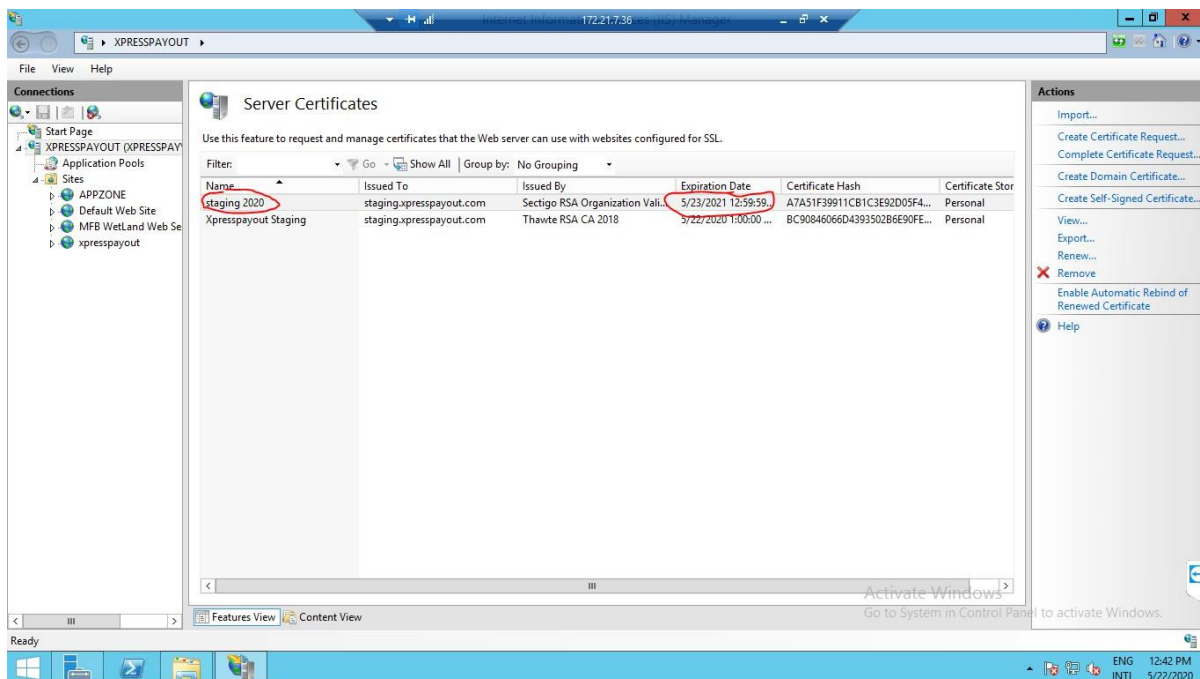
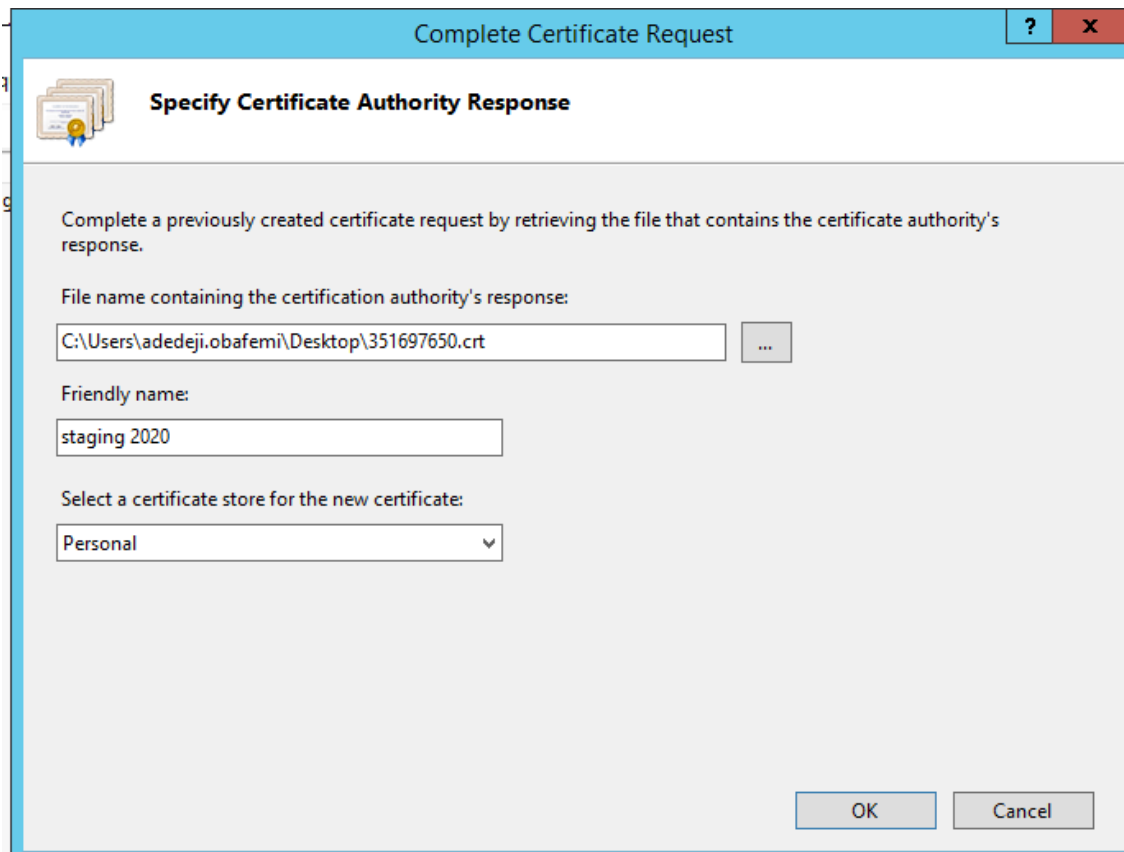


Copy the new certificate to the server

Goto IIS Manager>>>Server Certificate



Browse to where the certificate is located



Click on each of the site (Appzone,xpresspayout etc)

Click on Binding>>>>Edit>>>>select the current certificate>>>>ok

Restart the site IIS

Browse the site and view the certificate to be sure of the new date

11.0 How to create New User on Active Directory

11.0 Background

An active directory user is one whose username and password are stored on a domain controller rather than the computer the user is logging into. When you log in as an AD user, the computer asks the domain controller what privileges are assigned to you. When the computer receives an appropriate response from the domain controller, it logs you in with the proper permissions and restrictions. The purpose of this is to restrict permissions and monitor activities of the end users by the administrator.

11.1 Process Objective (Purpose)

This process is a guideline to create a new user on the Active directory. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

11.2 Process Owner

Group Head, IT

11.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	Staff creation on domain	1.1 Get the completed sign off User creation form 1.2 Remote to the domain server 1.3 Lunch the Active directory and computer 1.4 Right Click xpresspay.com.ng and click on NEW User 1.5 Supply the New User details 1.6 Supply the default password and enforce the user to change password at first login	15 Minutes

Support Staff/New User	Logging in to the User PC	2.1 Network team to connect the User to network. 2.2 Logging to the user PC with the new credentials 2.3 Ask the user to change is default password 2.4 Configure the User email via Outlook and Activate the Microsoft Applications. 2.5 Handover the system to the new user	30 Minutes
------------------------	---------------------------	---	------------

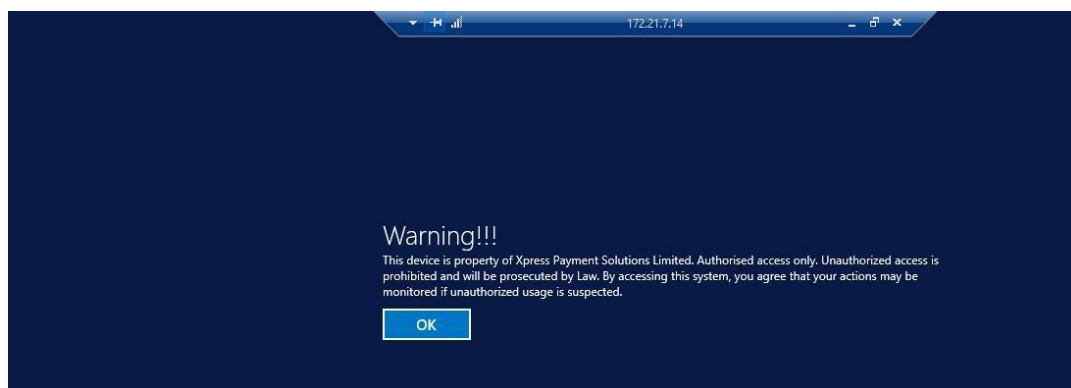
11.4 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

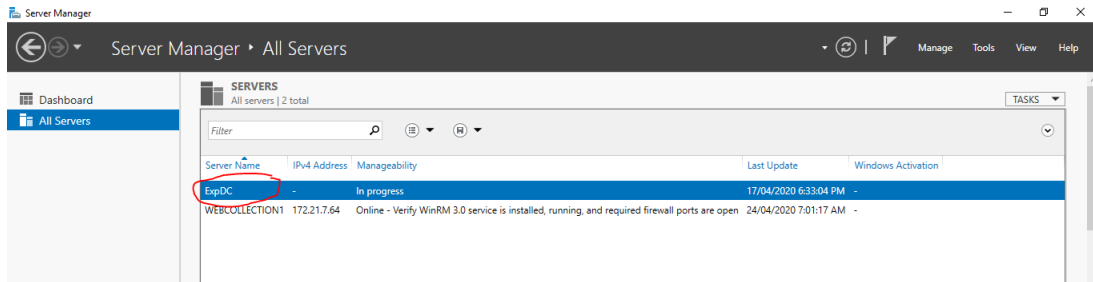
Key Risk	Control
Uncompleted User creation form	Escalate to the HCM
PC delay	Escalate to HCM and Admin team
Network related Issues	Escalate to the Network team

11.5 HOW TO CREATE NEW USER ON ACTIVE DIRECTORY

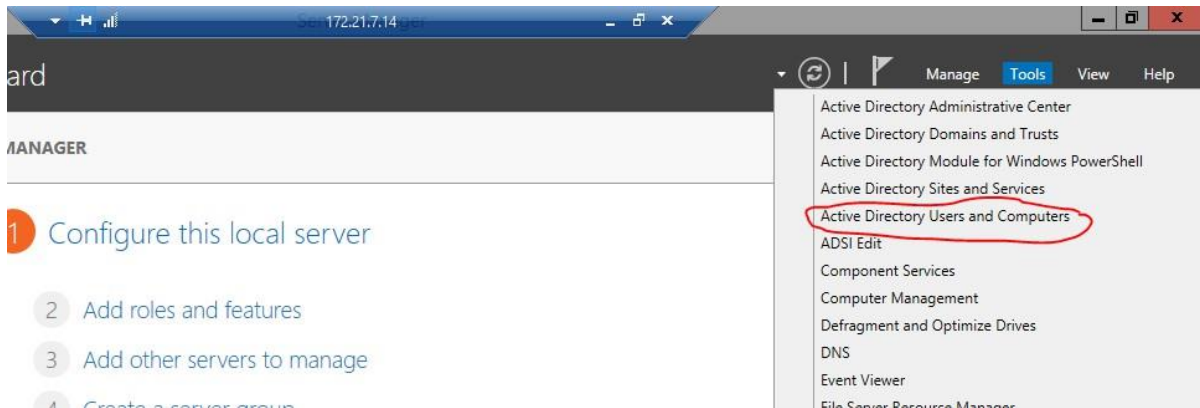
11.5.1 Login to the Active Directory Server or Use Server manager client



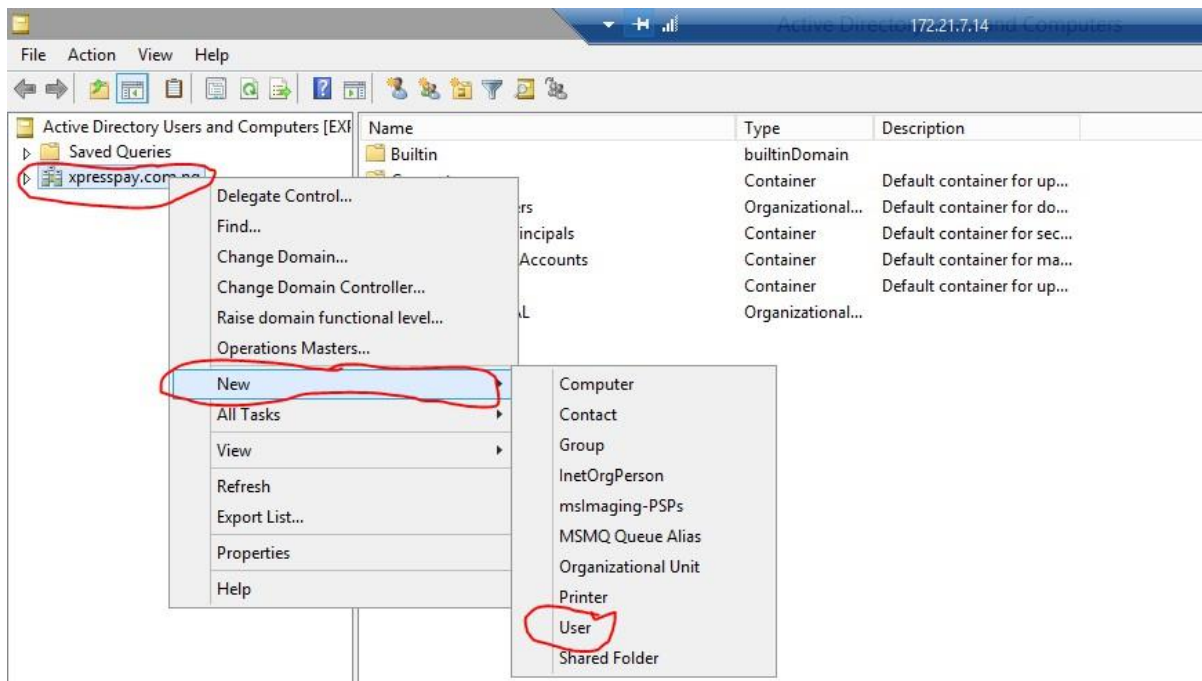
Or



11.5.2 Launch the Active directory and computer



11.5.3 Right Click >>>xpresspay.com.ng>>>NEW>>>User



11.5.4 Supply the New users details

The 'New Object - User' dialog box is shown with the following details entered:

- Create in: xpresspay.com.ng/
- First name: user
- Last name: user
- Full name: user user
- User logon name: user user | @xpresspay.com.ng
- User logon name (pre-Windows 2000): XPRESSPAY\ | user user

The 'Next >' button is highlighted.

11.5.5 Supply the default password and enforce the user to change password at first login

New Object - User

Create in: xpresspay.com.ng/

Password:

Confirm password:

☒ User must change password at next logon

☐ User cannot change password

☐ Password never expires

☐ Account is disabled

< Back Next > Cancel

11.5.6 Setup the user Email on Outlook

12.0 Terminal Management System Upgrade Process

12.0 Background

Terminal management system is used to deploy, maintain, monitor, and manage acceptance devices such as POS Terminals at Merchant locations.

12.1 Process Objective (Purpose)

This process is a guideline for us whenever we need to deploy or upgrade the terminal management system. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges

12.2 Process Owner

Group Head, IT

12.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	Deploy or upgrade terminal management	1.1 login to the TMS server via SSH 1.2 Supply Username and Password 1.3 Change to root by issuing the following command "<i>sudo su'username'</i>" and type password 1.4 Stop the Tomcat Service "<i>service tomcat stop</i>" 1.5 Create a new backup folder: "<i>mkdir /home/catalinalog/Upgrade/Backup/BackupDATE</i>" 1.6 Goto tomcat folder: "<i>cd /opt/apache-tomee-plume-7.1.0/webapps</i>" 1.7 Backup the TMS file: "<i>mv ROOT.war ROOT /home/catalinalog/Upgrade/Backup/BackupDATE</i>" 1.8 Backup TMS log files: "<i>mv /opt/apache-tomee-plume-7.1.0/logs/* /home/catalinalog/Upgrade/Backup/BackupDATE</i>" 1.9 Backup Catalina log: a) <i>cd /home/catalinalog/</i> b) <i>mv catalina.out Catalina.out.backupdate</i>	30 Minutes

		1.10 Create new Catalina log a) <i>touch catalina.out</i> b) <i>chown tomcat:tomcat catalina.out</i> c) <i>chmod 644 catalina.out</i> 1.11 Change directory to tomcat folder: <i>cd /opt/apache-tomee-plume-7.1.0/webapps/</i> 1.12 Rename TMS file on your PC to <i>ROOT.war</i> 1.13 Upload the tms ROOT.war file to tomcat folder: <i>/opt/apache-tomee-plume-7.1.0/webapps/</i> 1.14 Start tomcat: <i>service tomcat start</i>	
Server Administrator	To Start and Stop Postgres services	2.1 login to the TMS server via SSH 2.2 Supply Username and Password 2.3 Change to root by issuing the following command " <i>sudo su'username'"</i> and type password 2.4 <i>service postgresql-9.2 stop</i> 2.5 <i>service postgresql-9.2 start</i>	10 Minutes

12.4 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

Risks	Control
Service Failure	Scheduling upgrade of the TMS

13.0 Setting up and Configuring New User Laptop

13.0 Background

System installation or setup is the act of making the system or program ready for execution. Without the OS installation and configuration users will not be able to access the servers to perform any task.

13.1 Process Objective (Purpose)

This process is a guideline for setting up and configuring new user laptop. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

13.2 Process Owner

Group Head, IT

13.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Support /Shift Team	Pre-hand over	1.1 Check if there is network LAN port on the system 1.2 Power on the system and create user: xptestadmin 1.3 Check operating system if it is windows 10 professional 1.4 Rename the system 1.5 Create a folder in the c: drive as "Apps Install" folder 1.6 Copy all approved software's to the folder created 1.7 Copy Microsoft office folder "ODT" separately in drive c: 1.8 Install all the programs 1.9 Install TrendMicro Anti-virus 1.10 Join the system to the domain: "xpresspay.com.ng"	30 minutes

		1.11 Login to TrendMicro cloud and apply security policy on TrendMicro CloudOne	
	Post-handover after submission of User access creation form	2.1 Create domain account 2.2 Add staff to relevant organizational unit 2.3 Create email account 2.4 Add staff to appropriate distribution/security group 2.5 Hand over system to staff 2.6 Staff to log in and change password 2.7 Log in and change Email password 2.8 Setup Microsoft Outlook 2.9 Setup official signature on CodeTwo Portal 2.10 Log in to skype for business 2.11 Log in to Microsoft teams.	30 minutes

13.4 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/workaround, this should be stated as Not Applicable.

Risk	Control
Uncompleted User creation form	Escalate to HCM/Admin Team
Delay in laptop supply	Escalate to HCM/Admin Team

14.0 Veeam Backup on both Disk and Tape

14.1 Background

A Server backup is a computer in a network used to store copies of files from client machines or servers. The purpose of backing up most of our servers is as follows: -

14.1.1 Data Loss Prevention

14.1.2 Operation Plan B

14.1.3 Archiving

14.1.4 Improved Productivity

14.1.5 No Wasted Time

14.2 Process Objective (Purpose)

This process is a guideline for performing Veeam backup both on disk and tape. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges.

14.3 Process Owner

Group Head, IT

14.4 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
Server Administrator	To backup up a VM server to the repository	1.1 Remote to the Veeam backup server 1.2 Follow the screen shot below (From pg. 88 - 98)	1 hour
	To push a backup job to tape	2.1 Remote to the Veeam backup server 2.2 Follow the screen shot below (From pg. 98 - 140)	30 minutes

Support /Shift Team	To monitor backup and report generation.	3.1 Remote to the Veeam backup server 3.2 Follow the screen shot below (From pg. 103 - 105)	15 minutes
---------------------	--	--	------------

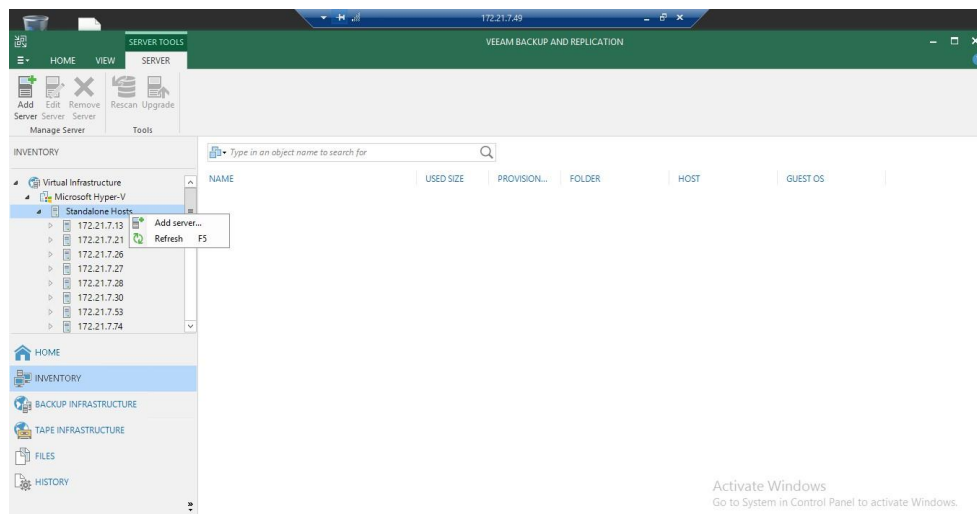
14.5 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/walkaround, this should be stated as Not Applicable.

Risks	Controls
Network related issues	Escalate to the Network team
Resource Availability	Escalate to Lead IT Infrastructure team.
Unknown Error	Escalate to Lead IT Infrastructure team.

14.6 To backup up a VM server to the repository

Step1: Add the host server to the inventory



New Hyper-V Server



Name
Specify DNS name or IP address of Microsoft Hyper-V server.

Name

Type

Credentials

Apply

Results

Summary

DNS name or IP address:

Description:

Created by XPRESSPAY\obafemi.adedeji at 5/11/2020 4:38 PM.

< Previous

Next >

Finish

Cancel

Type

Select the type of server you want to add.

Name

Type

Credentials

Apply

Results

Summary

☐ Microsoft System Center Virtual Machine Manager (SCVMM)

If you are using SCVMM to manage your Hyper-V infrastructure, we can pull all Hyper-V hosts and clusters information from SCVMM, and add them to managed servers automatically.

☐ Microsoft Hyper-V cluster

Use this option to register Hyper-V cluster that is not managed by SCVMM. Registering Hyper-V cluster instead of individual cluster nodes provides for Live Migration awareness.

☒ Microsoft Hyper-V server (standalone)

Use this option to register standalone Hyper-V server that is not a part of a cluster, and not managed by SCVMM.

< Previous

Next >

Finish

Cancel

New Hyper-V Server

Credentials

Select server administrator credentials.

Name

Type

Credentials

Apply

Results

Summary

Select an account with local administrator privileges on the server you are adding. Use DOMAIN\USER format for domain accounts, or HOST\USER for local accounts.

Credentials:

xpresspay\Administrator (xpresspay\Administrator, last edited: 63 days ago)

Add...

[Manage accounts](#)

Click Ports to customize network ports to be used by individual components

Ports...

< Previous

Next >

Finish

Cancel

New Hyper-V Server

Apply

Please review your settings and click Apply to continue.

Name

Type

Credentials

Apply

Results

Summary

Due to these modifications the following components will be installed or removed on the target host:

Component name	Status
Transport	already exists
Hyper-V Integration	already exists

This Hyper-V server will act as the backup proxy for jobs running in the on-host backup mode.

Task limit:

4

After you click Apply missed components will be installed on the target host.

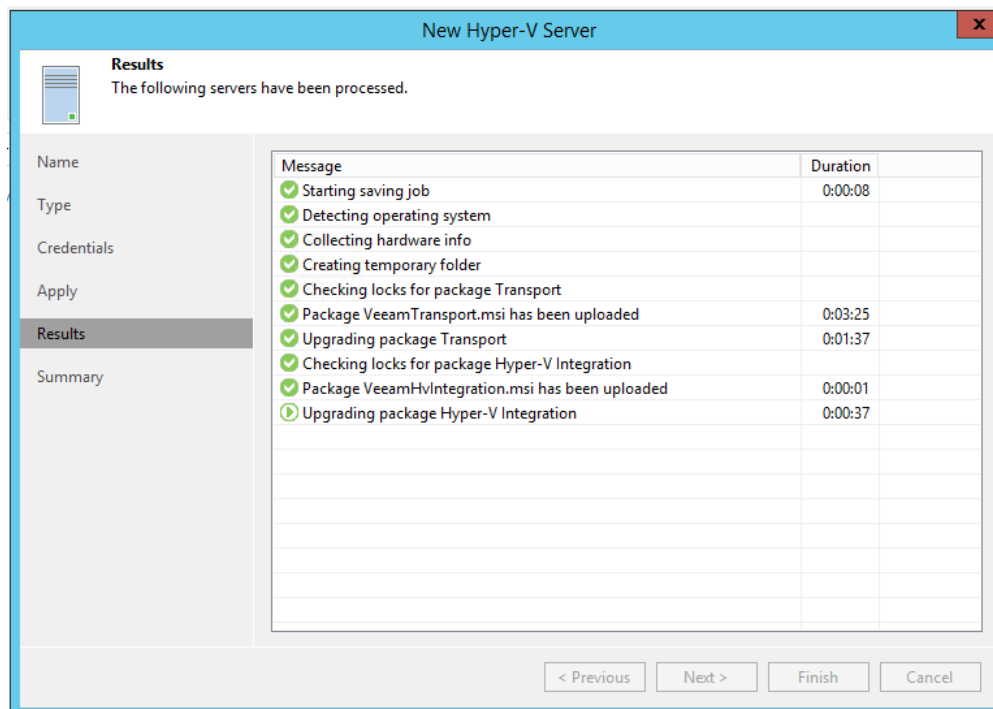
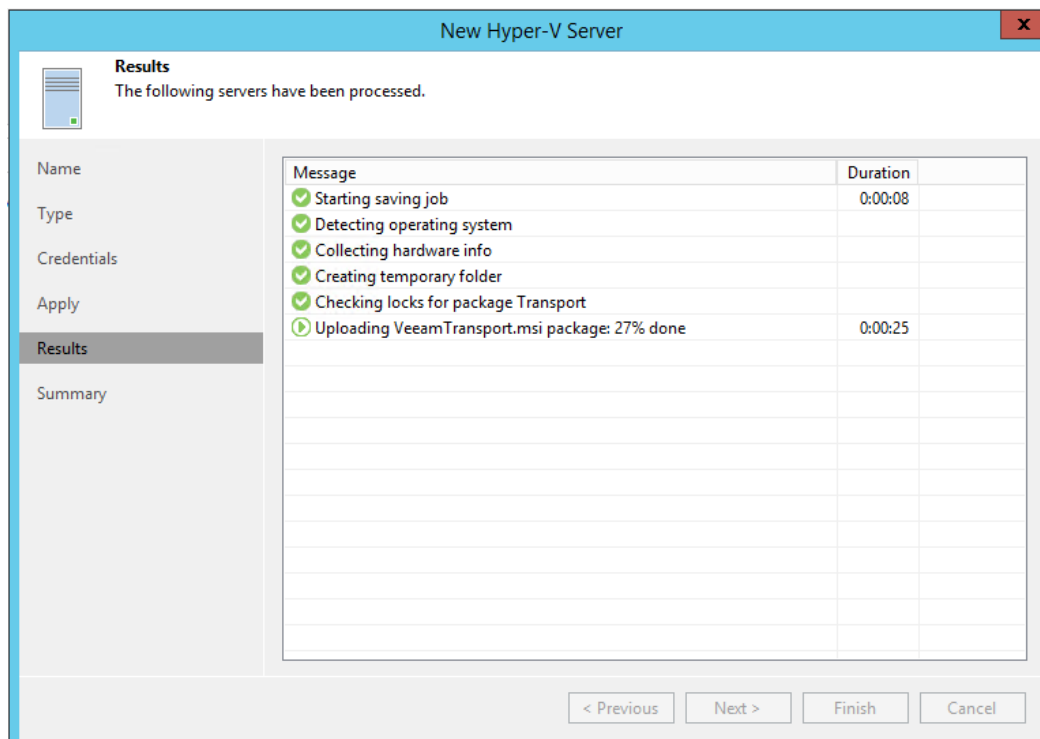
< Previous

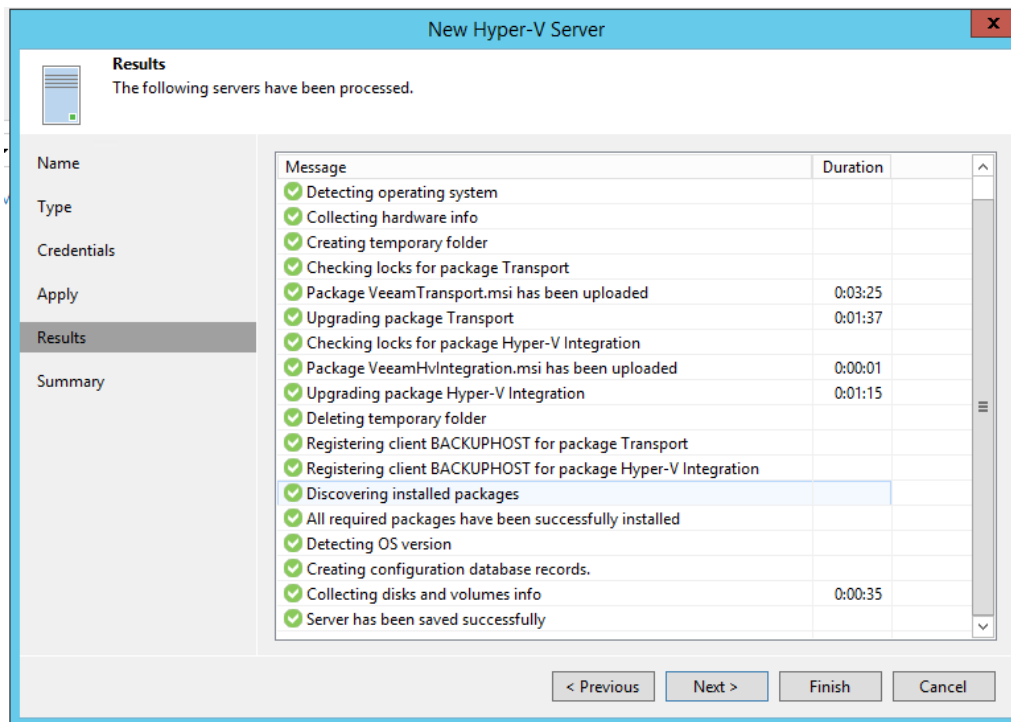
Apply

Finish

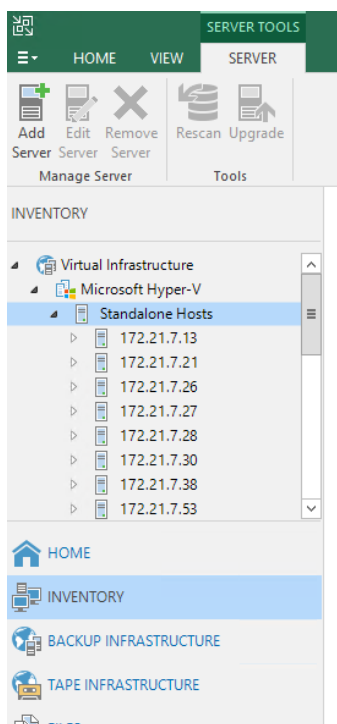
Cancel

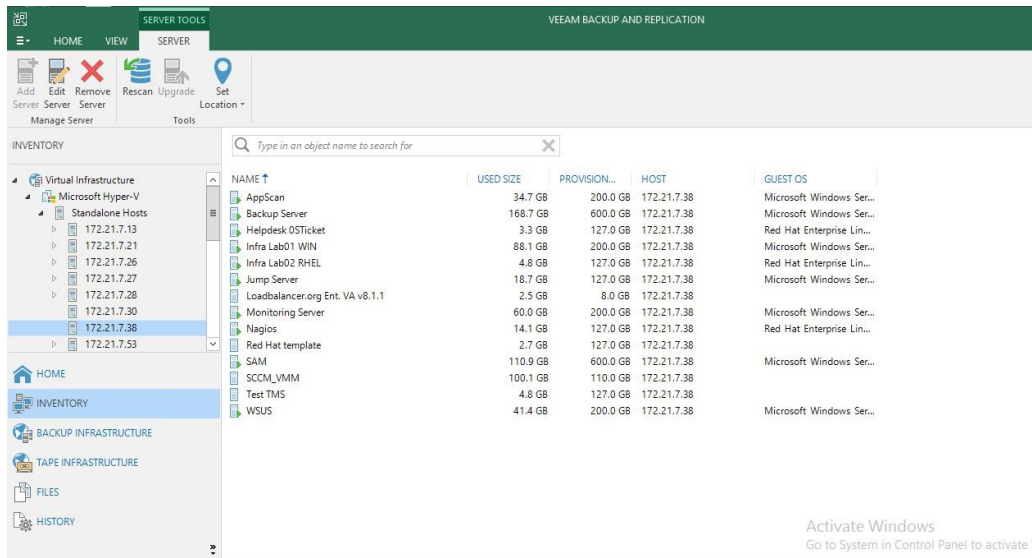
91



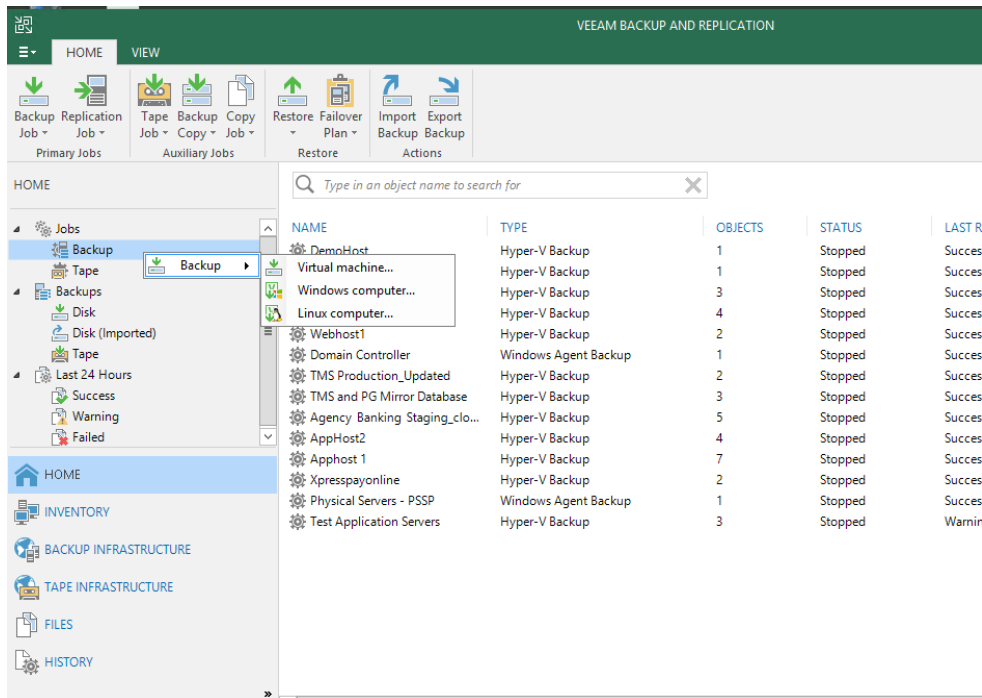


Step2:Confirm the Host Server has been added with the VM's on it





Step3: Create a backup job and schedule time to run



New Backup Job

Name
Type in a name and description for this backup job.

Name:
InfraLab01

Description:
Created by XPRESSPAY\obafemi.adedeji at 5/11/2020 4:54 PM.

< Previous Next > Finish Cancel

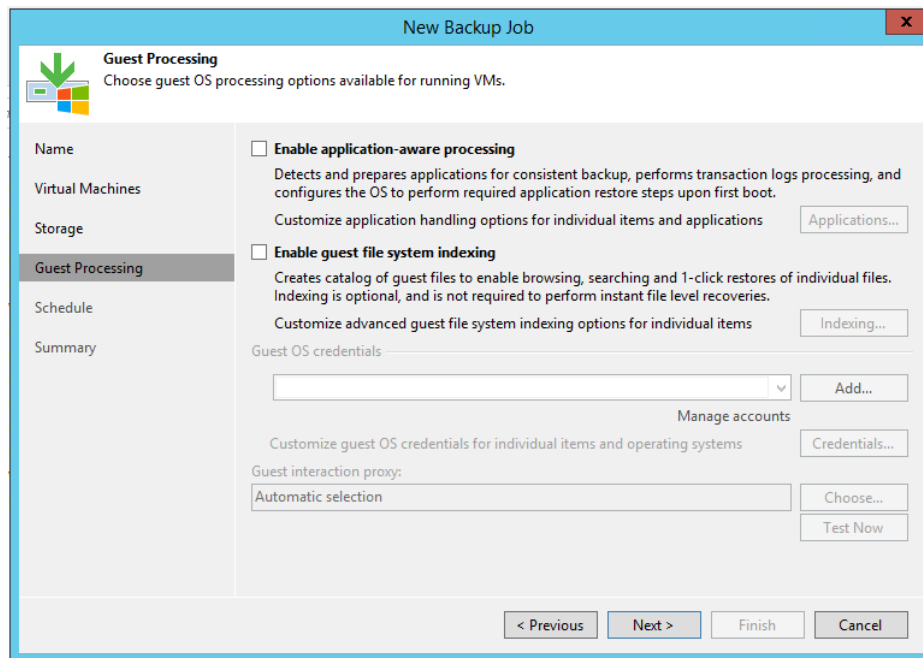
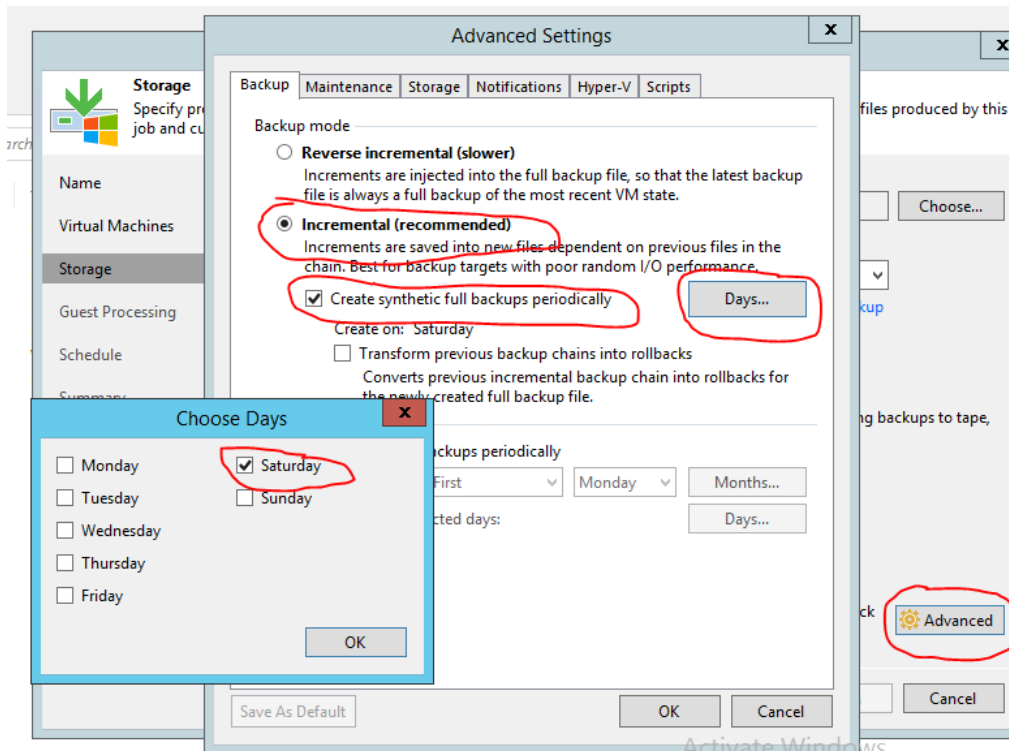
Add Objects

Select objects:

- 172.21.7.13
- 172.21.7.21
- 172.21.7.26
- 172.21.7.27
- 172.21.7.28
- 172.21.7.30
- 172.21.7.38
- AppScan
- Backup Server
- Helpdesk OSTicket
- Infra Lab01 WIN
- Infra Lab02 RHEL
- Jump Server
- Loadbalancer.org Ent. VA v8.1.1
- Monitoring Server
- Nagios
- Red Hat template

Add Cancel

Add... Remove Exclusions... Up Down Recalculate Total size: 0 B Finish Cancel



New Backup Job

Schedule
Specify the job scheduling options. If you do not set the schedule, the job will need to be controlled manually.

☒ Run the job automatically

☒ Daily at this time: 3:00 AM Everyday Days...
☐ Monthly at this time: 10:00 PM Fourth Saturday Months...
☐ Periodically every: 1 Hours Schedule...
☐ After this job: Agency Banking Staging_clone1 (Created by XPRESSPAY\omotayo.fi)

Automatic retry

☒ Retry failed items processing: 3 times
Wait before each retry attempt for: 10 minutes

Backup window

☐ Terminate job if it exceeds allowed backup window Window...
If the job does not complete within allocated backup window, it will be terminated to prevent snapshot commit during production hours.

< Previous Apply Finish Cancel

New Backup Job

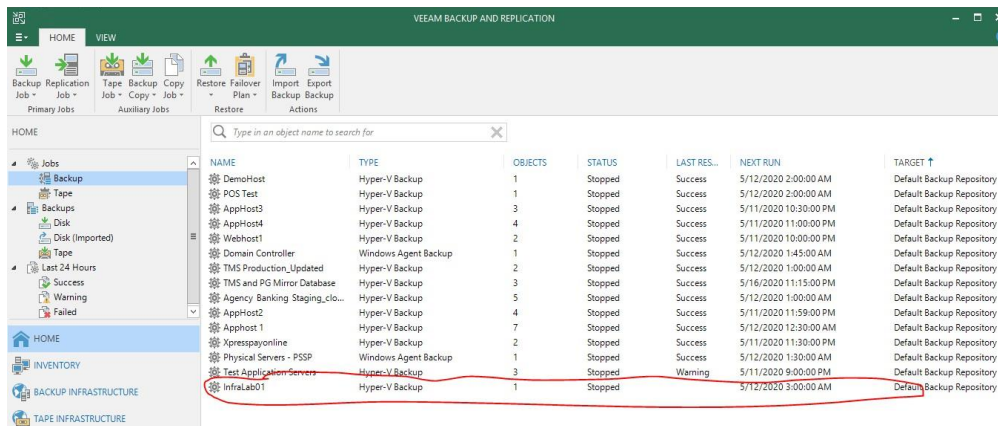
Summary
The job's settings have been saved successfully. Click Finish to exit the wizard.

Summary:

Name: InfraLab01
 Target Path: D:\Backup
 Type: Hyper-V Backup
 Source items:
 Infra Lab01 WIN (172.21.7.38)
 Target repository: Default Backup Repository
 Target repository host: BackupHost.xpresspay.com.ng
 Target repository path: D:\Backup
 Command line to start the job on backup server:
 "C:\Program Files\Veeam\Backup and Replication\Backup\Veeam.Backup.Manager.exe" backup 0f45f108-0c46-4f25-a0b4-a82871fe84a5

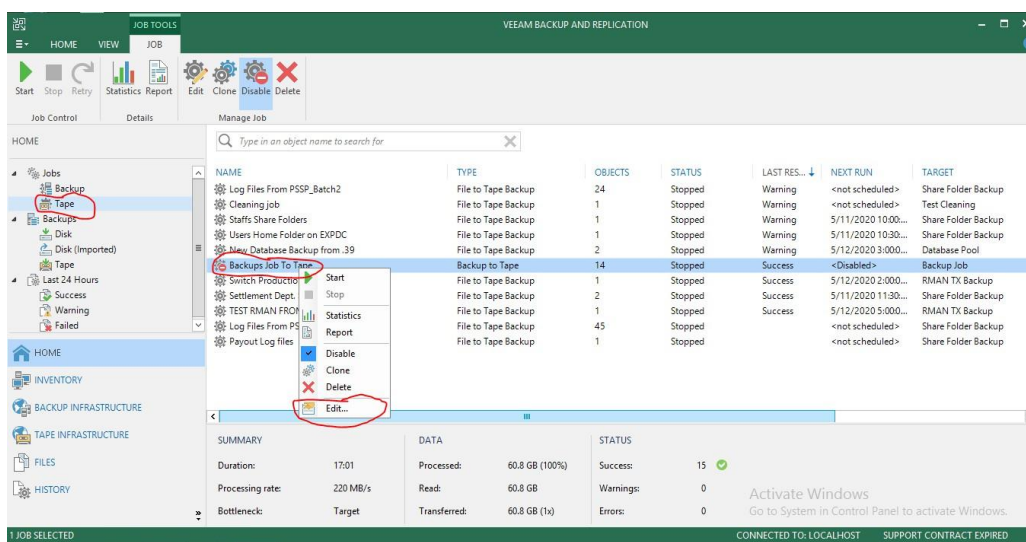
☐ Run the job when I click Finish

< Previous Next > Finish Cancel



14.7 How to push backup job to Tape Drive

Step1: Right click on Tape, select edit and follow the screenshot



Edit Backup to Tape Job [Backups Job To Tape]

Name
Type in a name and description for this tape job.

Name:
Backups Job To Tape

Description:
Created by XPRESSPAY\adedeji.obafemi at 11/23/2018 4:51 PM.

Backups
Media Pool
Incremental Backup
Options
Schedule
Summary

< Previous Next > Finish Cancel

Backups
Specify objects

Select Jobs

Select Jobs

- Agency Banking Staging_clone1
- Apphost 1
- AppHost2
- AppHost3
- AppHost4
- DemoHost
- Domain Controller
- InfraLab01
- Physical Servers - PSSP
- POS Test
- Test Application Servers
- TMS and PG Mirror Database
- TMS Production_Updated
- Webhost1
- Xpresspayonline

Type in an object name to search for

OK Cancel

Add...
Remove
Up
Down
Full: 2.86 TB
Incremental: 994 GB
Finish Cancel

Backups

Specify objects to be processed by this tape job.

Name

Backups

Media Pool

Incremental Backup

Options

Schedule

Summary

Name	Type	Size
AppHost3	Hyper-V Backup	96.8 GB
POS Test	Hyper-V Backup	37.0 GB
Xpresspayonline	Hyper-V Backup	133 GB
Agency Banking Staging_clone1	Hyper-V Backup	433 GB
Webhost1	Hyper-V Backup	255 GB
TMS Production_Updated	Hyper-V Backup	212 GB
Apphost 1	Hyper-V Backup	345 GB
AppHost2	Hyper-V Backup	196 GB
Domain Controller	Windows Agent...	572 GB
Physical Servers - PSSP	Windows Agent...	271 GB
AppHost4	Hyper-V Backup	173 GB
DemoHost	Hyper-V Backup	26.9 GB
Test Application Servers	Hyper-V Backup	868 GB
TMS and PG Mirror Database	Hyper-V Backup	299 GB
InfraLab01	Hyper-V Backup	0 B

Add...

Remove

Up

Down

Full: 2.86 TB

Incremental: 994 GB

< Previous

Next >

Finish

Cancel

Media Pool

Specify the media pool to perform backup to.

Name

Backups

Media Pool

Incremental Backup

Options

Schedule

Summary

Media pool:

Backup Job (HP MSL G3 Series 7.00)

Add New...

Schedule...

Tapes: 16

Free space: 1.38 TB

Media set: Do not create, always continue using current media set

Retention: Never overwrite data

Parallel processing: Disabled

Encryption: Disabled

Export to vault: do not export

WORM: False

< Previous

Next >

Finish

Cancel

Edit Backup to Tape Job [Backups Job To Tape]

Incremental Backup

Specify media pool for incremental backups.

Name

Backups

Media Pool

Incremental Backup

Options

Schedule

Summary

☒ Archive incremental backups to tape

Media pool for incremental backups:

Backup Job (HP MSL G3 Series 7.00)

Add New...

Tapes: 16

Free space: 1.38 TB

Media set: Do not create, always continue using current media set

Retention: Never overwrite data

Parallel processing: Disabled

Encryption: Disabled

Export to vault: do not export

WORM: False

< Previous

Next >

Finish

Cancel

Edit Backup to Tape Job [Backups Job To Tape]

Options

Specify tape job options.

Name

Backups

Media Pool

Incremental Backup

Options

Schedule

Summary

Media automation

☐ Eject media upon job completion

This option makes the job automatically eject tape from drives upon completion, so that tape does not stay in the drive, which is a best practice.

☐ Export current media set upon job completion

Days...

This option makes the job automatically close and export the current media set on specific days.

Advanced job settings include compression, notification settings, automated post-job activity and other settings.

⚙️ Advanced

< Previous

Next >

Finish

Cancel

102



Schedule

Specify the job scheduling options. If you do not set the schedule, the job will need to be controlled manually.

Name

Backups

Media Pool

Incremental Backup

Options

Schedule

Summary

☒ Run the job automatically

☒ Daily at this time:

6:00 AM

Everyday

Days...

☐ Monthly at this time:

10:00 PM

Fourth

Saturday

Months...

☐ After this job:

Agency Banking Staging_clone1 (Created by XPRESSPAY\omot)

Schedule...

☐ As new backup files appear:

Schedule...

Wait for backup jobs

If some linked backup jobs are still running, wait for up to:

3

hours

☐ Prevent this job from being interrupted by source backup jobs

If the source backup job starts while the tape job is still running, the source job will be placed on hold instead of interrupting the tape job and resulting in incomplete tape backups. Using this option may result in source backup jobs starting significantly later than their scheduled start time, thus potentially missing recovery point objectives.

< Previous

Apply

Finish

Cancel



Summary

You can copy the job settings below for future reference.

Name

Backups

Media Pool

Incremental Backup

Options

Schedule

Summary

Summary:

Name: Backups Job To Tape

Media pool for full backups: Backup Job

Media pool for incremental backups: Backup Job

Command line to start the job on backup server:

"C:\Program Files\Veeam\Backup and Replication\Backup\Veeam.Backup.Manager.exe" backup 9b7711a9-5946-4b69-86b7-12a58d68c52f

☐ Run the job when I click Finish

< Previous

Next >

Finish

Cancel

14.8 Backup monitoring and Reporting

- 1) Login to the backup web console via
"<https://backuphost.xpresspay.com.ng:9443/login.aspx>"
- 2) Supply Username and Password

Welcome! Please log in.

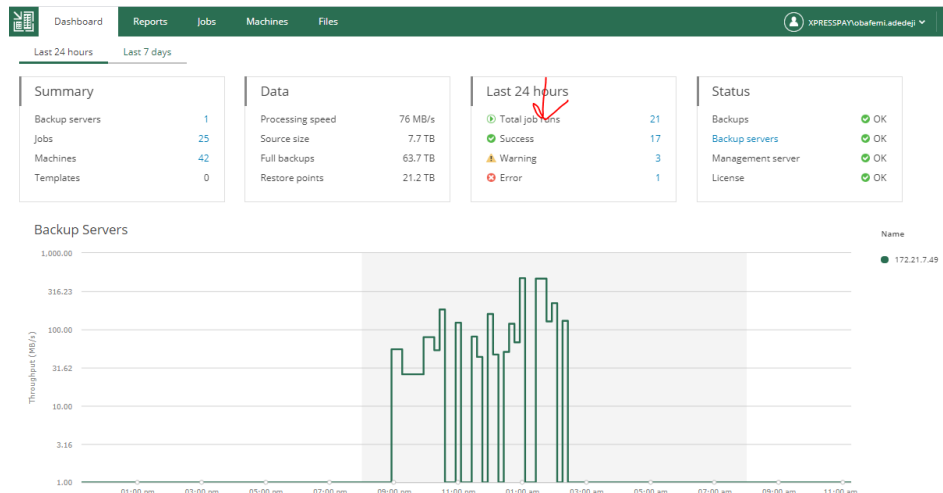
Username: xpresspay\lobafemi.adedeji

Password:

☒ Remember me

Login

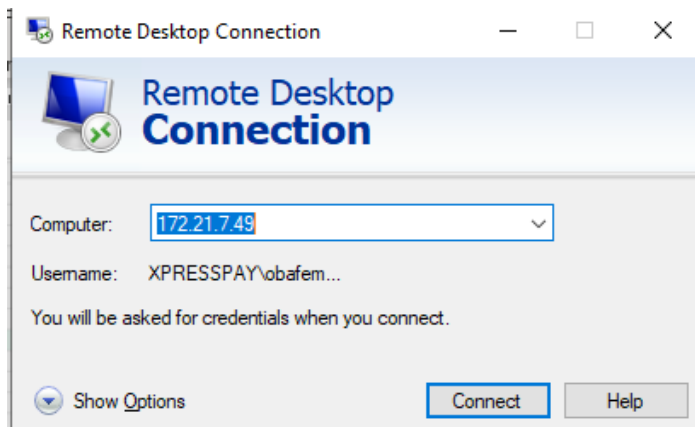
- 3) Click on total job run to see the scheduled job



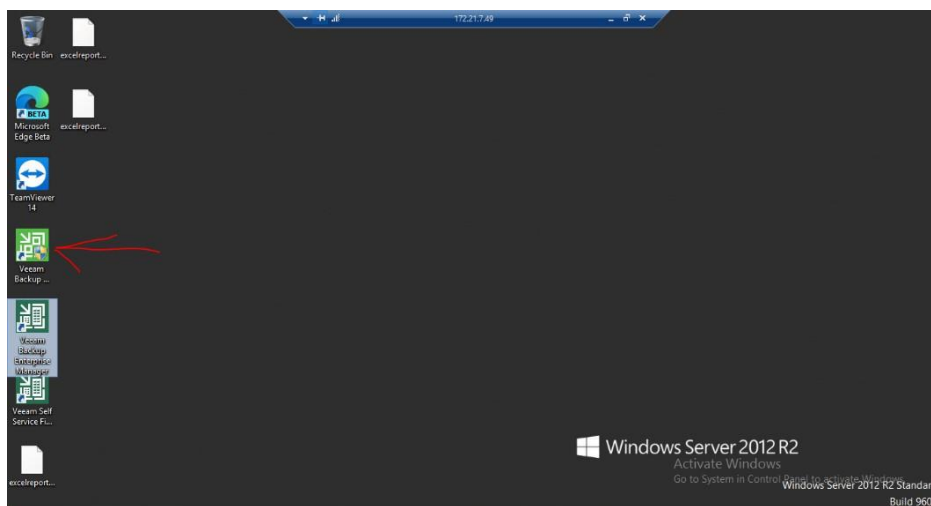
Job name	Backup server	Start Time	End Time	Status
Users Home Folder on EXFDC	172.21.7.49	4/26/2020 10:30:14 pm	4/26/2020 10:32:28 pm	Success
AppHost3	172.21.7.49	4/26/2020 10:30:14 pm	4/26/2020 10:35:39 pm	Success
AppHost4	172.21.7.49	4/26/2020 11:00:18 pm	4/26/2020 11:07:36 pm	Success
Settlement Dept. Share Folder	172.21.7.49	4/26/2020 11:30:03 pm	4/26/2020 11:33:48 pm	Success
Xpresspayonline	172.21.7.49	4/26/2020 11:30:03 pm	4/26/2020 11:41:28 pm	Success
AppHost2	172.21.7.49	4/26/2020 11:59:20 pm	4/27/2020 12:11:55 am	Success
AppHost1	172.21.7.49	4/27/2020 12:30:13 am	4/27/2020 01:00:31 am	Success
TMS Production_Updated	172.21.7.49	4/27/2020 01:00:16 am	4/27/2020 01:02:02 am	Success
Agency Banking Staging_clone1	172.21.7.49	4/27/2020 01:00:16 am	4/27/2020 01:07:36 am	Success
Physical Servers - PSSP	172.21.7.49	4/27/2020 01:30:26 am	4/27/2020 01:41:31 am	Success
Domain Controller	172.21.7.49	4/27/2020 01:45:16 am	4/27/2020 01:54:20 am	Success
POS Test	172.21.7.49	4/27/2020 02:00:05 am	4/27/2020 02:01:54 am	Success
DemoHost	172.21.7.49	4/27/2020 02:14:46 am	4/27/2020 02:19:25 am	Success
Switch Production RMAN From .46	172.21.7.49	4/27/2020 02:00:05 am	4/27/2020 02:47:34 am	Success
TEST RMAN FROM .46	172.21.7.49	4/27/2020 05:00:05 am	4/27/2020 05:07:46 am	Success
Backups Job To Tape	172.21.7.49	4/27/2020 06:28:57 am	4/27/2020 06:45:16 am	Success
New Database Backup from .39	172.21.7.49	4/27/2020 03:00:14 am	4/27/2020 03:48:39 am	Warning
Test Application Servers	172.21.7.49	4/26/2020 09:00:20 pm	4/26/2020 10:12:41 pm	Warning
Staffs Share Folders	172.21.7.49	4/26/2020 10:00:08 pm	4/26/2020 10:21:50 pm	Warning

14.9 To push Backup jobs to Tape

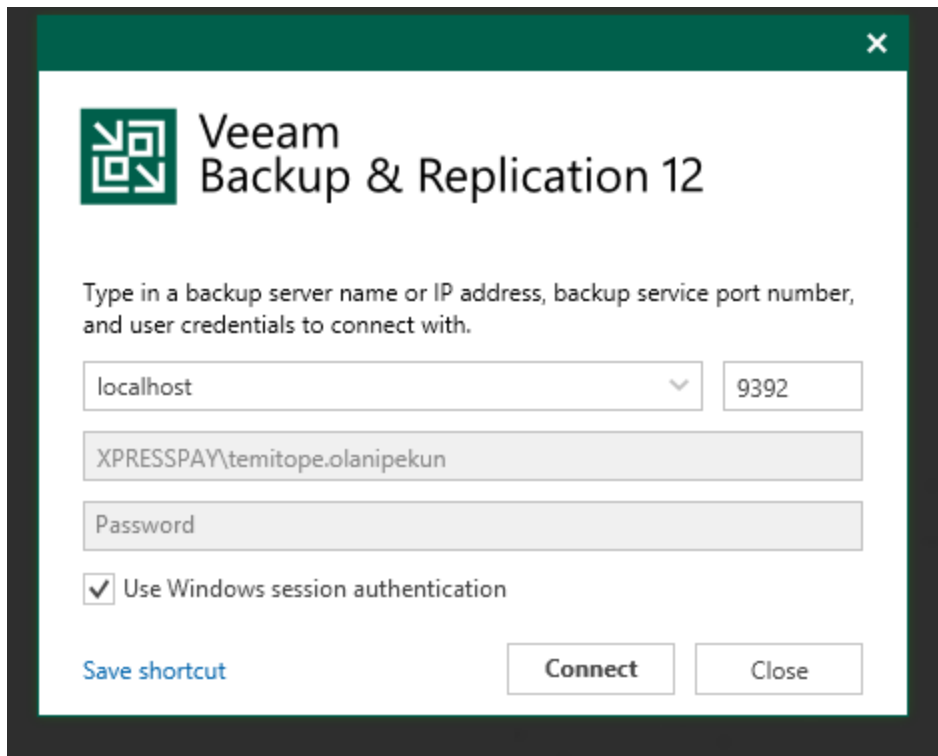
1) Login to the Veeam backup server (172.21.7.49)



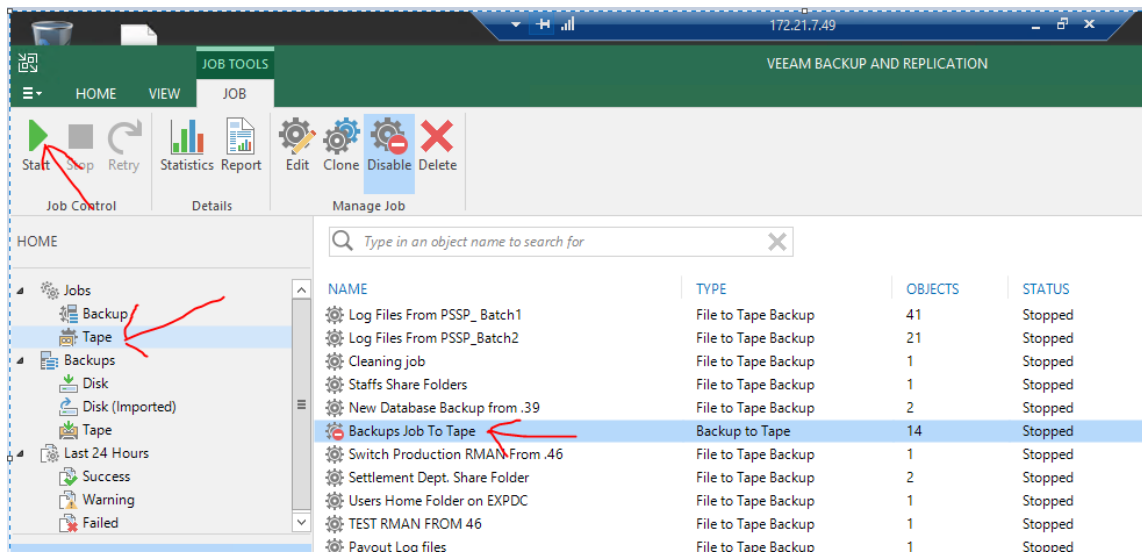
2) Double Click on the Veeam backup and replication icon



3) Click on Connect



4) Select Tape>>>Backupjob to Tape>>> Start



15.0 How to replicate VM servers to DR site using Nutanix Data Protection

15.0 Background

VM replication creates exact copies of source VM's (as we have in at the head office) and puts those copies (called VM Replicas) on the target Hyper-V host/storage (DR Site). Each time a replication job runs (every 5 minutes), VM replicas are updated, so they are identical to the source VMs. VM replicas are fully functional VMs which are stored in the powered-off state at the DR site, so they do not consume compute resources.

15.1 Process Objective (Purpose)

The main purpose of VM replication is to avoid loss of data and give a minimum down time during any unexpected disasters.

This process is a guideline to assist us perform VM replications either to the DR site or to another host machine. If well followed, you can know the amount of time, steps, challenges you might likely encounter and how to solve the challenges, even before you start.

15.2 Process Owner

Group Head, IT

15.3 Process Attribute

The process attribute captures the input and output activities. The steps required in implementing the process, the approving officers and estimated turnaround time (TAT).

Role	Task	Process Steps	Estimated TAT
		1.1 Connect to the Nutanix Prism Central web console at the Head office. 1.2 Navigate to the VM section and add the VMs to respective categories. 1.3 With the Categories of the VMs, create a Protection Policy and Recovery Plan. 1.4 Navigate to Data Protection and create a Protection Policy and Recovery Plan respectively Follow the screen shot attached below.	1 hours

Server Administrator	To replicate a new VM server to the DR site	2.1 Upon successful creation of Protection Policy and Recovery Plan. 2.2 VM replications will begin automatically. 2.3 If any issue or skipped replication warning is reported by the cluster, escalate. 2.4 Verify the VMs recovery points are created from the VM recovery point console.	(Depending on the size of the Sources server)
----------------------	---	---	---

15.4 Key Risks/Control

In event a process cannot be completed. The workaround process should be defined, which may include escalation to line supervisor or resort to manual process of the exceptions notice. The details as it relates to specific process should be captured and stated here. Where there are no exceptions/workaround, this should be stated as Not Applicable.

Risk	Controls
Network related issues	Escalate to the Network team
Resource Availability	Escalate to Lead IT Infrastructure team.

15.5 Sample Process Documentation

HQ Prsim Central : <https://172.22.51.38:9440>

DR Prsim Central : <https://172.22.80.31:9440>

← → ↻ 🏠 https://172.22.51.38:9440/console/#page/explore/s/vms?entity_type=vm&term_1=vm&term_2=vm

Congratulations! 90 days of Prism Ultimate is on us, enjoy! [See what's new](#)

☰ 🔍 **VMs** VM Type=User VM × ▶ List ☆ Prism

VM ECashier

- Disable Emergency Measurements
- Disable Anomaly Detection
- Protect
- Unprotect
- Create Recovery Point
- Migrate
- Add to Recovery Plan
- Run Playbook
- Manage Categories
- Quarantine VMs
- Unquarantine VMs
- Install NGT
- Manage NGT Applications
- Upgrade NGT

Alerts Events Metrics ▾ External VMs

CPU	Memory	IP Addresses	Cluster	Hypervisor	OS
	6 GiB	172.21.7.47	hqhci.xpresspay.com.ng	AHV	-
	4 GiB	172.21.7.48	hqhci.xpresspay.com.ng	AHV	Wi
	2 GiB	172.21.7.51	hqhci.xpresspay.com.ng	AHV	-
	7 GiB	172.21.7.43	hqhci.xpresspay.com.ng	AHV	Wi
	5 GiB	172.21.7.52	hqhci.xpresspay.com.ng	AHV	Wi
	0 GiB	172.21.7.33	hqhci.xpresspay.com.ng	AHV	-

Manage VM Categories

Set Categories

 You have selected ECashier. Selected categories will be applied to the VM.

PROD-APPS: Apps-FullF... 

Search 🔍

Possible Associated Policies

For all categories associated with a policy, please go to the policy page.

Policy Type	Policy Name	Applies Because Of
Recovery Plan	Apps-FullFailover1	PROD-APPS: Apps-FullFailover1
Protection Policy	Apps-FullFailover1	PROD-APPS: Apps-FullFailover1

Protection Policies

VMS

Templates

OVA's

Images

Catalog Items

Storage Containers

Volume Groups

Storage Policies

New

Network & Security

Data Protection

Protection Summary

Protection Policies

Recovery Plans

VM Recovery Points

Hardware

Lock Navigation Bar

Create Protection Policy

Actions

Type name to filter by

Viewing all 11 Protection Policies

☐ Policy Name	Schedules
☐ Apps-CrashConsistent1	1 Schedule
☐ Apps-CrashConsistent2	1 Schedule
☐ Apps-CrashConsistent3	1 Schedule
☐ Apps-FullFailover1	1 Schedule
☐ Apps-FullFailover2	1 Schedule
☐ Apps-FullFailover3	1 Schedule
☐ Apps-FullFailover4	1 Schedule
☐ DB-CrashConsistent1	1 Schedule

Create Protection Policy

1 Configure Schedule

2 Add Entities

Protection Policy will be automatically synced to all connected locations and protect entities from all the locations. Upon failover, Entities will be automatically replicated back using the same policy.

Policy name Smile

+ Add Recovery Location

Start Protection Immediately

Primary Location

Local AZ : hqhcl.xpresspay.com.ng

+ Add Local Schedule

+ Add Schedule

PC_172.22.80.31 : drhcl.xpresspay.com...

+ Add Local Schedule

Cancel

Next

Create Protection Policy

1 Configure Schedule

2 Add Entities

Protection Policy will be automatically synced to all connected locations and protect entities from all the locations. Upon failover, Entities will be automatically replicated back using the same policy.

Policy name similw

Primary Location

Location

Local AZ

Cluster

hqhcl.xpresspay.com.ng

Save

+ Add Schedule

- 1 Configure Schedule 2 Add Entities

protect entities from all the locations. Upon failover, Entities will be automatically replicated back using the same policy.



+ Add Recovery Location Start Protection **Immediately**

+ Add Schedule

Recovery Location

Location

PC_172.22.80.31

Cluster

drhci.xpresspay.com.ng

Cancel Save

Cancel

Next

Add Schedule

Primary Location

Recovery Location

Local AZ : hqhci.xpress...

PC_172.22.80.31 : drhci....

Protection Type

☒ Asynchronous ☐ Synchronous

Take Snapshot Every

Hour(s) : 12

Retention Type

☒ Linear ☐ Roll-up

Retention on Local AZ : hqhci.xpresspay.com.ng

12 Recovery Point(s)

Retention on PC_172.22.80.31 : drhci.xpresspay.com.ng

24 Recovery Point(s)

☒ Reverse retention for entities on recovery location [Learn More](#)

For VMs protected on PC_172.22.80.31 : drhci.xpresspay.com.ng

Retention on PC_172.22.80.31 : drhci.xpresspay.com.ng : 12 Recovery Point(s)

Retention on Local AZ : hqhci.xpresspay.com.ng : 24 Recovery Point(s)

Cancel

Save Schedule

Save Schedule

Click Next and Add the VM Category

← → ↻ 🏠 https://172.22.51.38:9440/console/#/page/explore/protection_rule 80% ☆ 📧 📞 ⚙️ 🔍

Create Protection Policy ? ✕

1 Configure Schedule 2 Add Entities

VMs only on Primary and Recovery Location will be protected

Categories

- ☐ Key: Value
- ☐ Search for a category
- ☐ VirtualNetworkType : Test
- ☐ VirtualNetworkType : Tenant
- ☐ TemplateType : Vm
- ☐ TemplateType : Application
- ☐ Quarantine : Forensics
- ☐ Quarantine : Default
- ☐ Environment : Testing
- ☐ Environment : Staging
- ☐ Environment : Production

Cancel Add

Primary Location

Local AZ : hghcl.xpresspay.co... 0 VM(s)

RPO: 12 hour(s)
Loc... : hghcl.xpres... 12 Recovery Point(s)
PC_17... : drhcl.xpr... 24 Recovery Point(s)
Crash consistent, reversed retention

PC_172.22.80.31 : drhcl.xpress... 0 VM(s)

Back  Create

Sample of a Protection Policy

Apps-CrashConsistent2 : Update Clone Delete ? ✕

Summary Alerts

Total Violations (0 VM(s)) 0 0 0 Last Week

0

Nov 15 Nov 16 Nov 17 Nov 18 Nov 19 Nov 20 Nov 21 Nov 22

Critical Warning Info

Policy Overview 1 total schedules

Primary Location

Local AZ : hghcl.xpresspay.co... 9 VM(s)

RPO: 1 day(s)
Loc... : hghcl.xpres... 5 Recovery Point(s)
PC_17... : drhcl.xpro... 8 Recovery Point(s)
Crash consistent, reversed retention

PC_172.22.80.31 : drhcl.xpress... 9 VM(s)

Create a Recovery Plan

← → ↺ 🏠 https://172.22.51.38:9440/console/#/page/explore/ebrowser?entity_type=recovery_plan

Congratulations! 90 days of Prism Ultimate is on us, enjoy! [See what's included](#)

☰ 🔍 Recovery Plans ☆ Prism

VMS
 Templates ☆
 OVAs ☆
 Images ☆
 Catalog Items ☆
 Storage Containers ☆
 Volume Groups ☆
 Storage Policies **New** ☆
 Network & Security
 Data Protection
 Protection Summary ☆
 Protection Policies ☆
Recovery Plans ☆
 VM Recovery Plans
 Hardware
 Clusters ☆
 Hosts ☆
 Disks ☆
 GPUs ☆
 Activity
 Lock Navigation Bar

Create Recovery Plan Actions ▾

Type text to filter by

Viewing all 13 Recovery Plans

☐	Name	Primary Location	Recovery Location	Entities
☐	Apps-FullFailover1	Local AZ	PC_172.22.80.31	9
☐	Apps-FullFailover2	Local AZ	PC_172.22.80.31	9
☐	Apps-FullFailover3	Local AZ	PC_172.22.80.31	9
☐	Apps-FullFailover4	Local AZ	PC_172.22.80.31	7
☐	CRASH_DB_RECOVERY	Local AZ	PC_172.22.80.31	-
☐	FULL_DB_FAILOVER2	Local AZ	PC_172.22.80.31	6
☐	FULL_DB_FAILOVER	Local AZ	PC_172.22.80.31	5
☐	PayGateWay	Local AZ	PC_172.22.80.31	2
☐	Prod-VMs	Local AZ	PC_172.22.80.31	4
☐	Staging_Entity	Local AZ	PC_172.22.80.31	3
☐	tempfollover	Local AZ	PC_172.22.80.31	1

Apps-FullFailover1

1 General 2 Power On Sequence 3 Network Settings

Recovery Plan Name

Apps-FullFailover1

Recovery Plan Description

Recovery Plan Description

Locations

Primary Location Recovery Location

Local AZ PC_172.22.80.31

Next

: Apps-FullFailover1

1 General2 Power On Sequence3 Network Settings

1 Stages 1 Total | 1 Categories + Add New Stage

Stage 1 1 Total | 1 Categories Actions

Name

☐ CATEGORY PROD-APPS Apps-FullFailover1

< Back

Next

: Apps-FullFailover1

1 General2 Power On Sequence3 Network Settings

Static IP addresses will be preserved post recovery. Dynamic IPs will not be preserved for VMs on ESX or VMs using non-IPAM networks.

Network Type

Are all L2 production networks in the recovery plan, stretched? If networks are stretched, then live migration of VMs is possible.

☒ Non-stretch networks ☐ Stretch networks ?

How it works? + Add Network Mapping

Local AZ (Primary)

Production

Virtual Network or Port Group

VLAN_7

Gateway IP / Prefix Length

172.21.71

24

Test Failback

Virtual Network or Port Group

VLAN_7

Gateway IP / Prefix Length

172.21.71

24

PC_172.22.80.31 (Recovery)

Production

Virtual Network or Port Group

VLAN_7

Gateway IP / Prefix Length

172.21.71

24

Test Failover

Virtual Network or Port Group

VLAN_7

Gateway IP / Prefix Length

172.21.71

24

Advanced Settings

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Done

VM recovery Points

116

